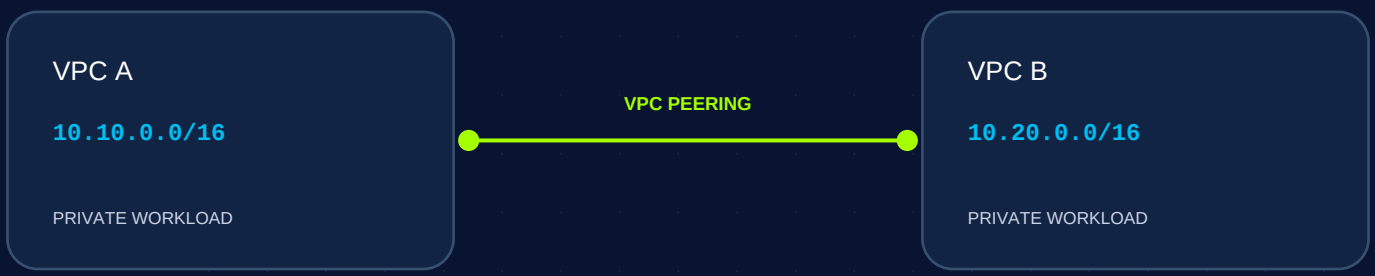
AWS

VPC Peering

Connect two VPCs.
Prove private connectivity.

A practical networking implementation using EC2,
custom route tables, security groups and a bastion host.



2 VPCs / 3 EC2 INSTANCES / 1 PRIVATE CONNECTION

AWS ARCHITECT ASSOCIATE

Name: Aniket Rajendra Kolapkar

Batch: AWS Internship Batch 1

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1. Executive Summary

This project demonstrates private IPv4 communication between two separate AWS VPCs by using VPC Peering. The implementation combines non-overlapping CIDR ranges, subnet design, route-table entries in both directions, and scoped security-group rules. App-A and App-B were tested from their private EC2 shells, without public IPv4 addresses on either application instance.

A controlled negative test removed the relevant inbound ICMP Echo Request rule on App-B. Ping from App-A then showed 100% packet loss. Restoring the rule returned the test to four replies with 0% packet loss. This shows how a security-group rule can control the tested traffic path.

2. Objectives & Scope

• Create two VPCs with non-overlapping IPv4 CIDR blocks.

• Place a bastion host in VPC-A and application instances in private subnets.

• Create VPC Peering and configure forward and return routes.

• Restrict administration and ICMP test traffic with security groups.

• Validate connectivity, controlled failure, and recovery.

• Collect evidence and clean up project resources after testing.

3. Architecture & Addressing

Component	CIDR / Private IP	Role
VPC-A	10.10.0.0/16	Bastion and App-A network
Public subnet A	10.10.1.0/24	Bastion access
Private subnet A	10.10.2.0/24	App-A: 10.10.2.10
VPC-B	10.20.0.0/16	App-B network
Private subnet B	10.20.1.0/24	App-B: 10.20.1.10
Region	ap-south-1	AWS Mumbai

Logical traffic path

Administrator → Bastion (SSH) → private application hosts for administration. App-A ↔ App-B application traffic uses the VPC Peering connection and private IP routes; the bastion is not the data path for the inter-VPC ping.

4. Routing & Security Controls

VPC Peering does not automatically add routes. Each VPC needs a route to the other VPC's CIDR, and the return route must be present. The relevant route tables must also be associated with the subnets that need connectivity.

Route table	Destination	Target / purpose
ewb-rt-a-public	10.20.0.0/16	VPC Peering
ewb-rt-a-private	10.20.0.0/16	VPC Peering
ewb-rt-b-private	10.10.0.0/16	VPC Peering
ewb-rt-a-public	0.0.0.0/0	Internet Gateway for public subnet

Security-group intent

• ewb-sg-bastion: SSH TCP/22 from the administrator client IP (/32).

• ewb-sg-app-a: SSH from bastion private IP 10.10.1.10/32; ICMP Echo allowed from App-B 10.20.1.10/32 for reverse-direction testing.

• ewb-sg-app-b: SSH from bastion private IP 10.10.1.10/32; ICMP Echo Request allowed from App-A 10.10.2.10/32.

Security principle

The application instances had no public IPv4 addresses. Administrative access used the bastion, and the tested ICMP permissions were restricted to the peer application's private IP rather than broadly opened to the internet.

5. Testing Methodology & Results

The following outcomes are based on the commands and results reported during the guided walkthrough. Commands were run from private EC2 shells and used private destination IP addresses.

Test	Action	Observed result	Interpretation
App-A → App-B	ping -c 4 10.20.1.10	4 replies; 0% loss	Forward path worked
App-B → App-A	ping -c 4 10.10.2.10	4 replies; 0% loss	Reverse path worked
Negative test	Remove App-B inbound ICMP Echo Request from 10.10.2.10/32; ping App-B	100% loss	Rule blocked tested traffic
Recovery	Restore ICMP rule and repeat ping	4 replies; 0% loss	Connectivity recovered

Interpretation & limitation

These tests show that the configured private ICMP path worked in both directions and that removing the relevant inbound rule blocked the tested traffic. Ping success does not by itself validate every application protocol. The initial pre-peering baseline ping was not captured, so this report does not present it as a verified result.

6. Cleanup & Lessons Learned

After evidence collection, the user reported deleting the three project EC2 instances, VPC Peering connection, project security groups, Internet Gateway, project subnets, custom route tables, both project VPCs, and the project key pair through the AWS Console.

Final cost hygiene

Verify in the AWS Console that no project EBS volumes, snapshots, Elastic IPs, NAT Gateways, or other billable resources remain. Review Billing and Cost Management because charges already incurred may remain visible after deletion. This report reflects user-reported cleanup actions and is not an AWS-generated audit log.

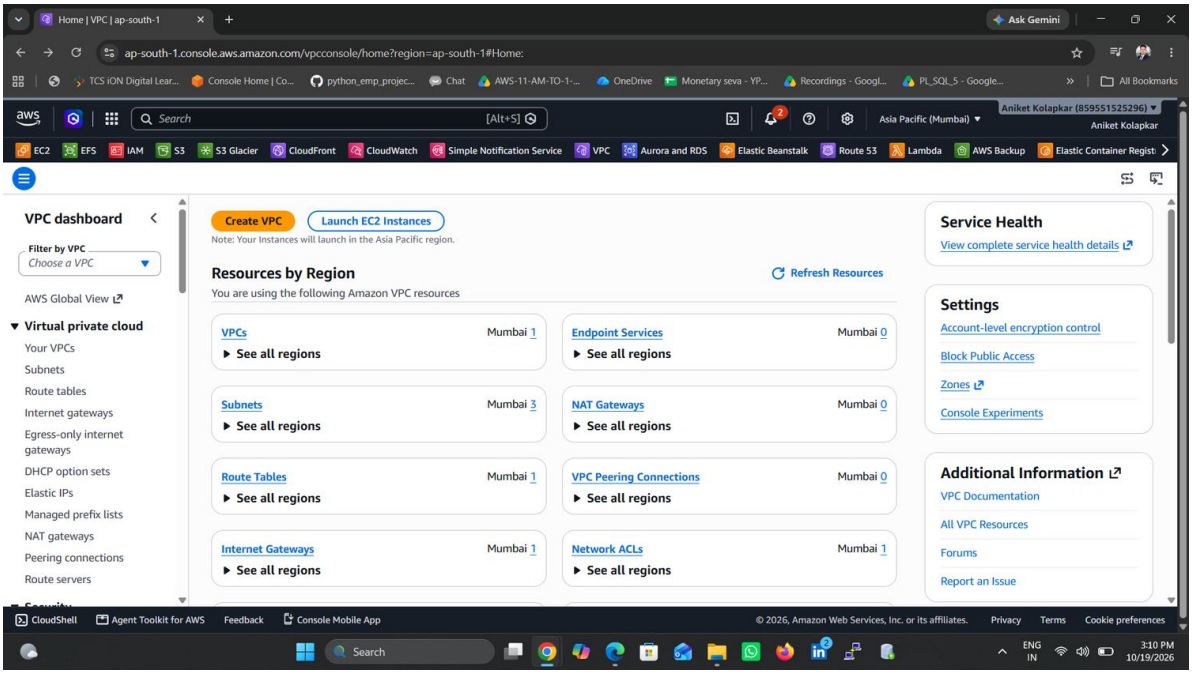
Key learning

Reliable private connectivity depends on CIDR planning, peering state, forward and return routes, route-table associations, and security-group permissions. Testing both directions and deliberately removing/restoring one permission provides a practical troubleshooting method.

Appendix A. Screenshot Evidence

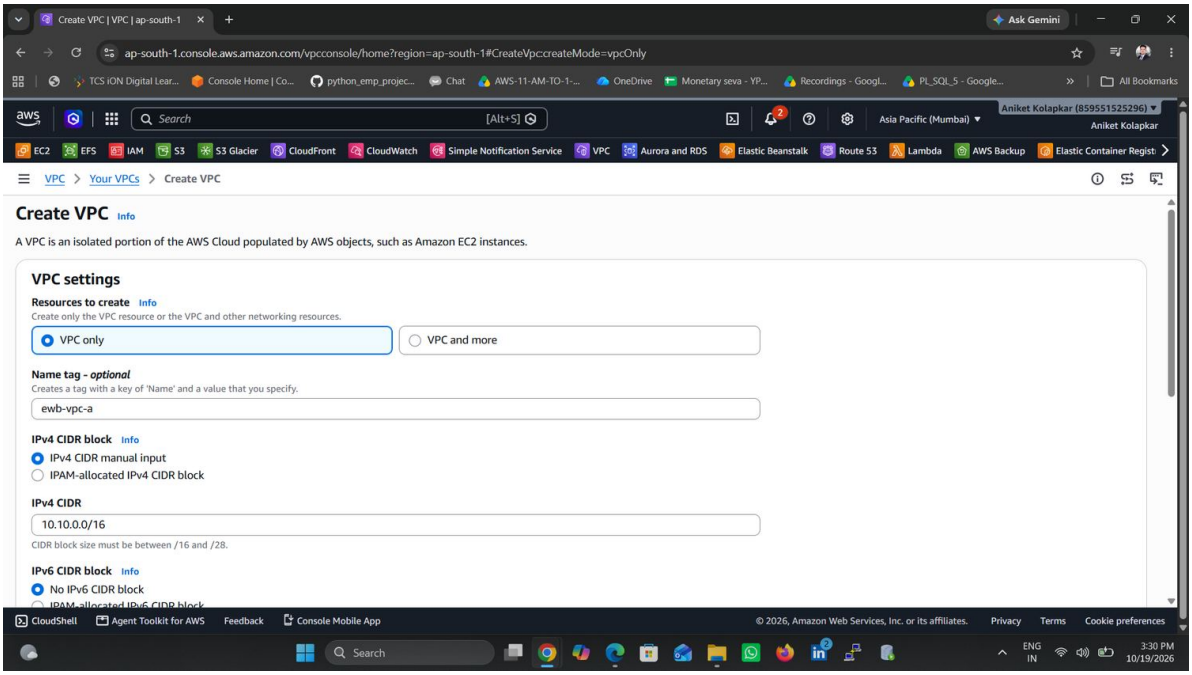
The following appendix includes the 97 supplied screenshots in numeric order. Each image has a short caption describing its apparent role in the implementation sequence. Review screenshots before external sharing and do not include the .pem private-key file itself.

Screenshot 01 — AWS Console overview before creating the project network



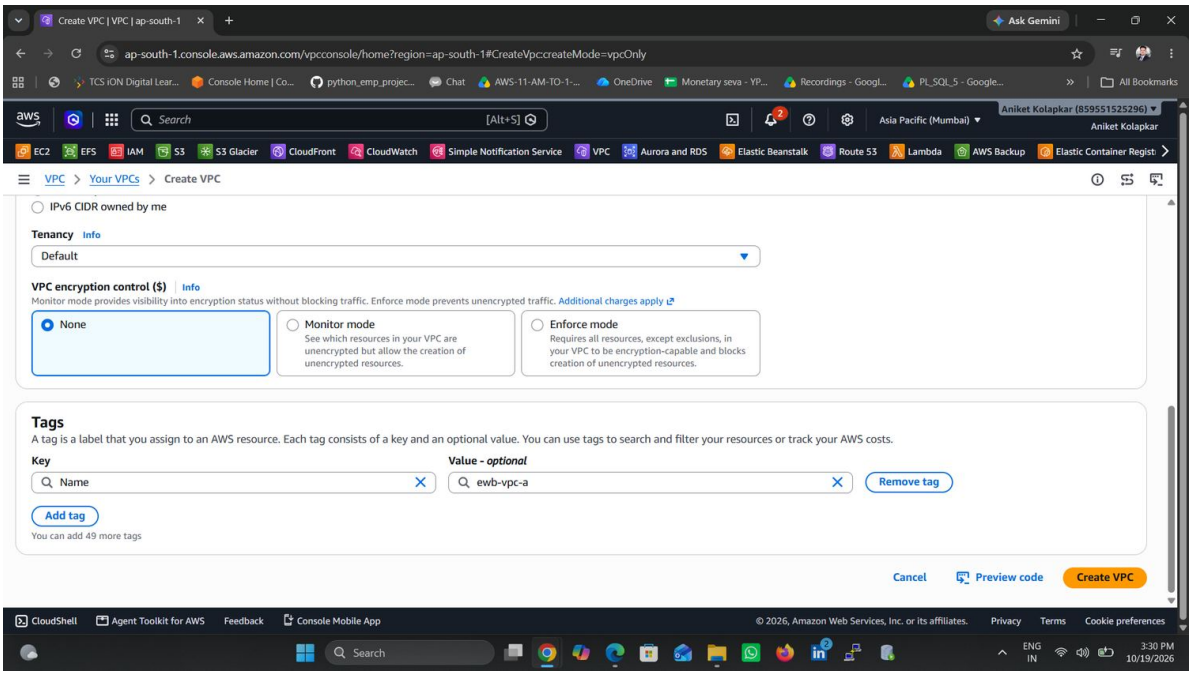
Evidence file: 1.png

Screenshot 02 — Create VPC-A and configure its IPv4 CIDR



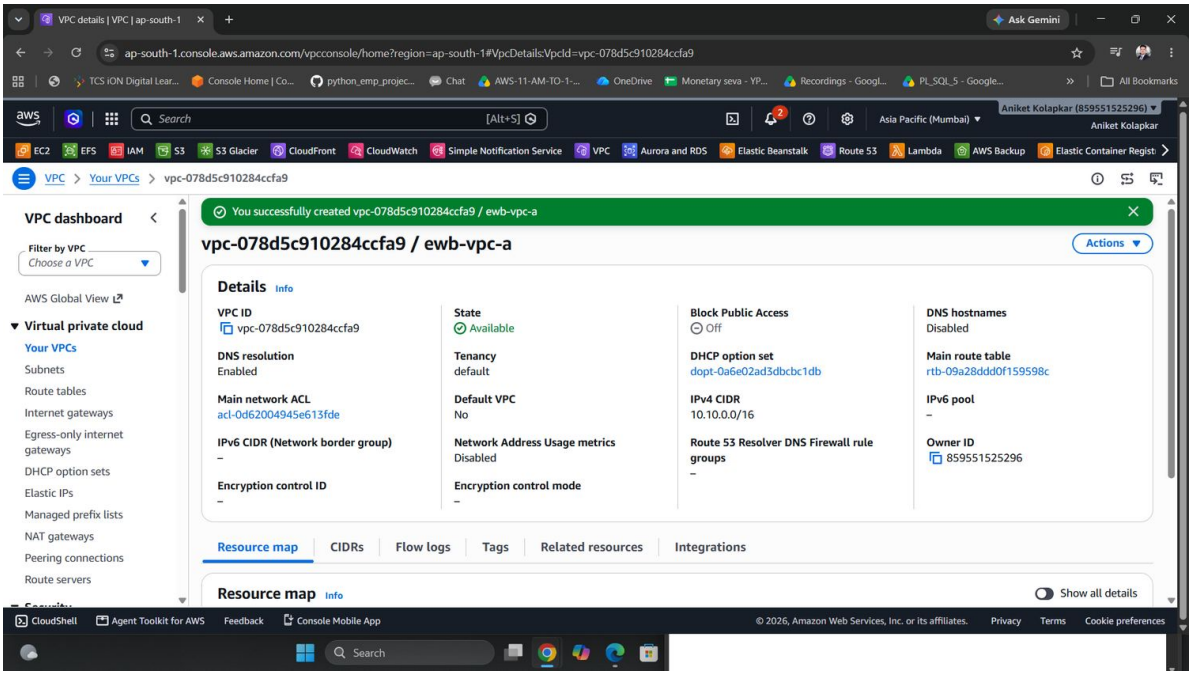
Evidence file: 2.png

Screenshot 03 — Create VPC-A and configure its IPv4 CIDR



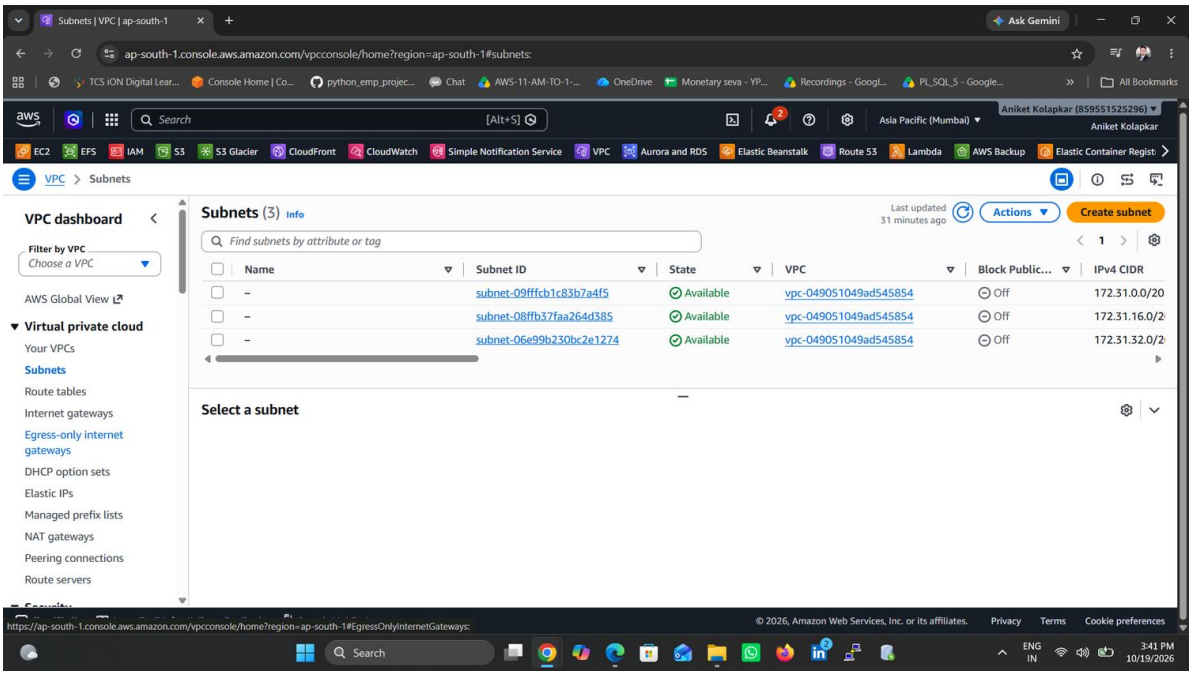
Evidence file: 3.png

Screenshot 04 — Review VPC details and subnet inventory



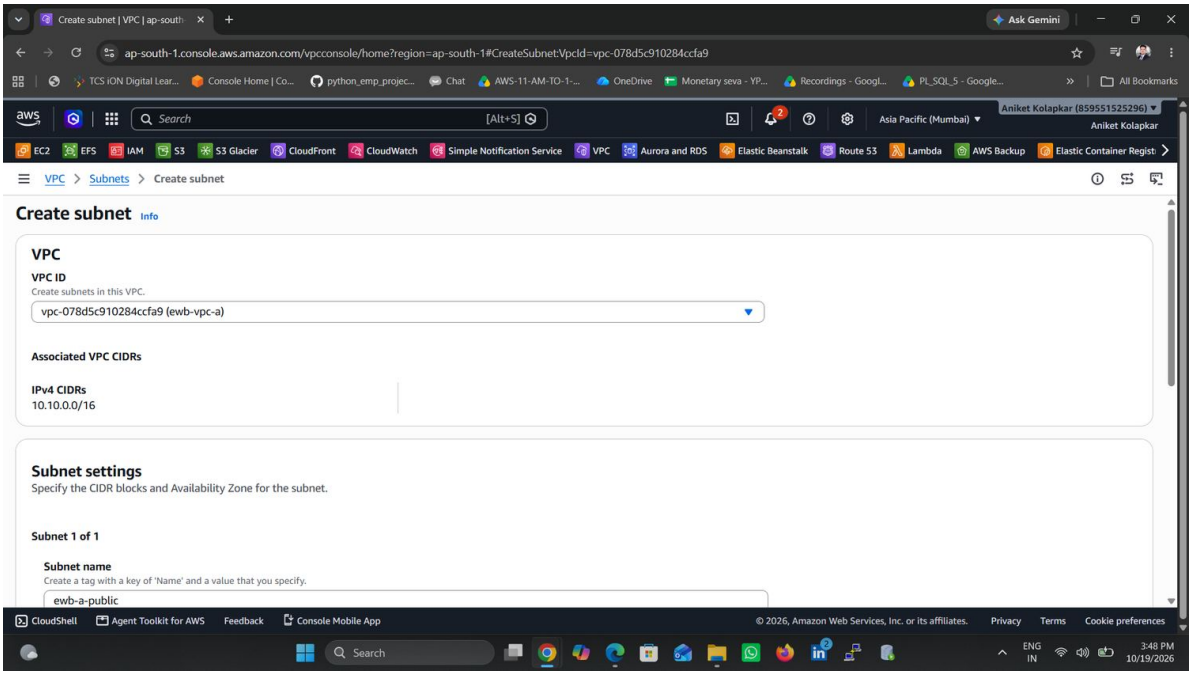
Evidence file: 4.png

Screenshot 05 — Review VPC details and subnet inventory



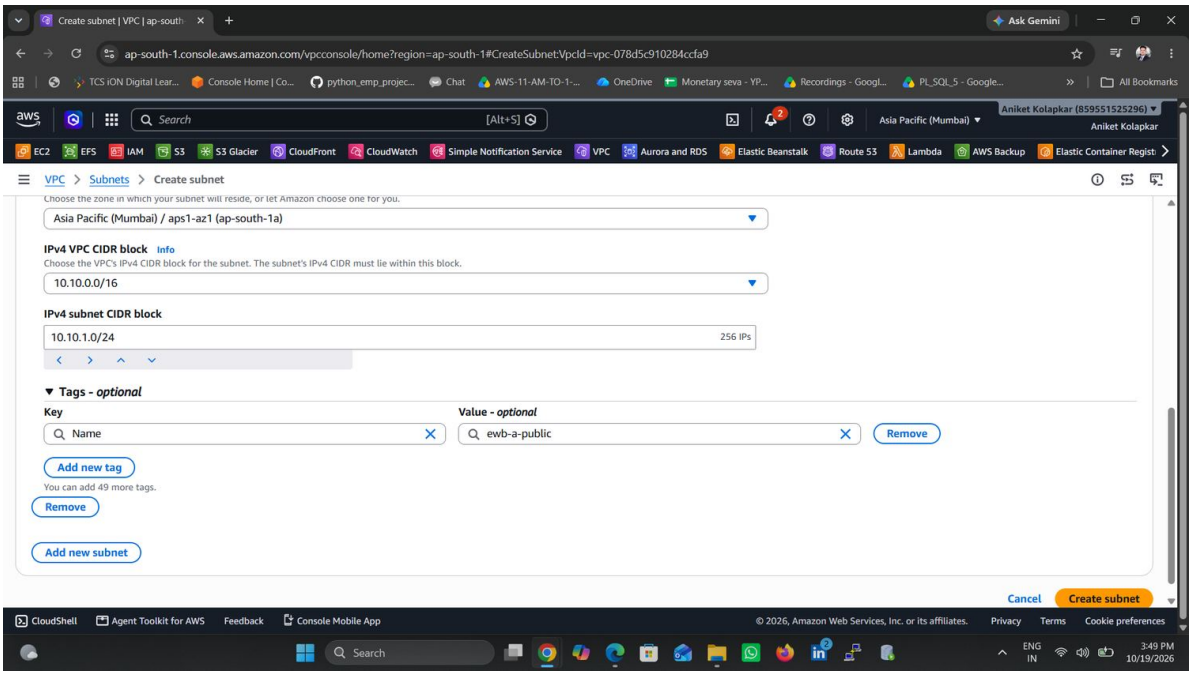
Evidence file: 5.png

Screenshot 06 — Create subnet and configure subnet CIDR and AZ



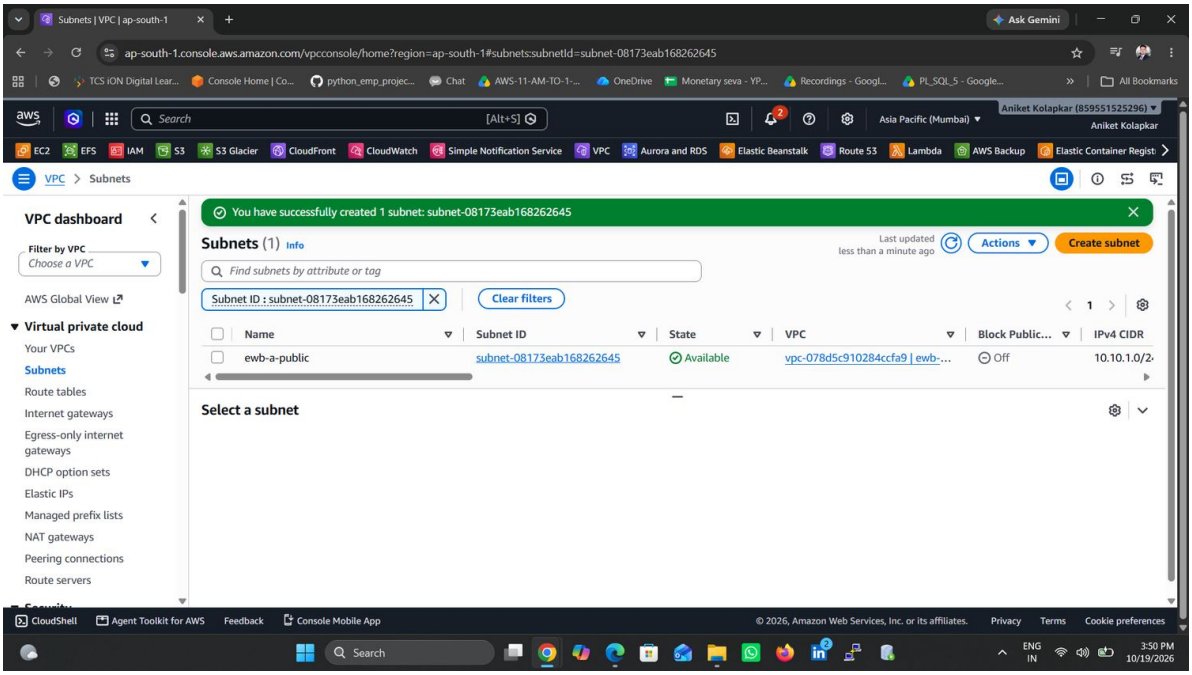
Evidence file: 6.png

Screenshot 07 — Create subnet and configure subnet CIDR and AZ



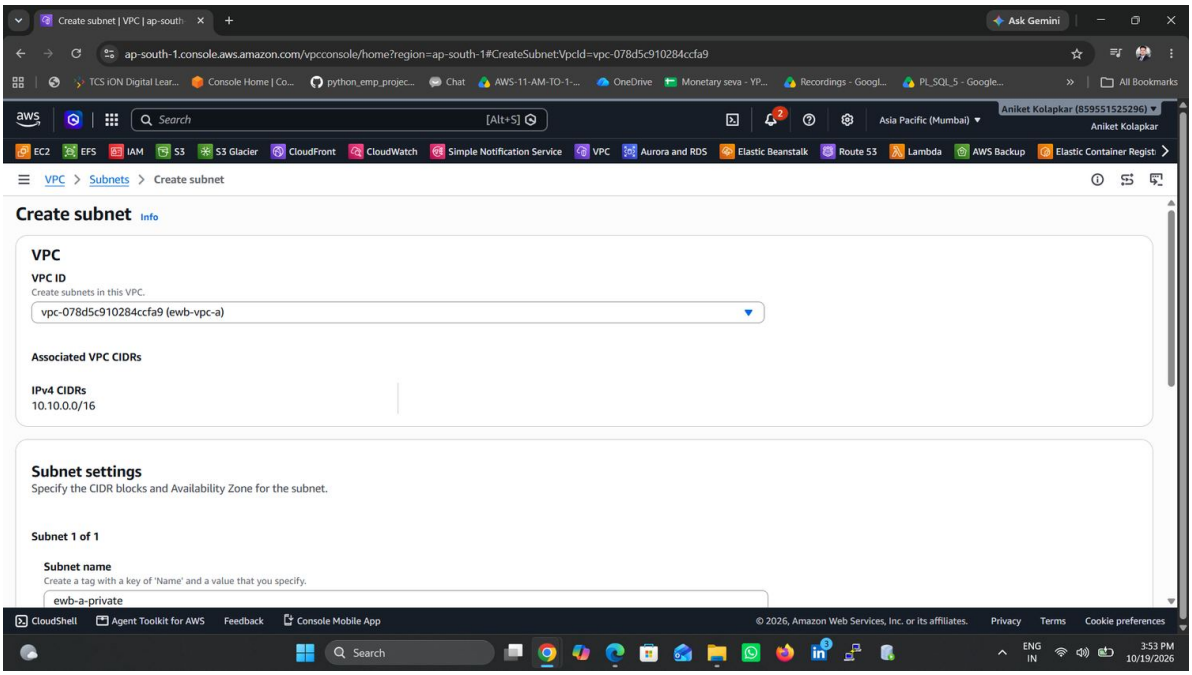
Evidence file: 7.png

Screenshot 08 — Create subnet and configure subnet CIDR and AZ



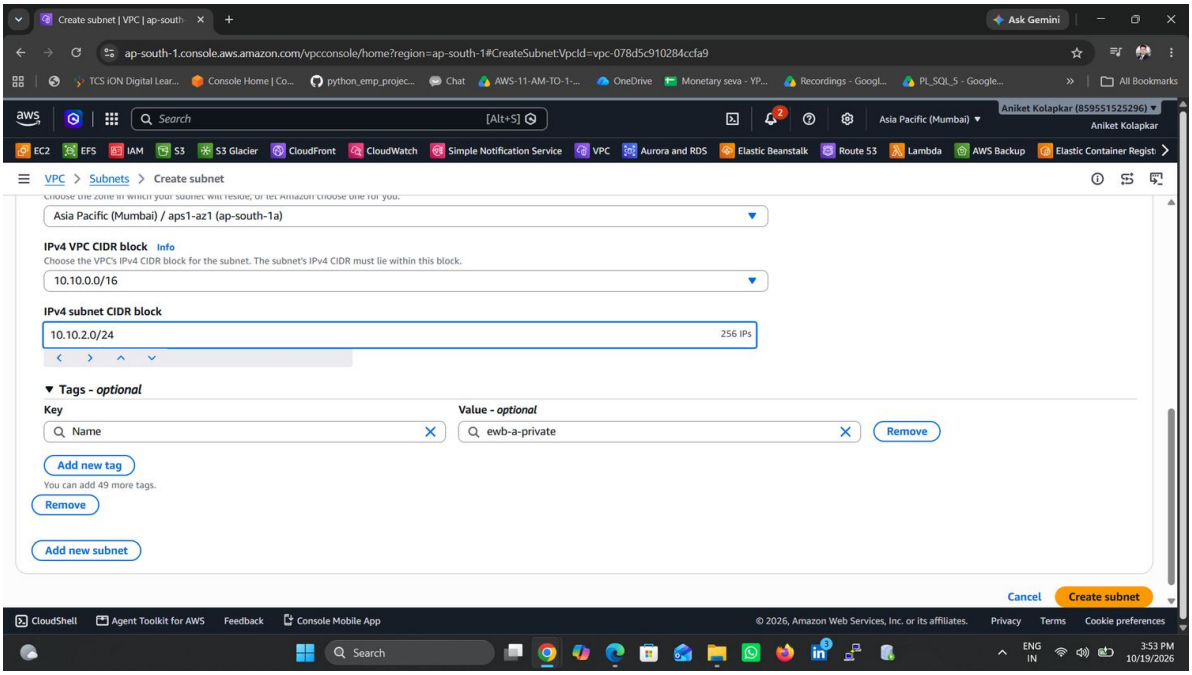
Evidence file: 8.png

Screenshot 09 — Create subnet and configure subnet CIDR and AZ



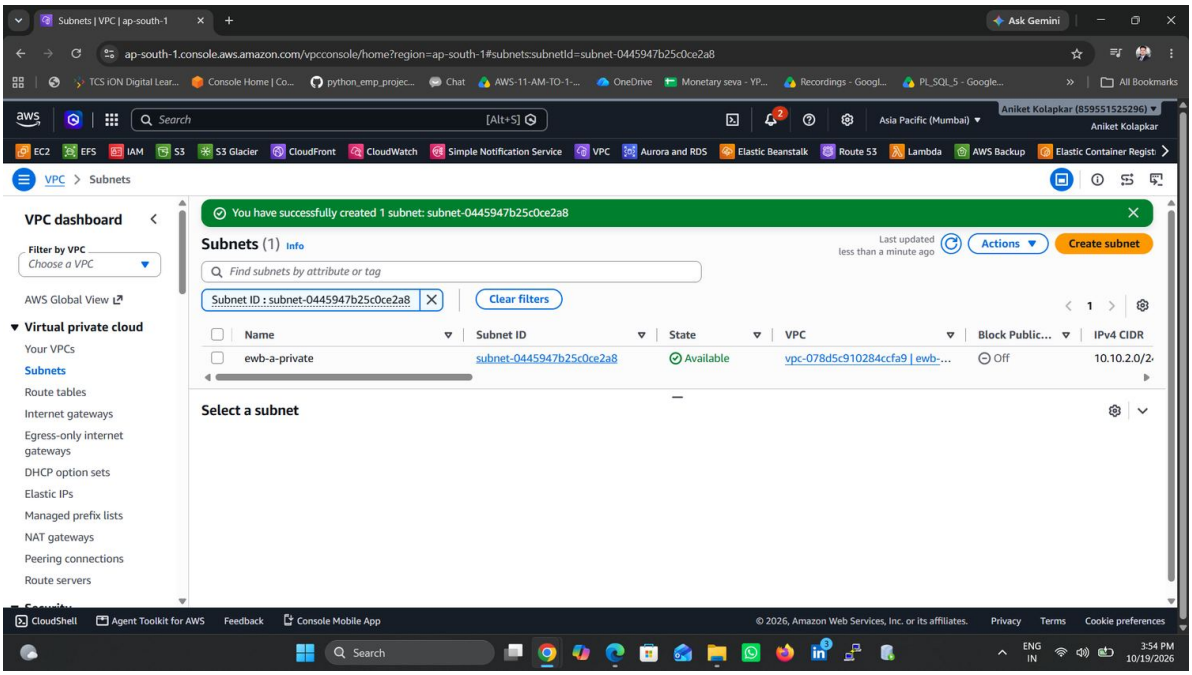
Evidence file: 9.png

Screenshot 10 — Create subnet and configure subnet CIDR and AZ



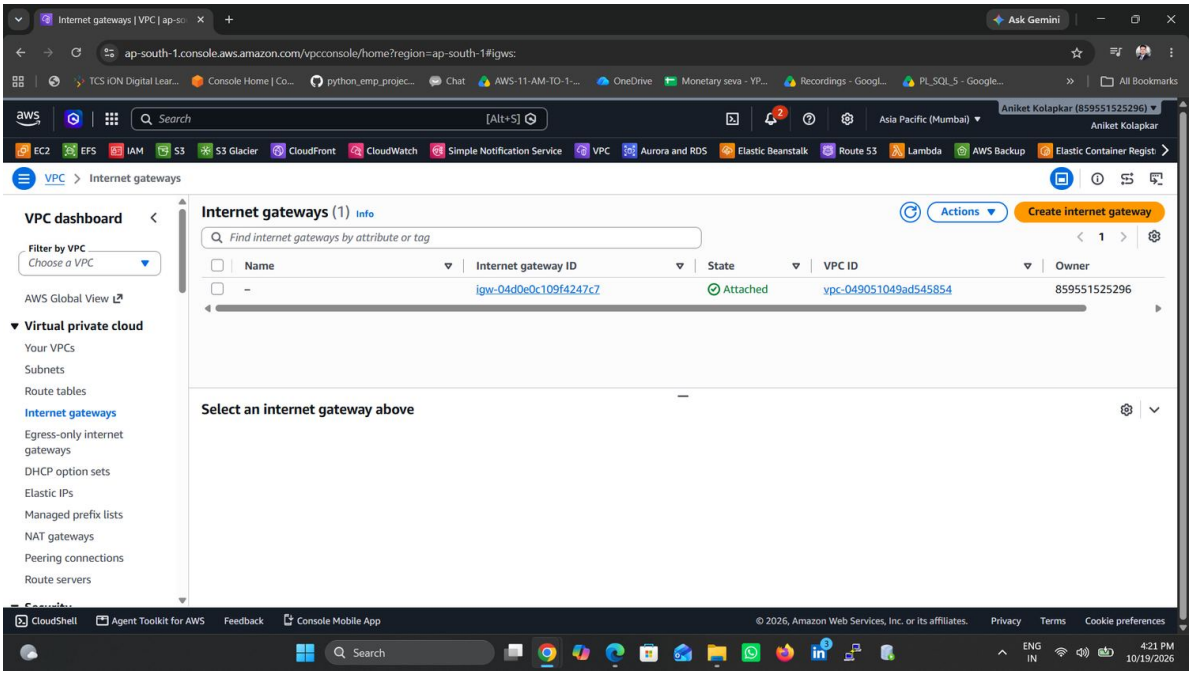
Evidence file: 10.png

Screenshot 11 — Create and attach the Internet Gateway to VPC-A



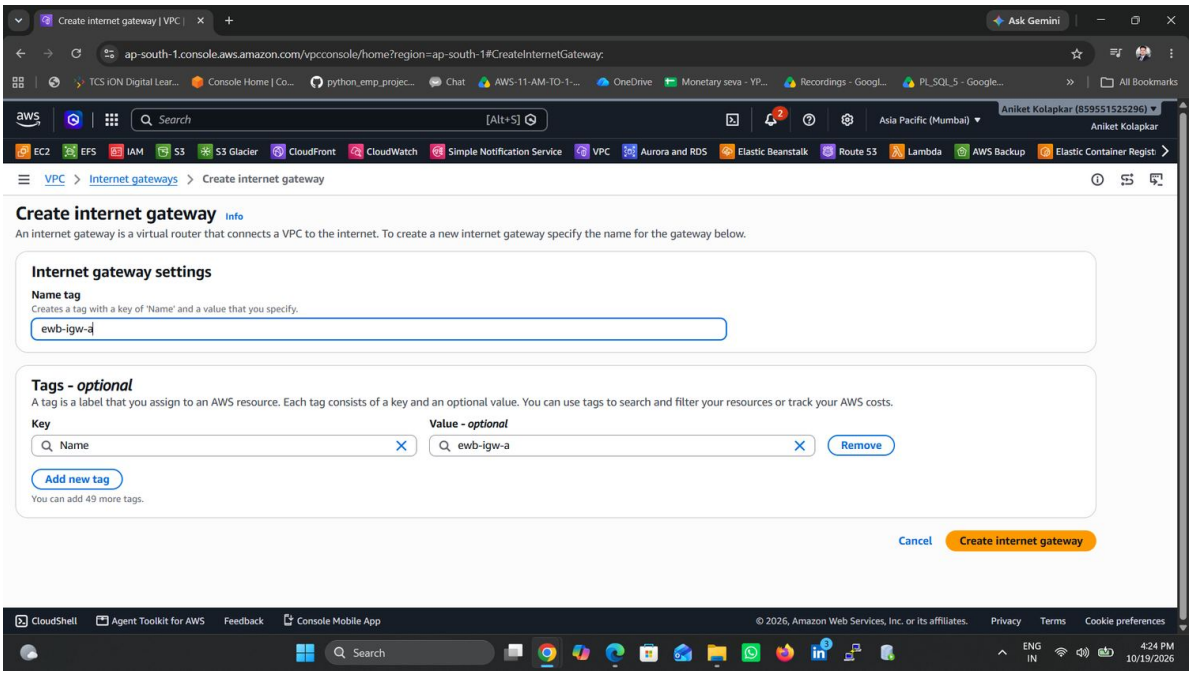
Evidence file: 11.png

Screenshot 12 — Create and attach the Internet Gateway to VPC-A



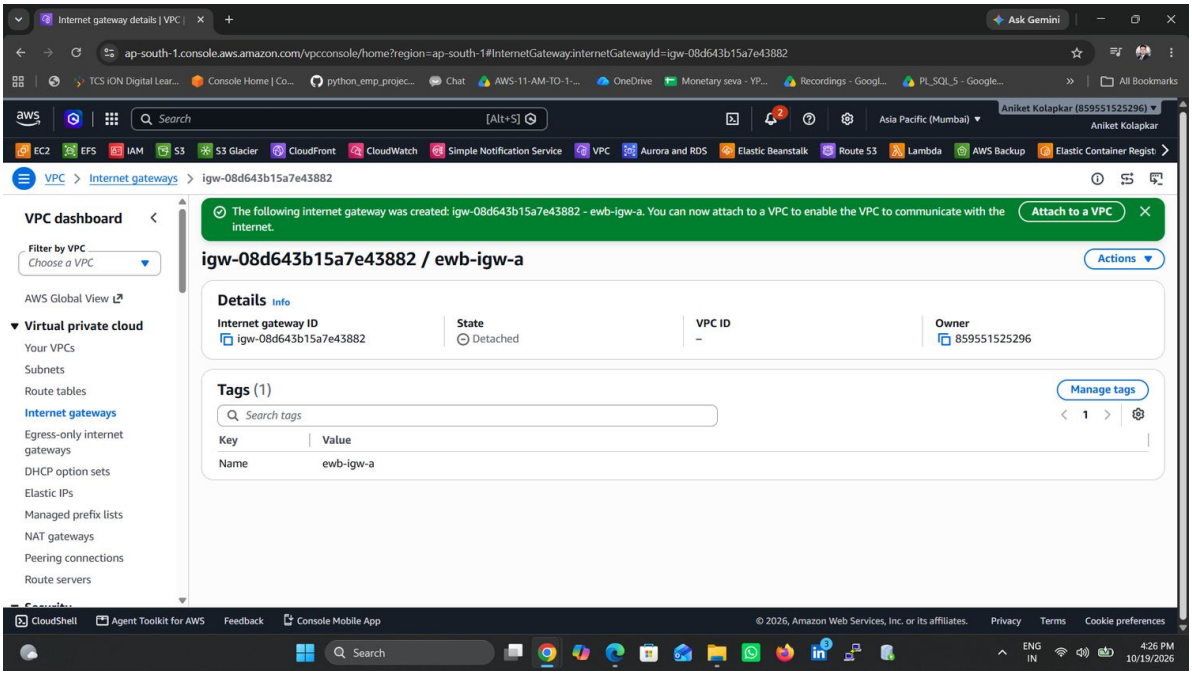
Evidence file: 12.png

Screenshot 13 — Create and attach the Internet Gateway to VPC-A



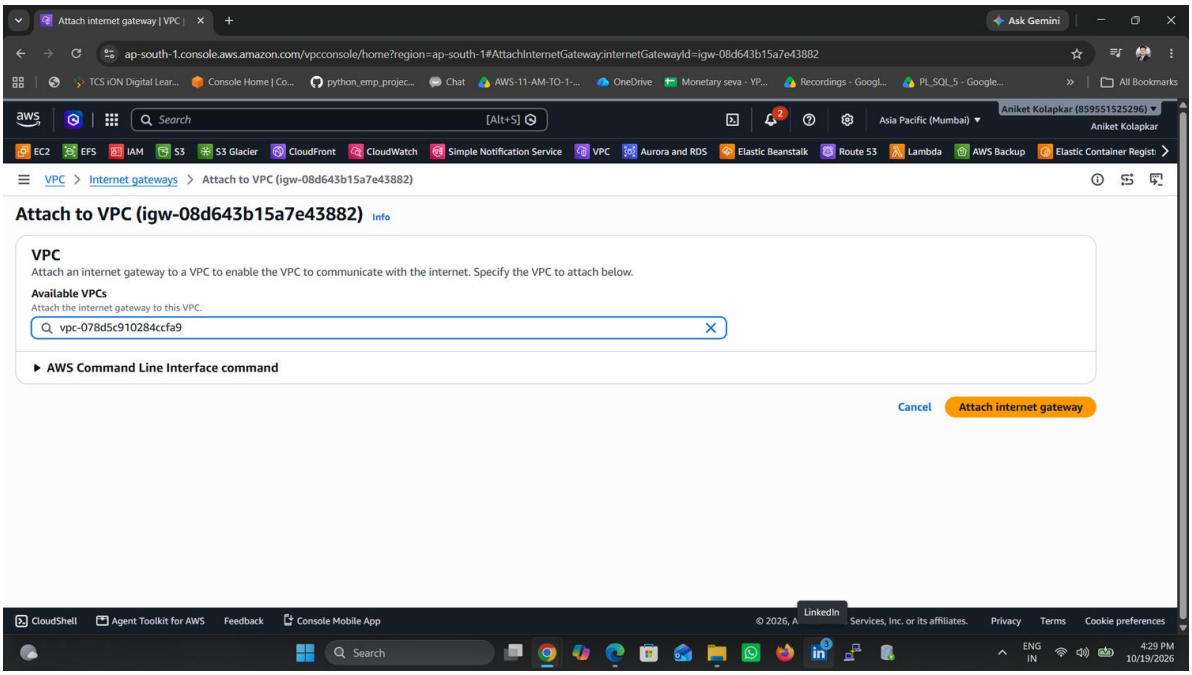
Evidence file: 13.png

Screenshot 14 — Create and attach the Internet Gateway to VPC-A



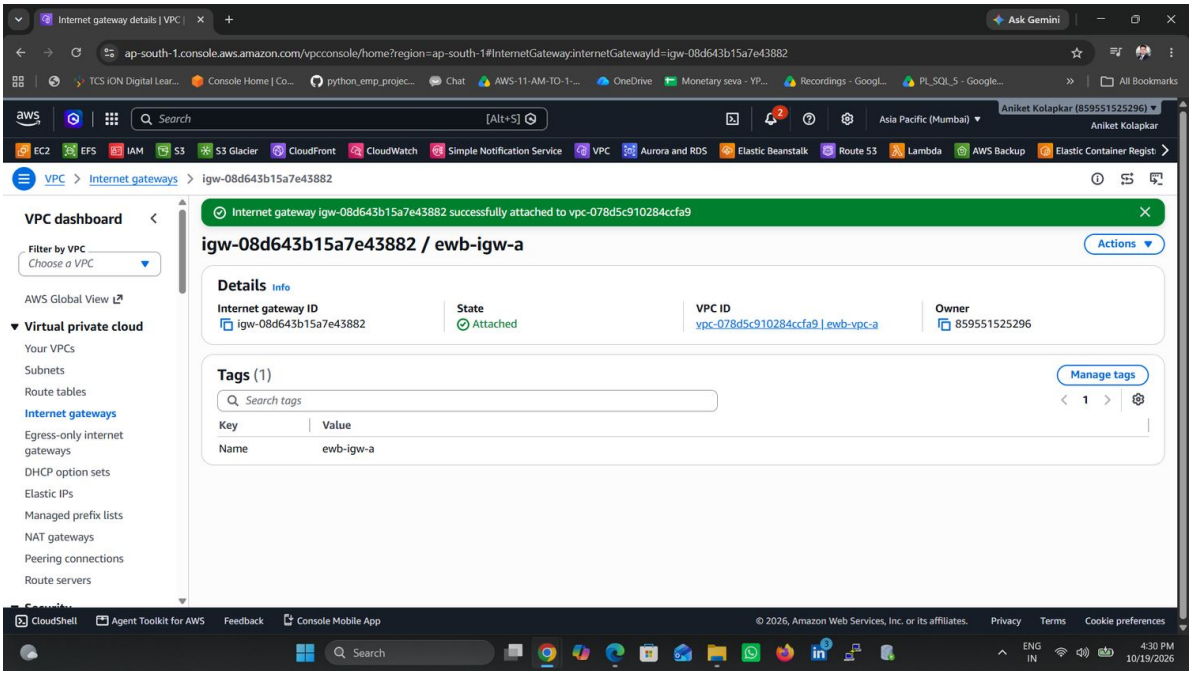
Evidence file: 14.png

Screenshot 15 — Create and attach the Internet Gateway to VPC-A



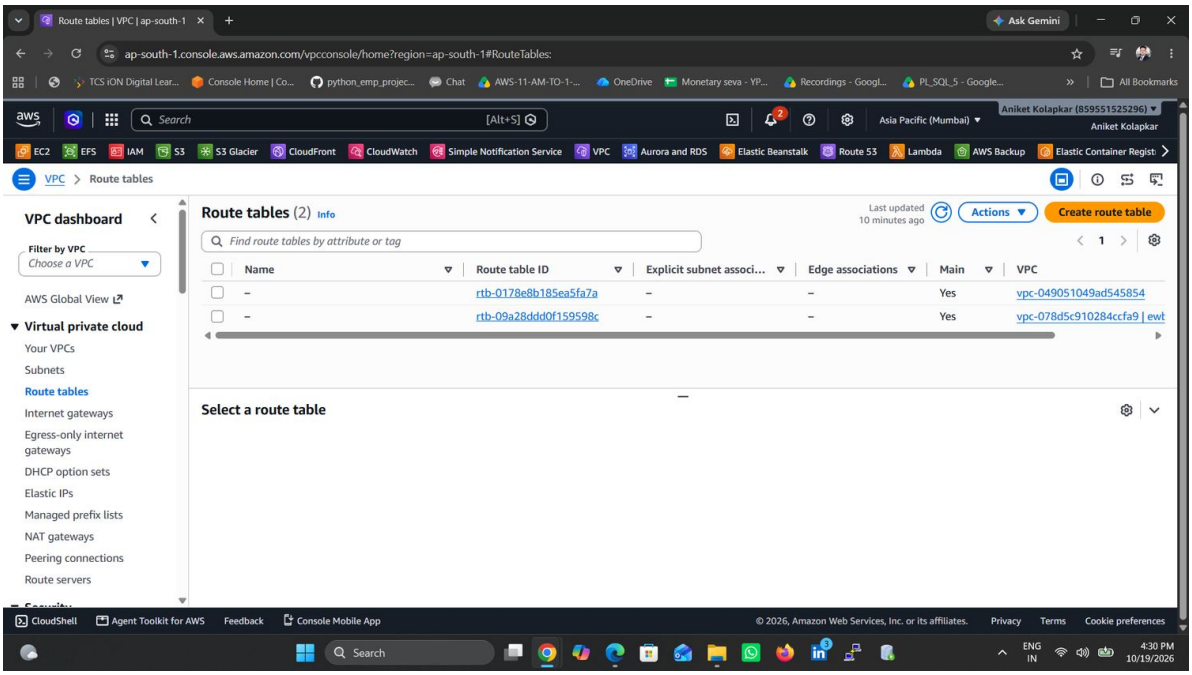
Evidence file: 15.png

Screenshot 16 — Create and attach the Internet Gateway to VPC-A



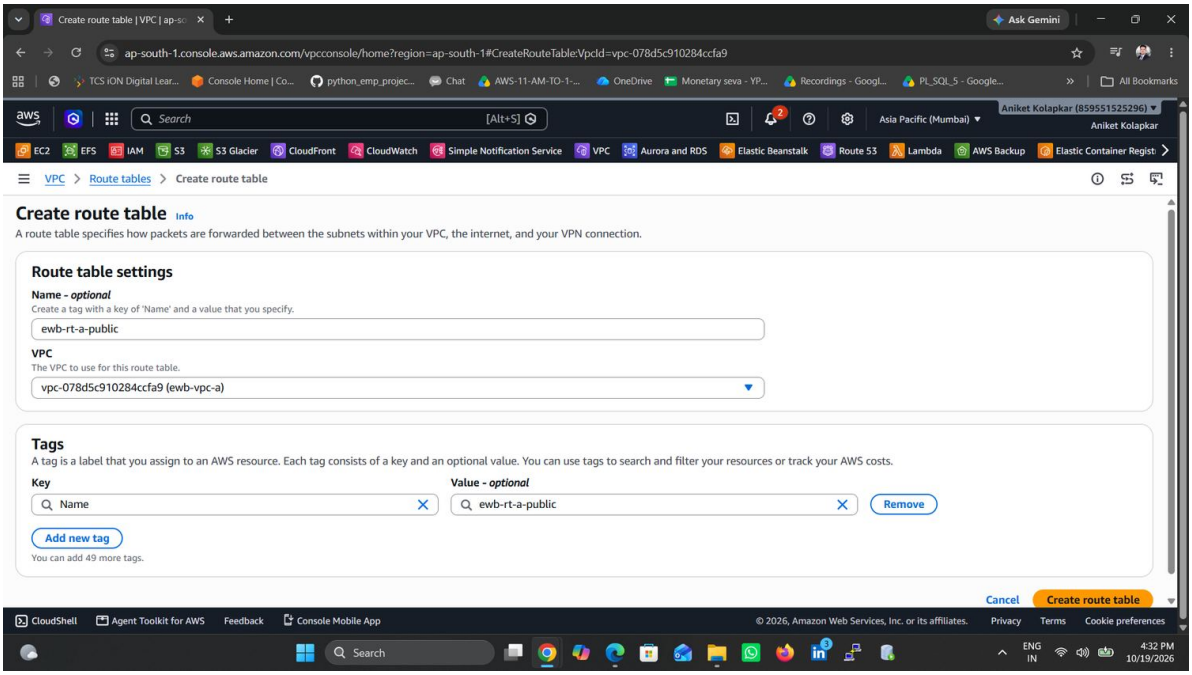
Evidence file: 16.png

Screenshot 17 — Create route table and review subnet associations



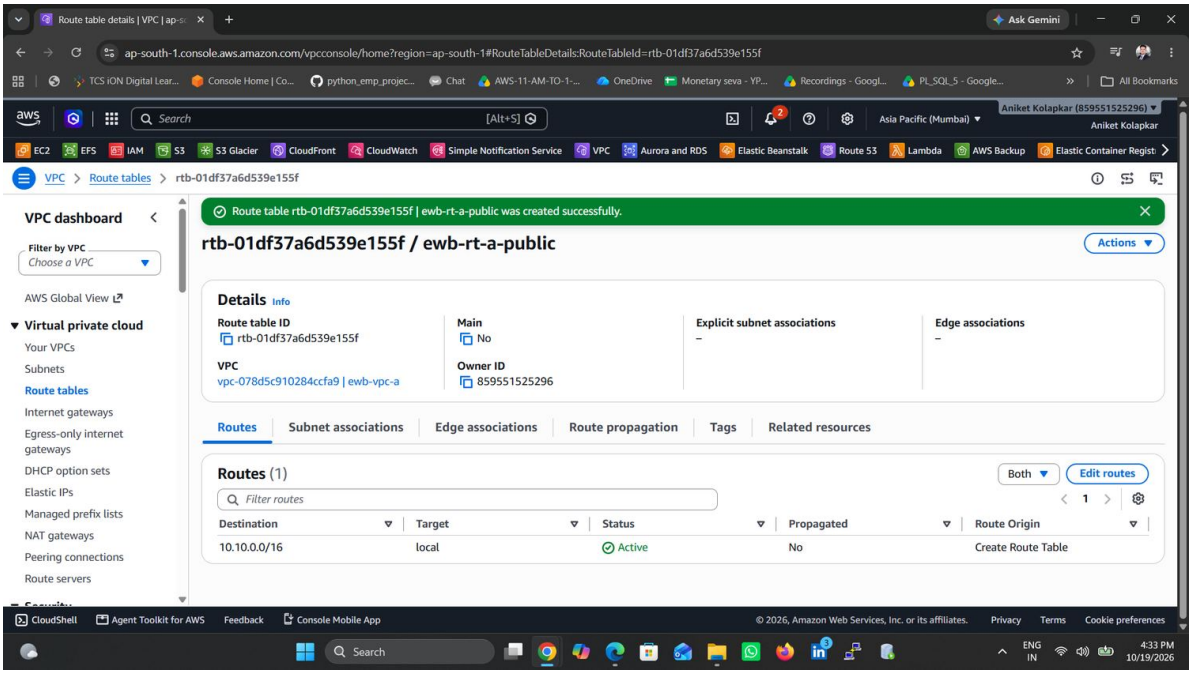
Evidence file: 17.png

Screenshot 18 — Create route table and review subnet associations



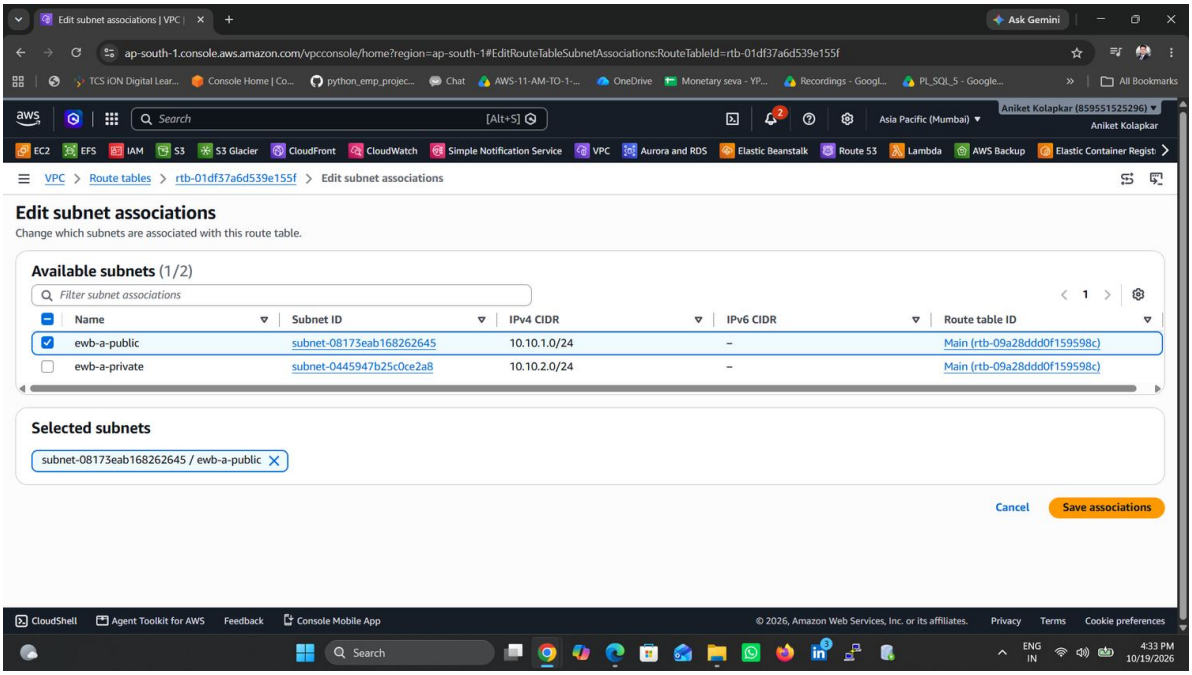
Evidence file: 18.png

Screenshot 19 — Create route table and review subnet associations



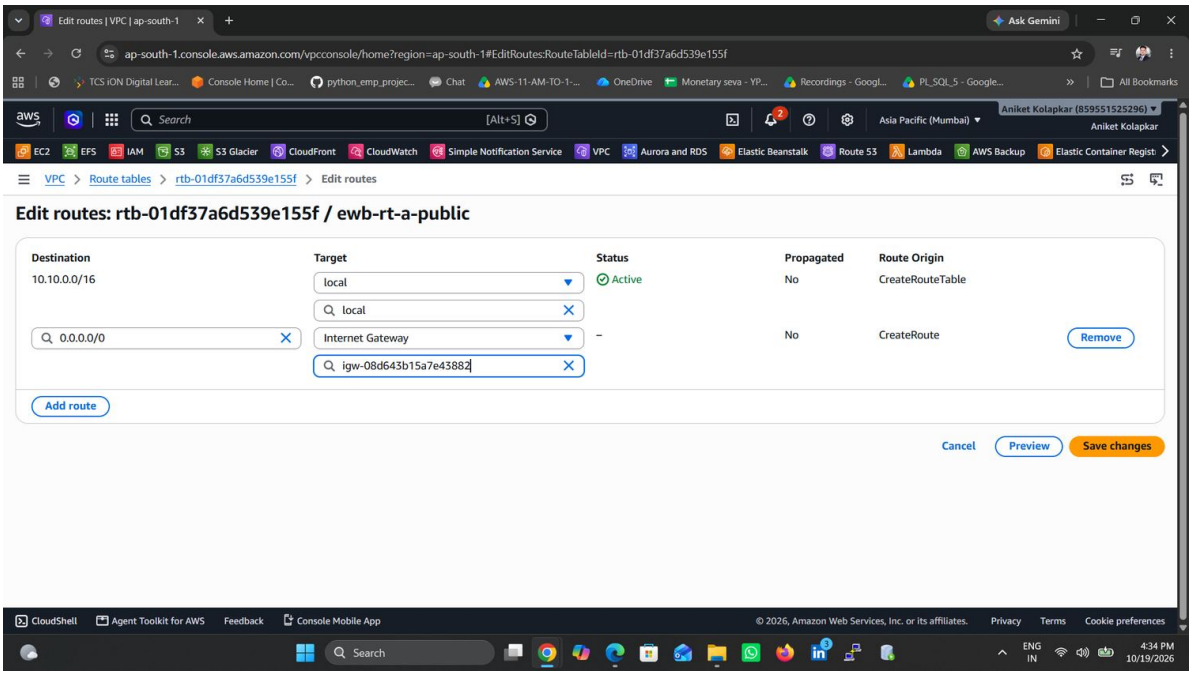
Evidence file: 19.png

Screenshot 20 — Create route table and review subnet associations



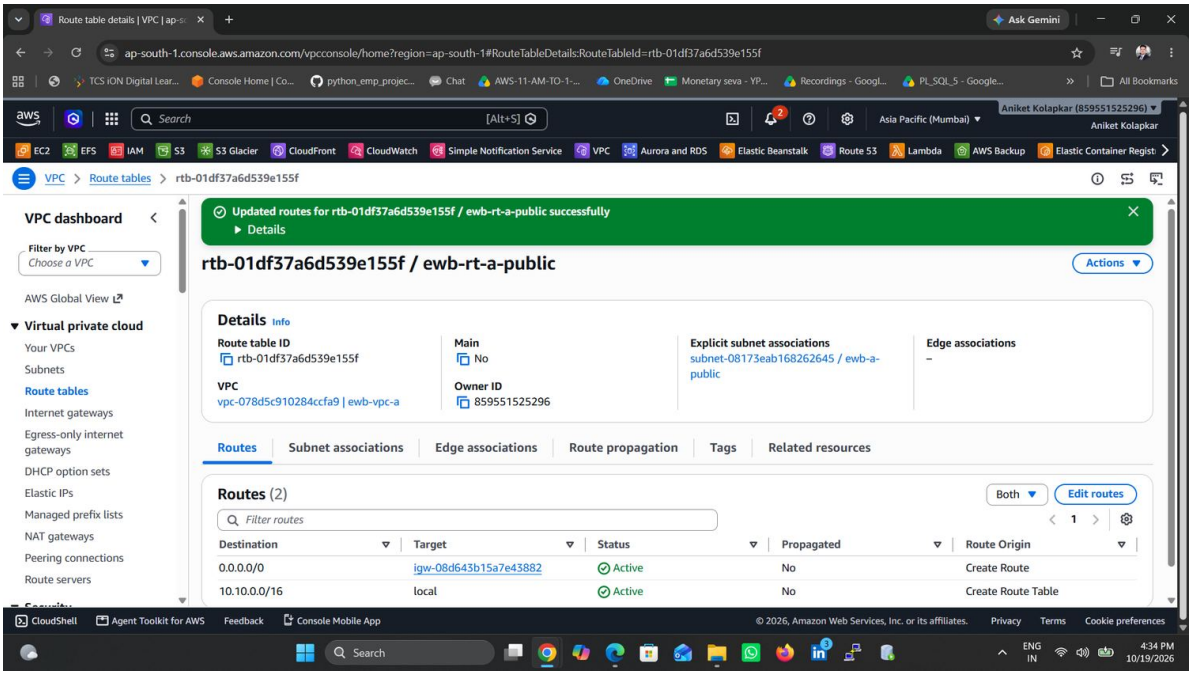
Evidence file: 20.png

Screenshot 21 — Create route table and review subnet associations



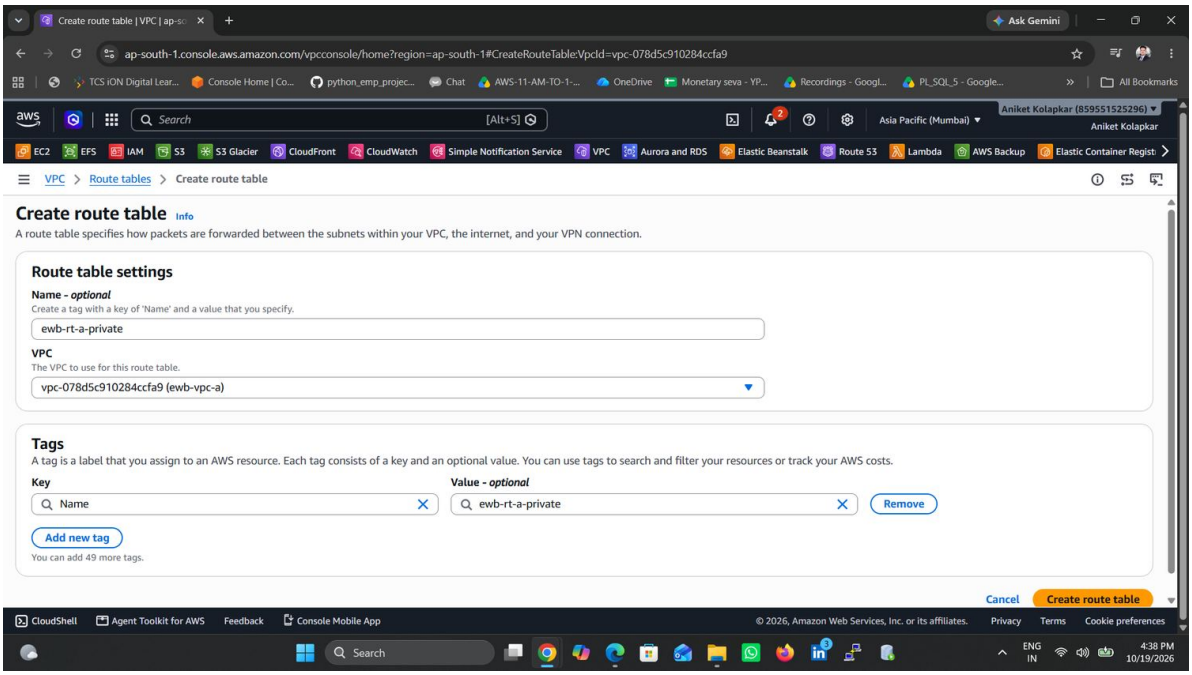
Evidence file: 21.png

Screenshot 22 — Create route table and review subnet associations



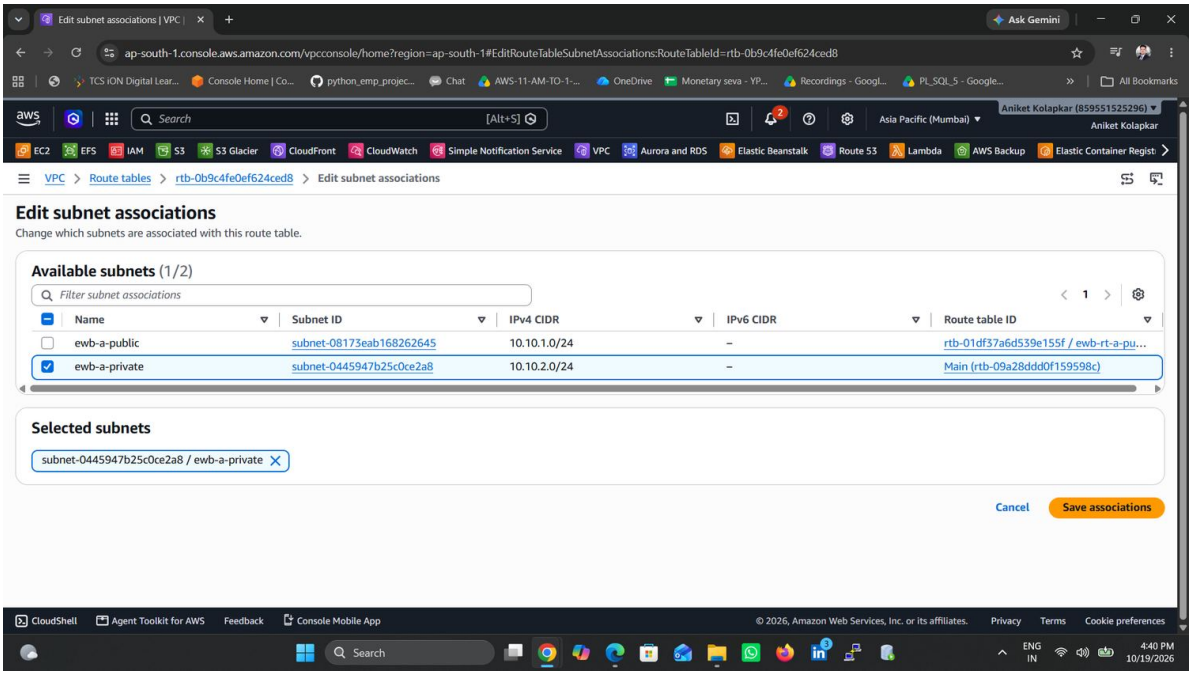
Evidence file: 22.png

Screenshot 23 — Create route table and review subnet associations



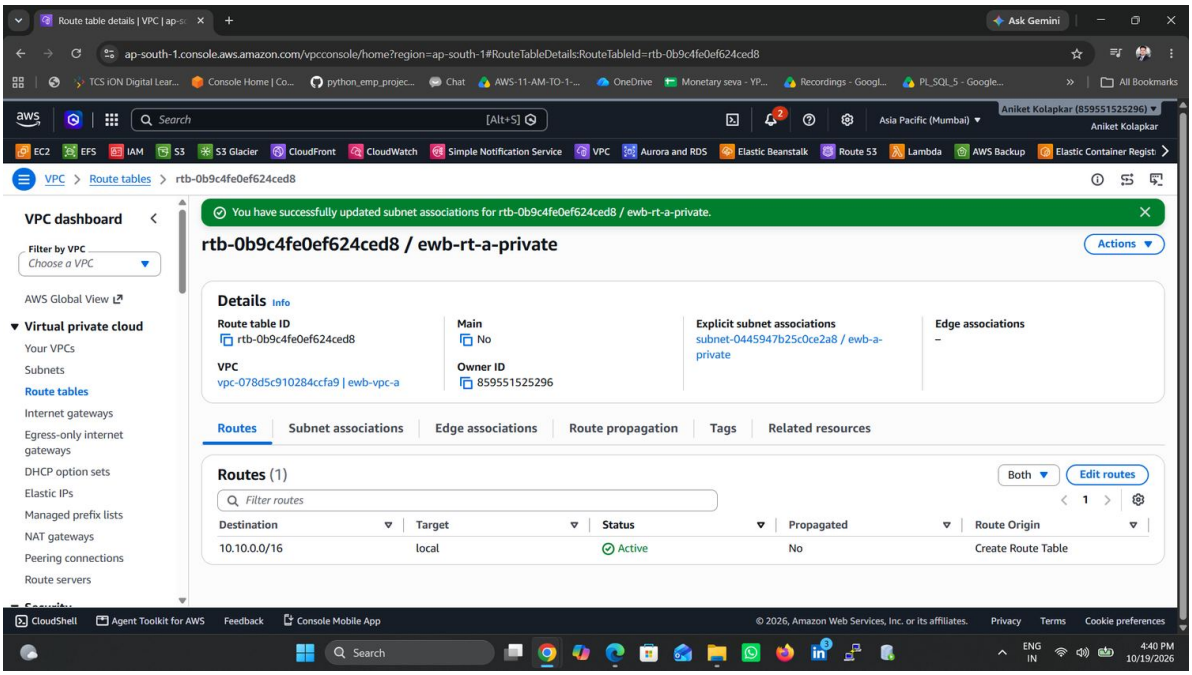
Evidence file: 23.png

Screenshot 24 — Create route table and review subnet associations



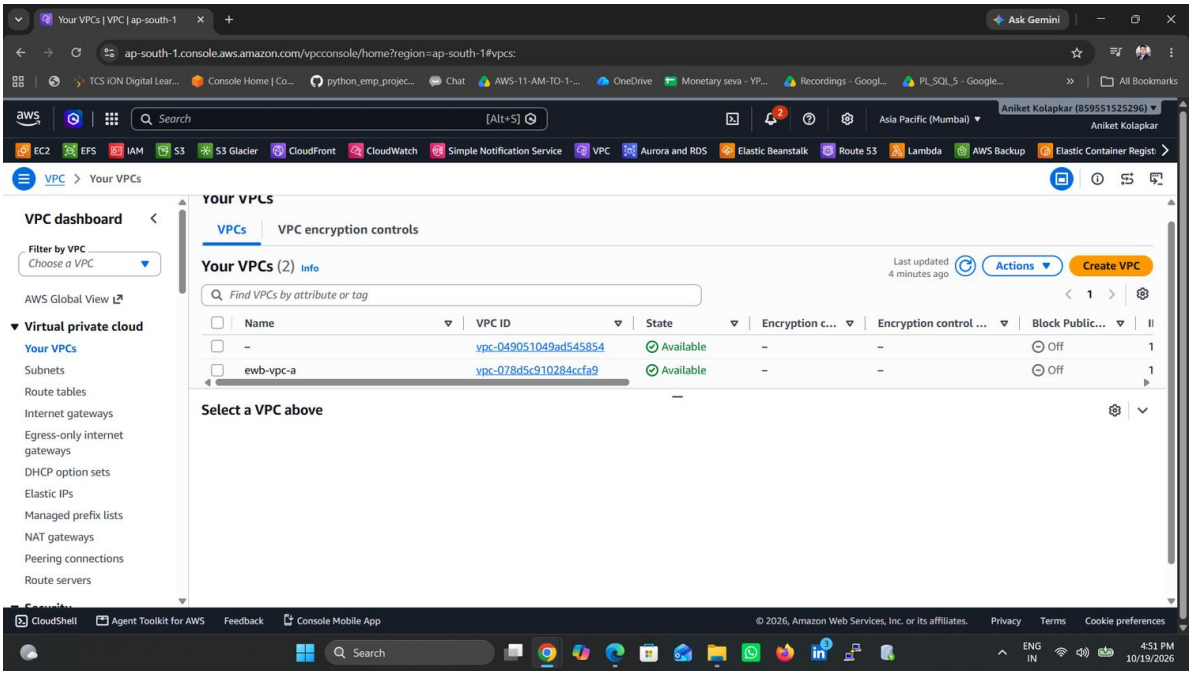
Evidence file: 24.png

Screenshot 25 — Create route table and review subnet associations



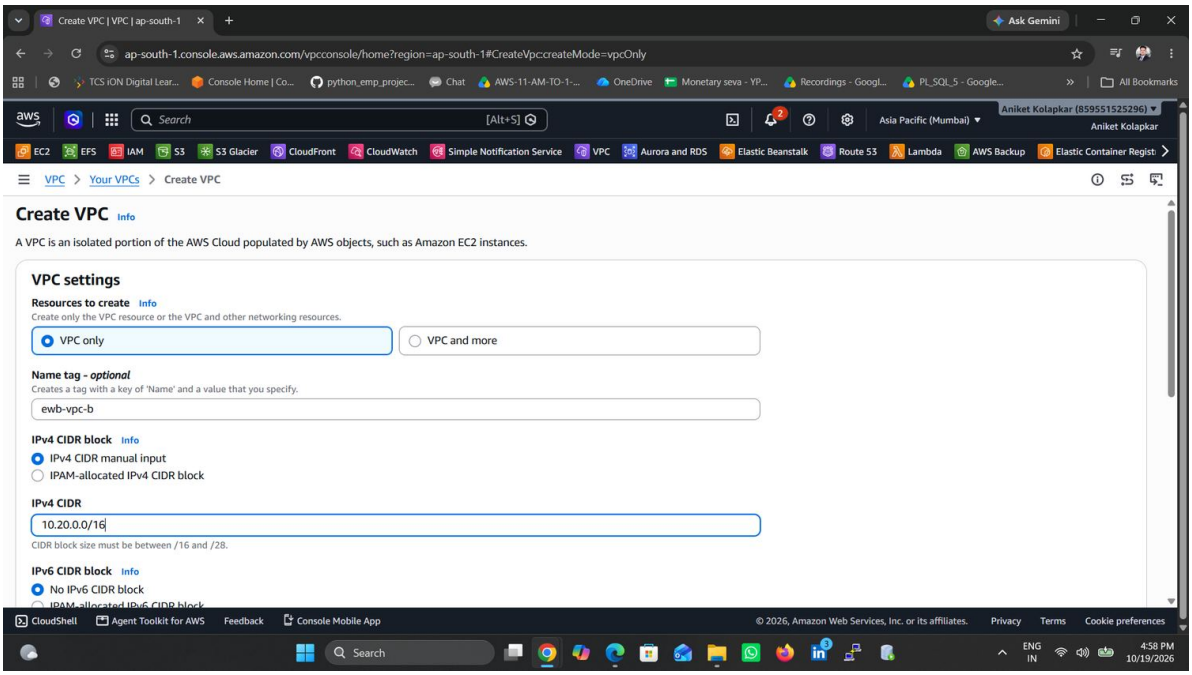
Evidence file: 25.png

Screenshot 26 — Create route table and review subnet associations



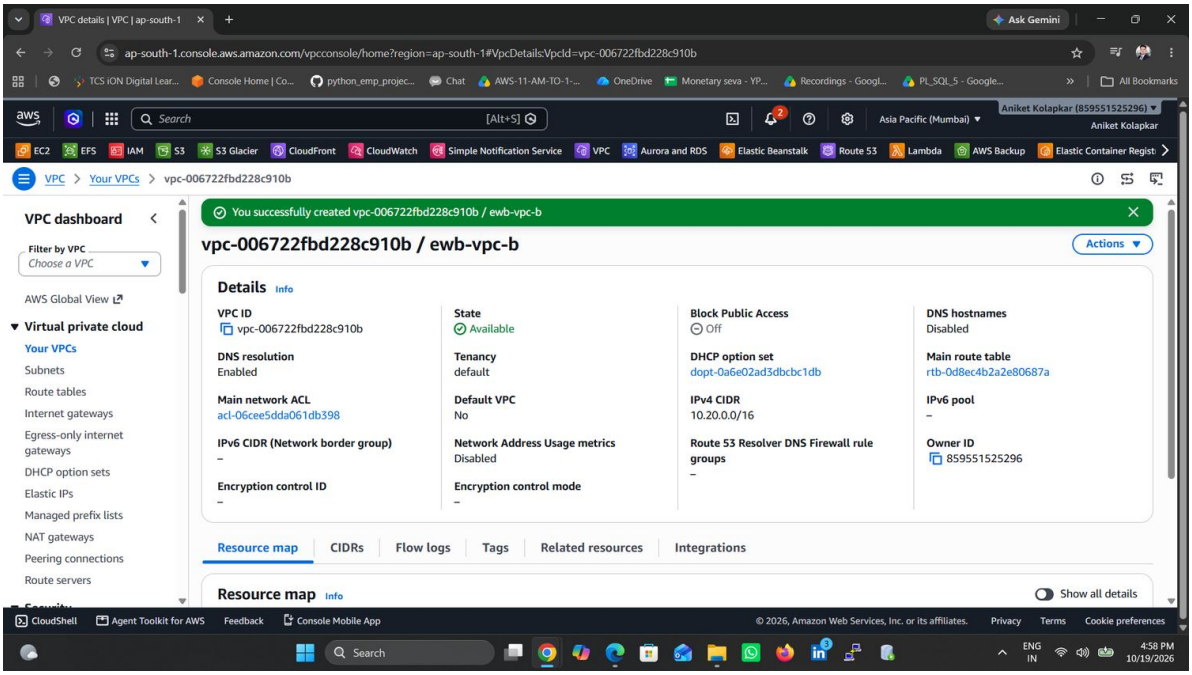
Evidence file: 26.png

Screenshot 27 — Configure VPC-B and its private subnet



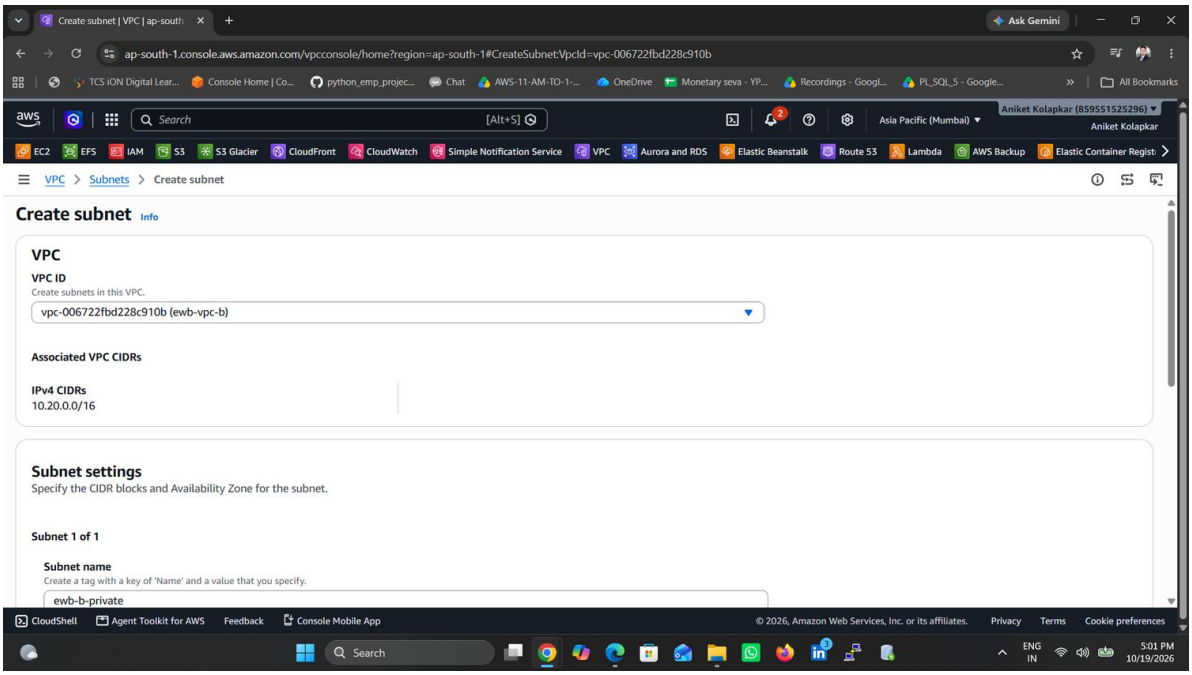
Evidence file: 27.png

Screenshot 28 — Configure VPC-B and its private subnet



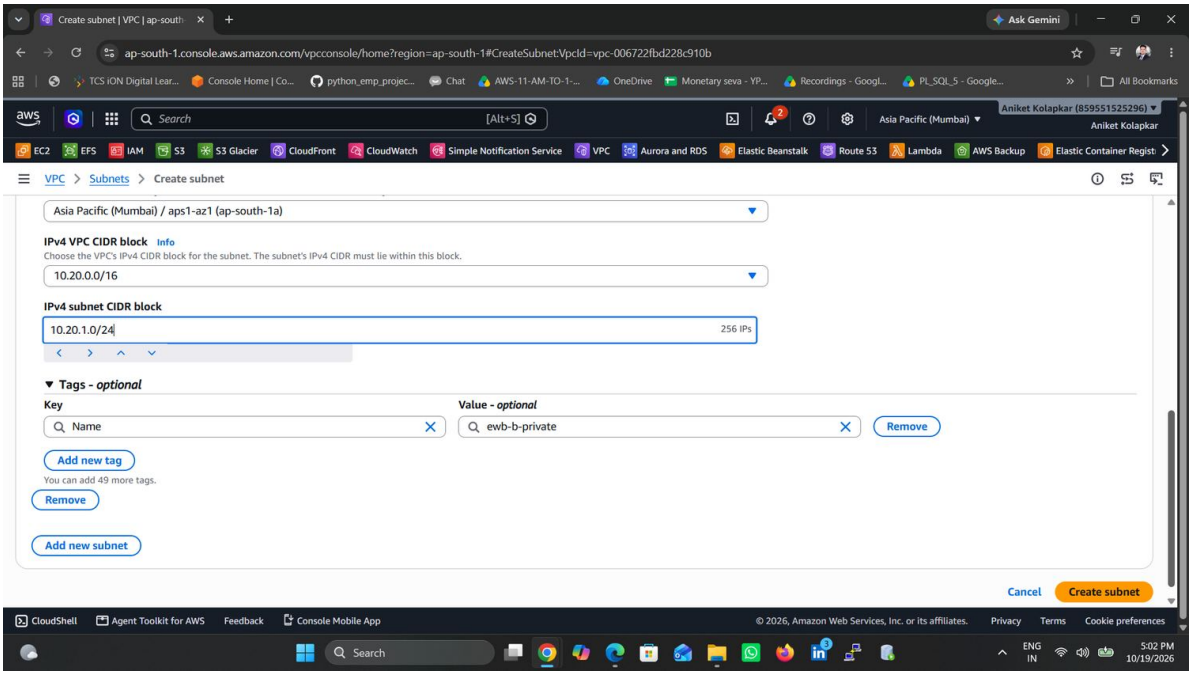
Evidence file: 28.png

Screenshot 29 — Configure VPC-B and its private subnet



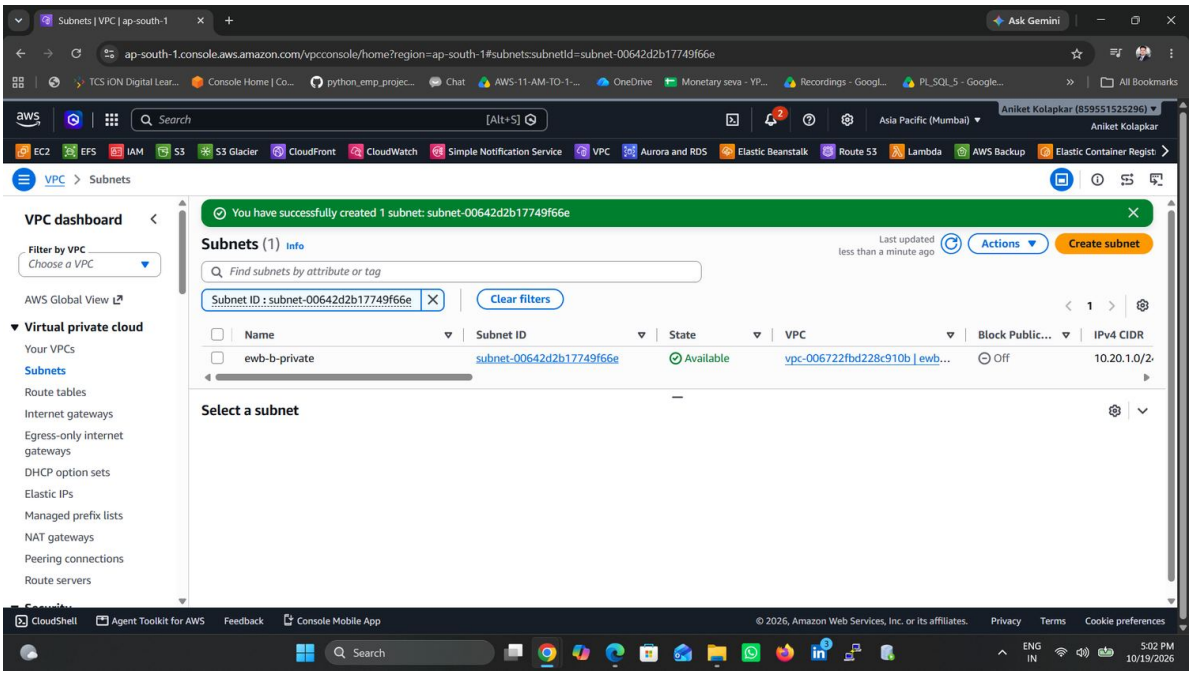
Evidence file: 29.png

Screenshot 30 — Configure VPC-B and its private subnet



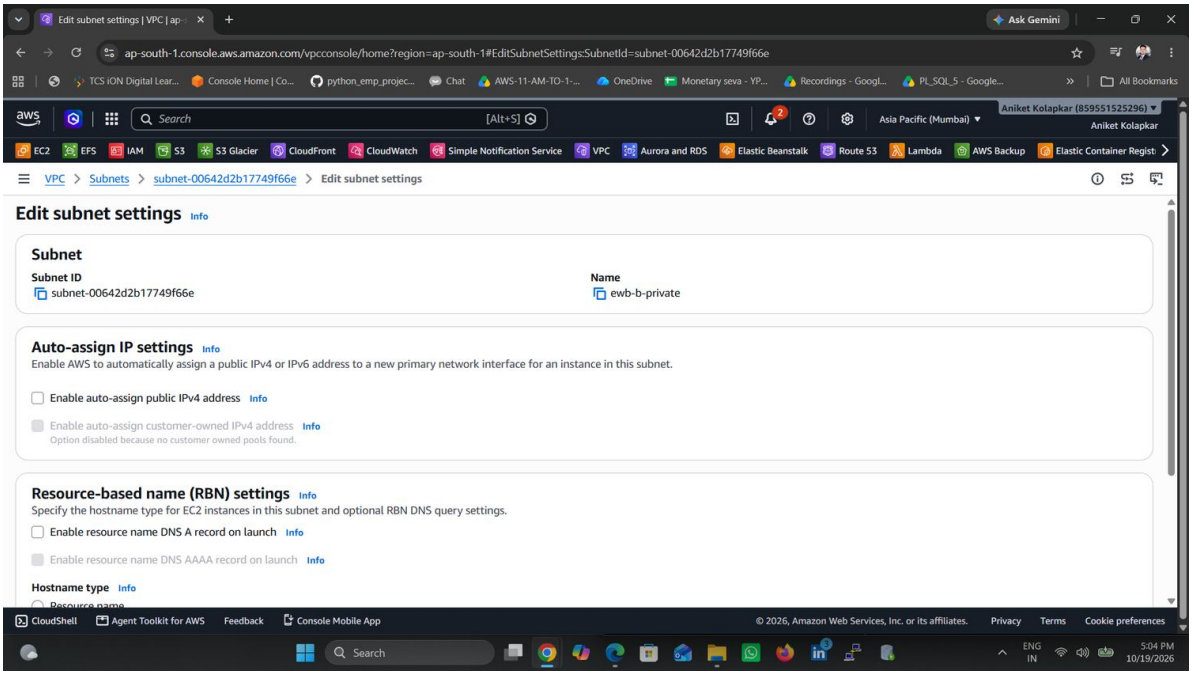
Evidence file: 30.png

Screenshot 31 — Configure VPC-B and its private subnet



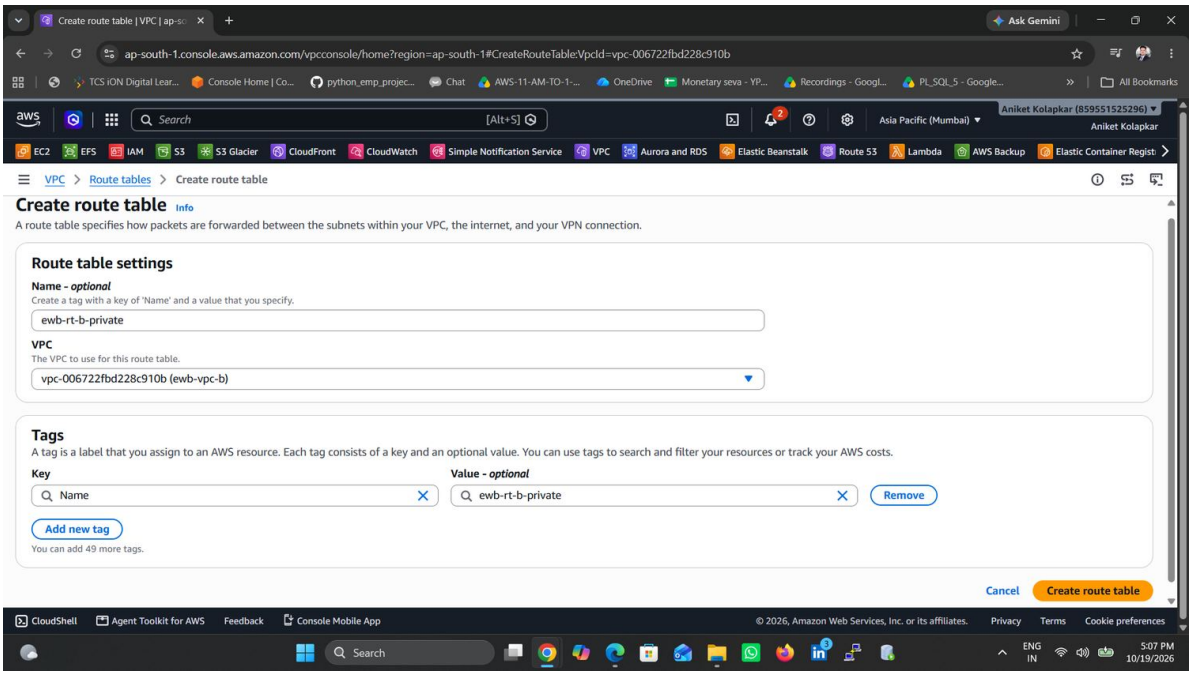
Evidence file: 31.png

Screenshot 32 — Configure VPC-B and its private subnet



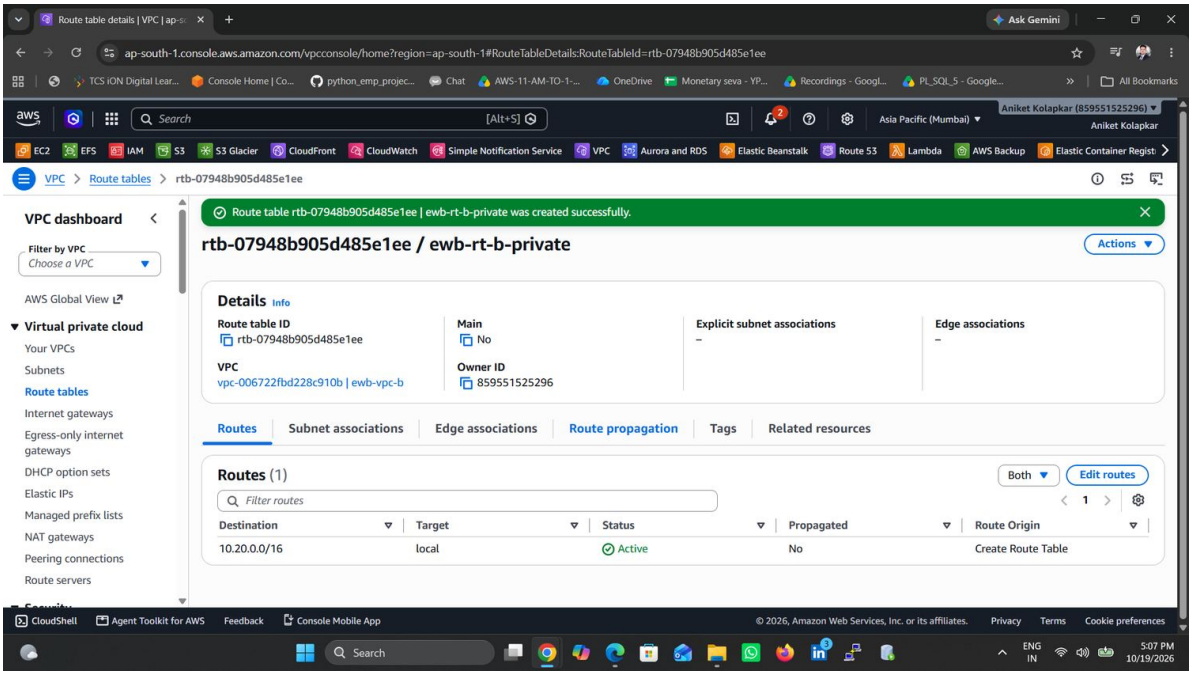
Evidence file: 32.png

Screenshot 33 — Configure VPC-B and its private subnet



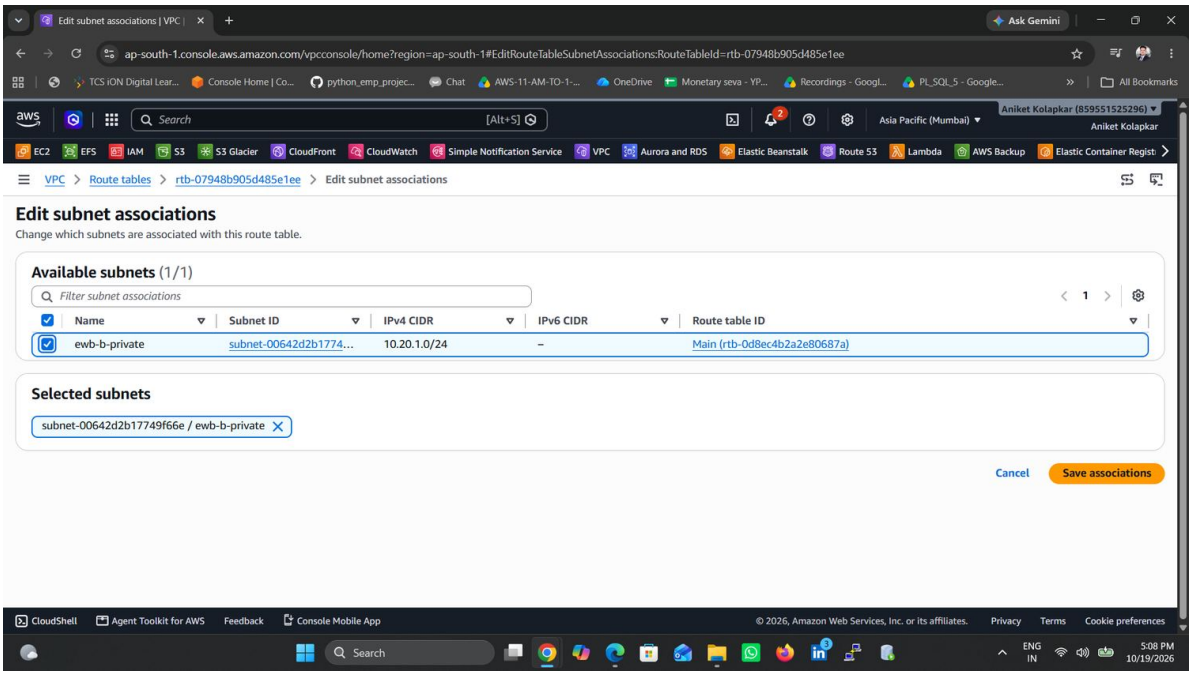
Evidence file: 33.png

Screenshot 34 — Configure VPC-B and its private subnet



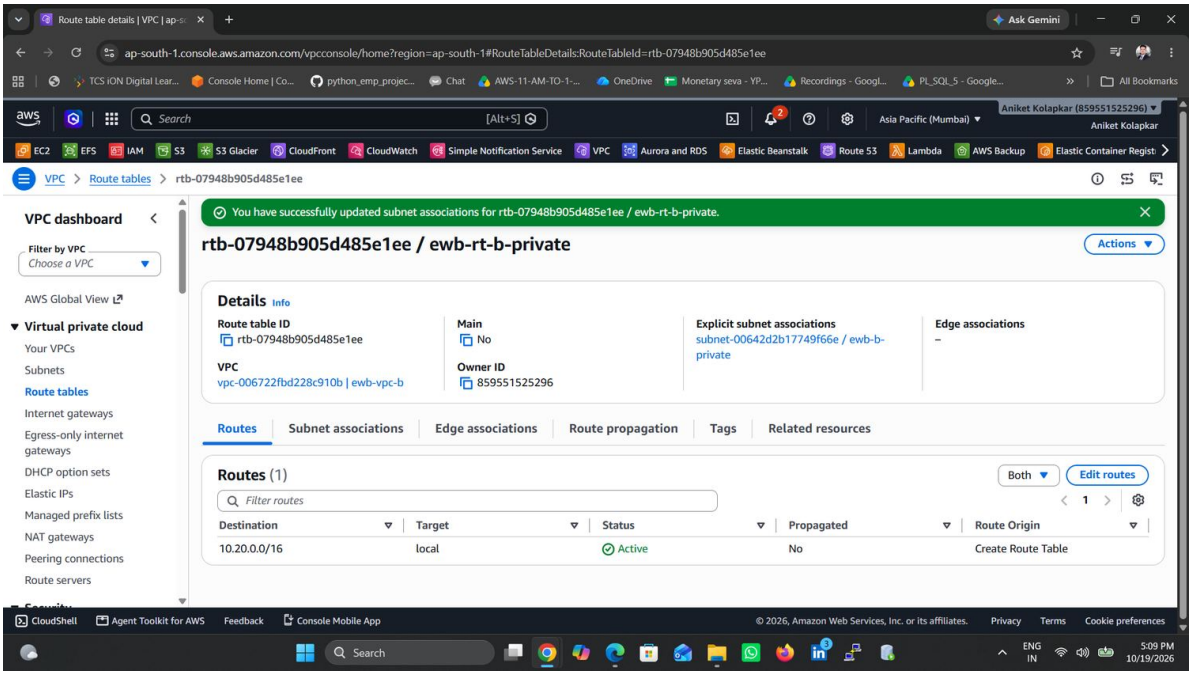
Evidence file: 34.png

Screenshot 35 — Configure VPC-B and its private subnet



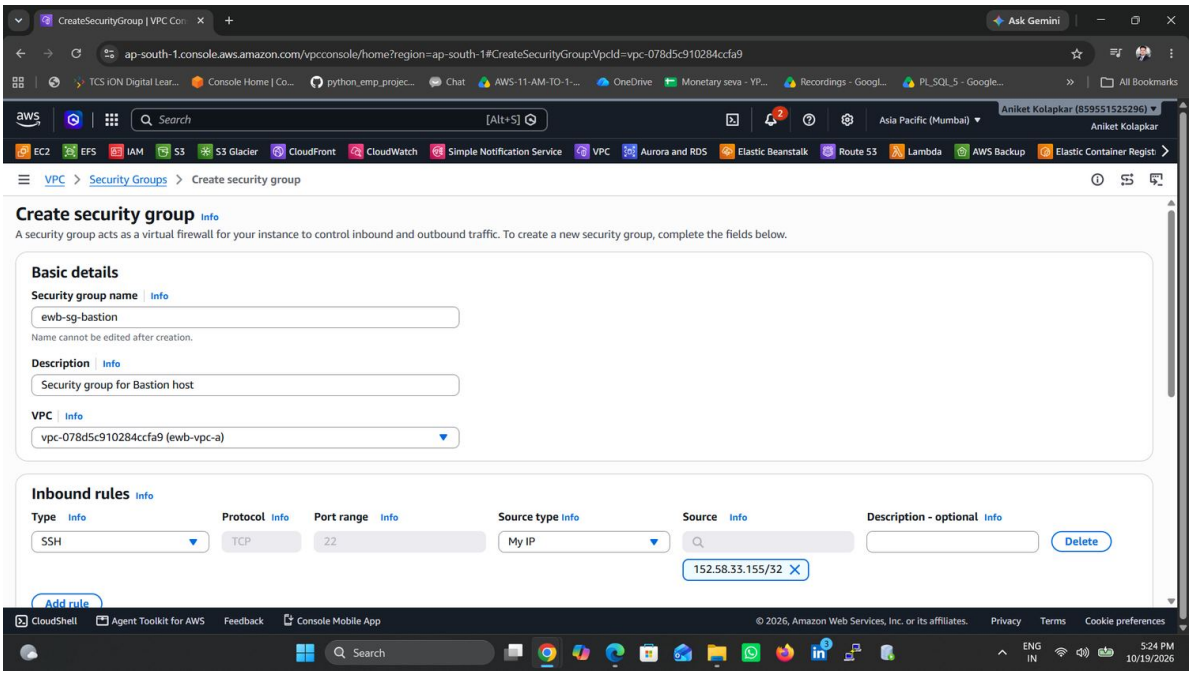
Evidence file: 35.png

Screenshot 36 — Create and configure EC2 security groups



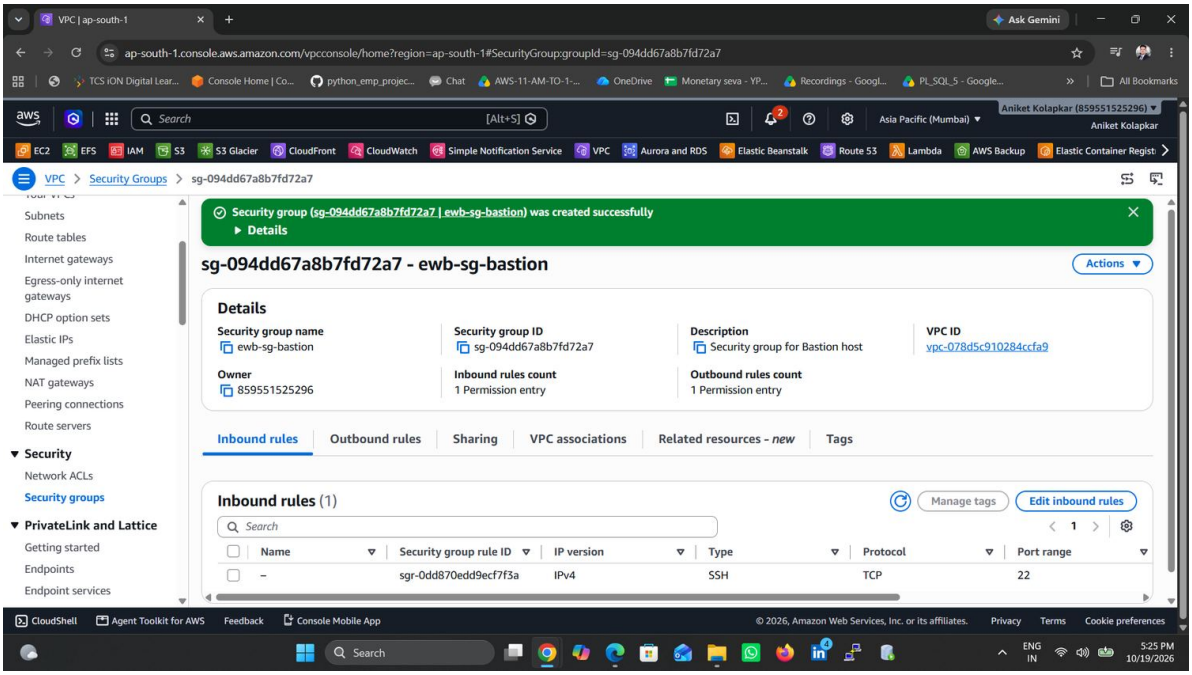
Evidence file: 36.png

Screenshot 37 — Create and configure EC2 security groups



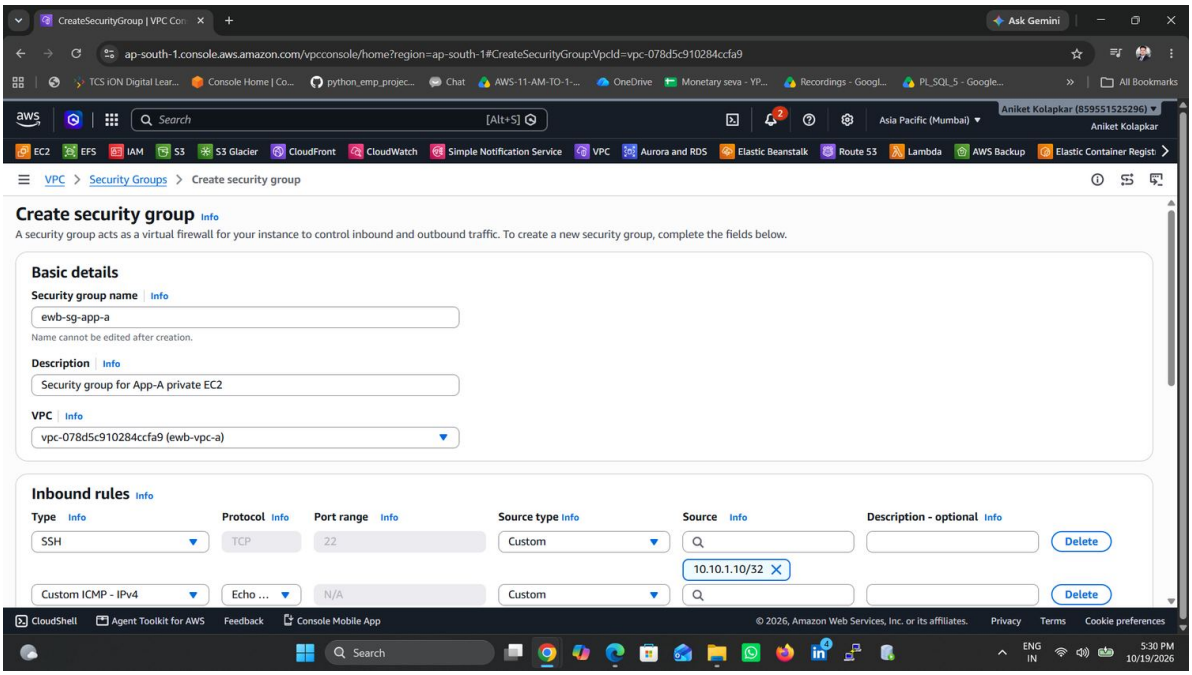
Evidence file: 37.png

Screenshot 38 — Create and configure EC2 security groups



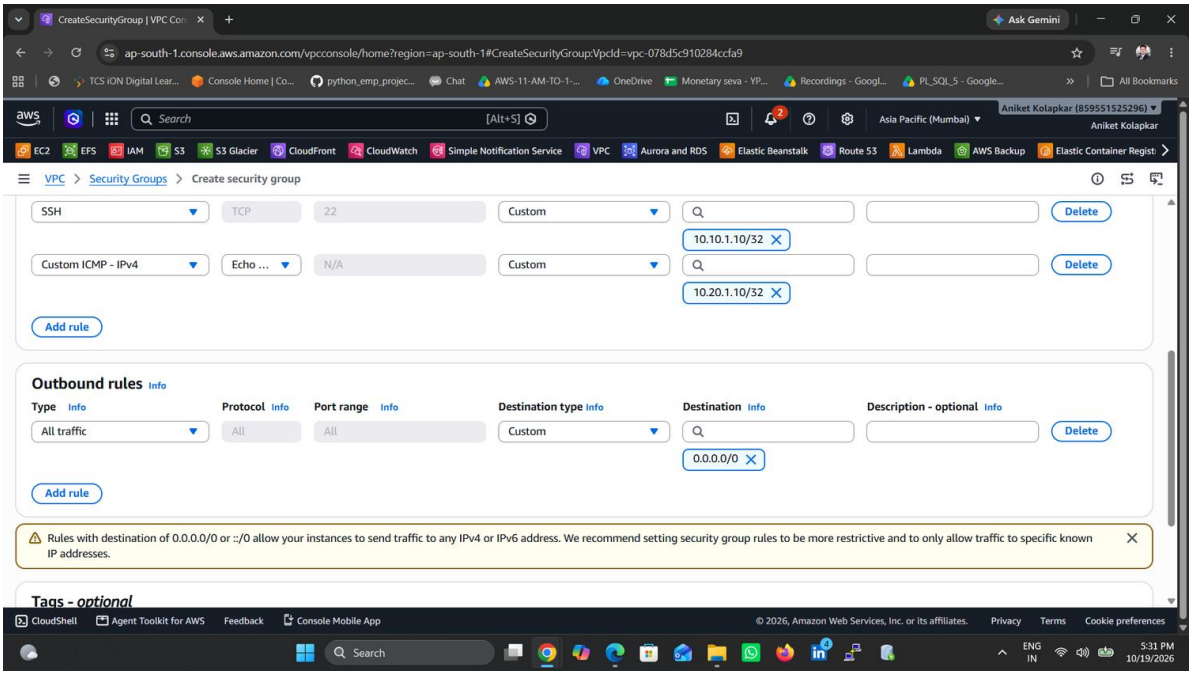
Evidence file: 38.png

Screenshot 39 — Create and configure EC2 security groups



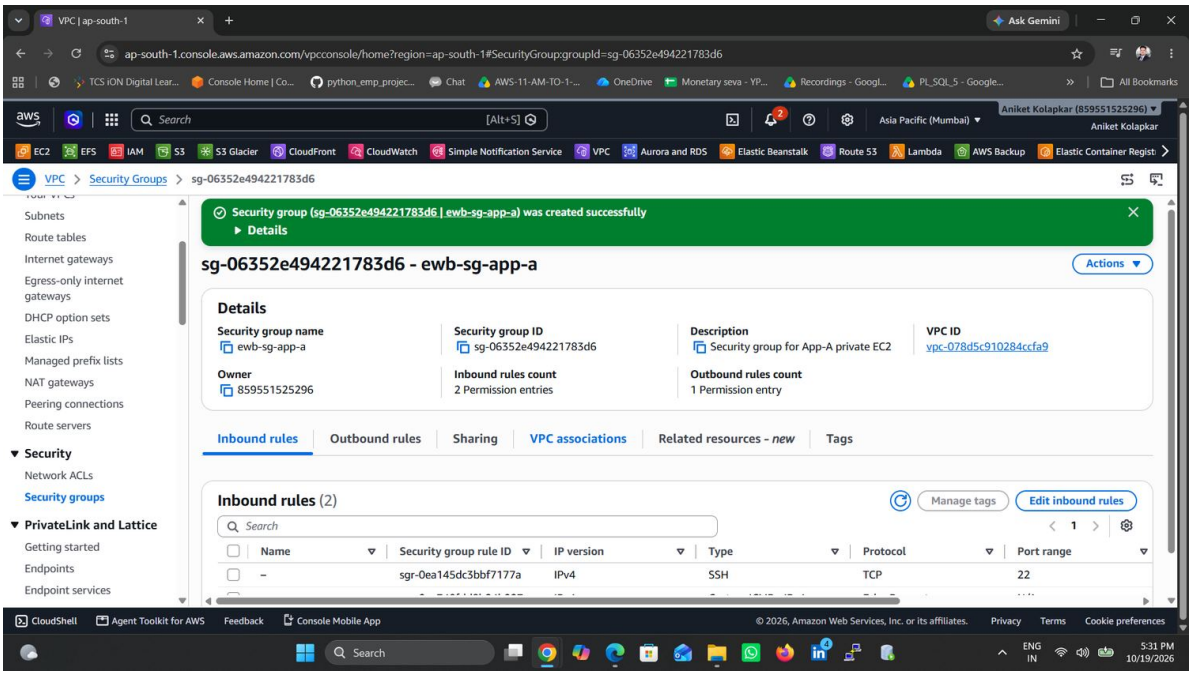
Evidence file: 39.png

Screenshot 40 — Create and configure EC2 security groups



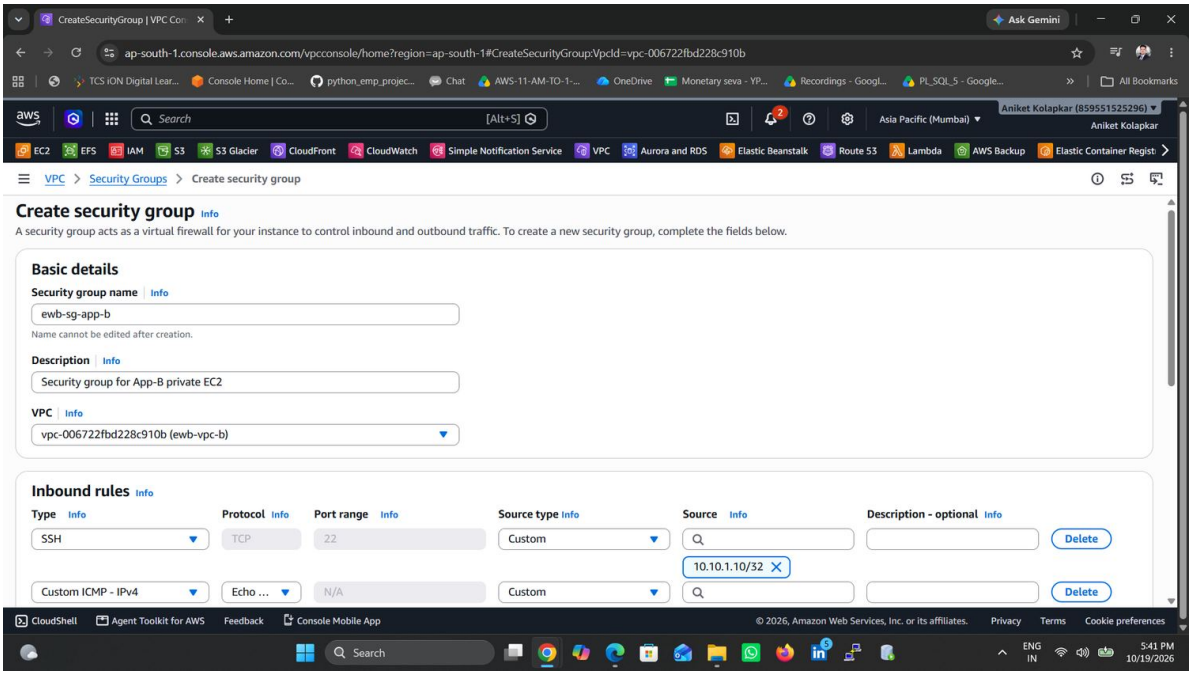
Evidence file: 40.png

Screenshot 41 — Create and configure EC2 security groups



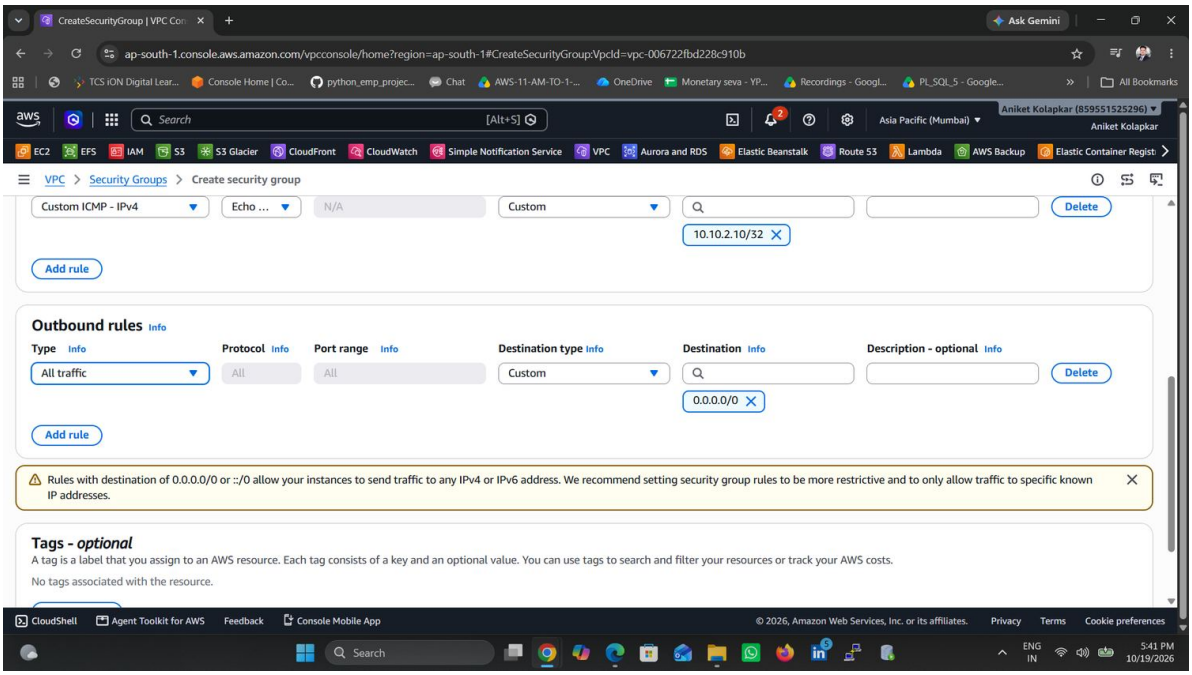
Evidence file: 41.png

Screenshot 42 — Create and configure EC2 security groups



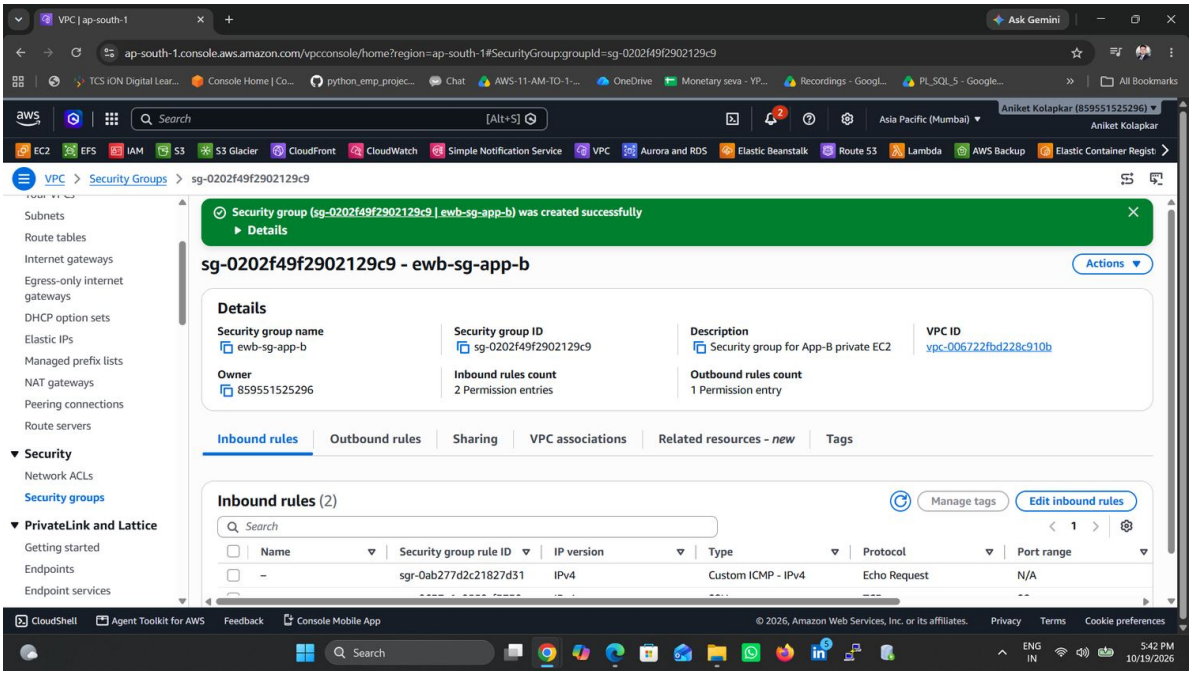
Evidence file: 42.png

Screenshot 43 — Create and configure EC2 security groups



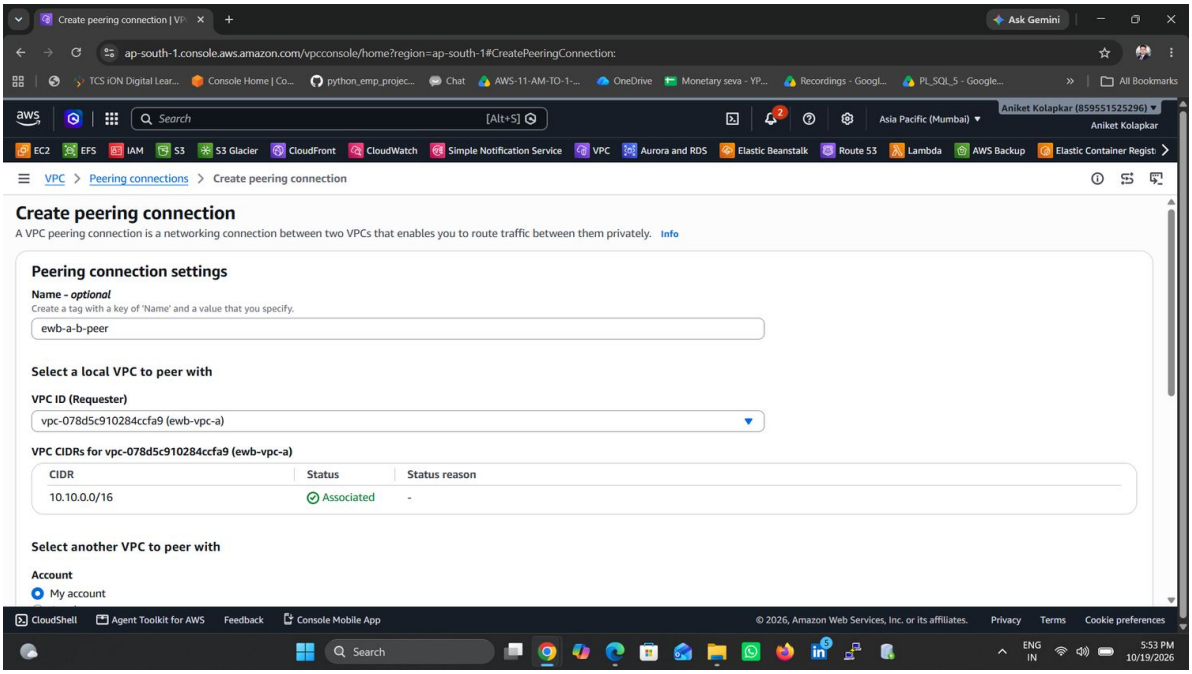
Evidence file: 43.png

Screenshot 44 — Create and configure EC2 security groups



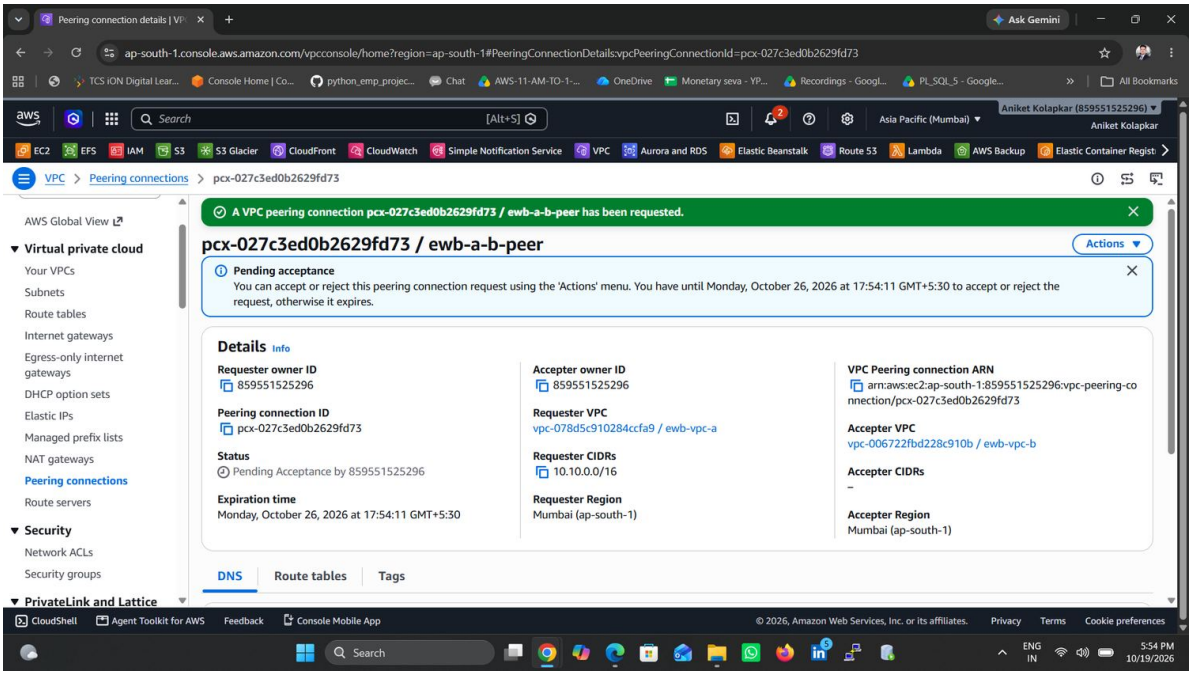
Evidence file: 44.png

Screenshot 45 — Create and verify the VPC Peering connection



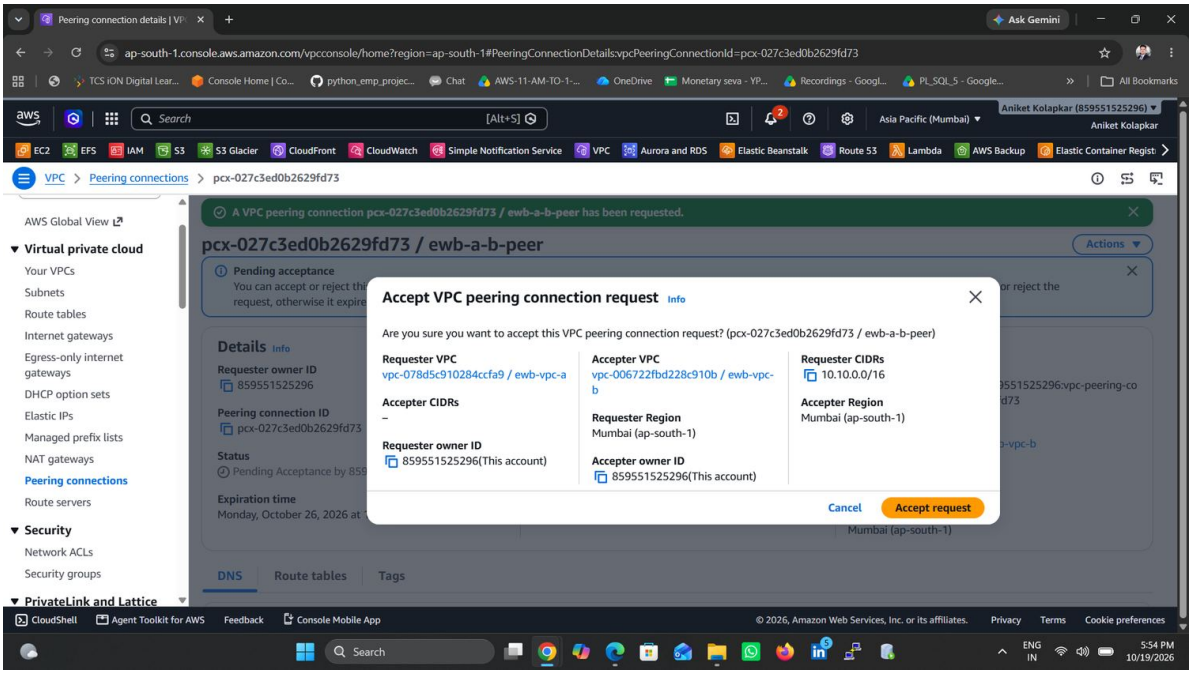
Evidence file: 45.png

Screenshot 46 — Create and verify the VPC Peering connection



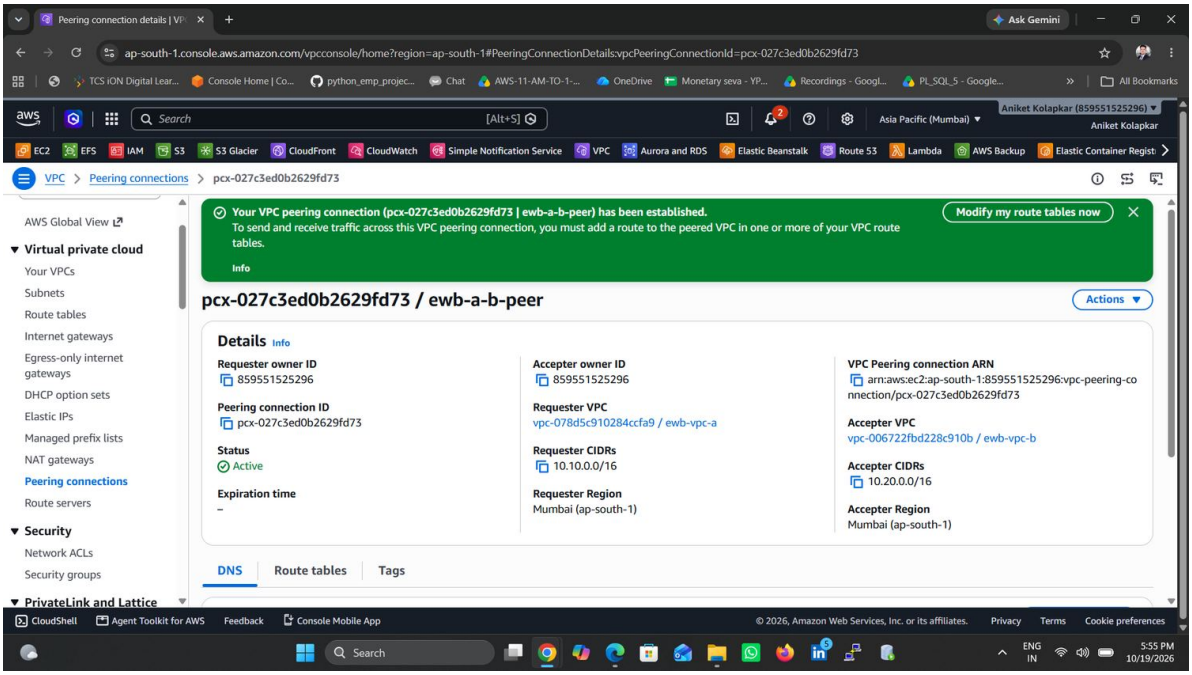
Evidence file: 46.png

Screenshot 47 — Create and verify the VPC Peering connection



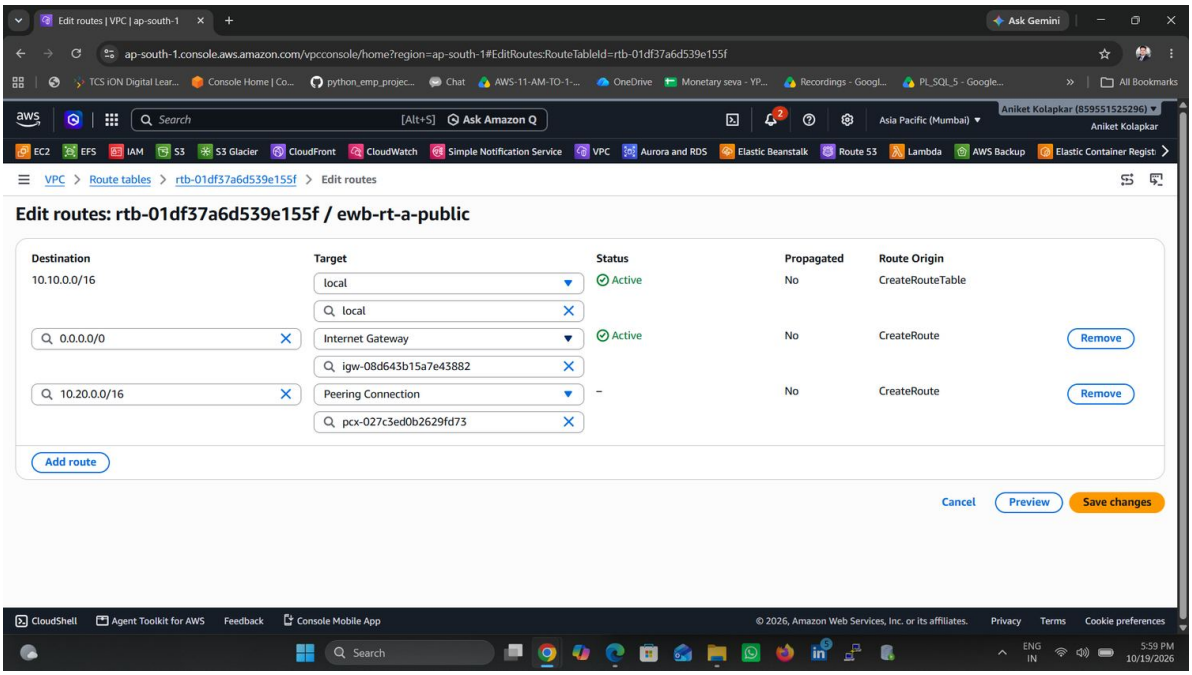
Evidence file: 47.png

Screenshot 48 — Create and verify the VPC Peering connection



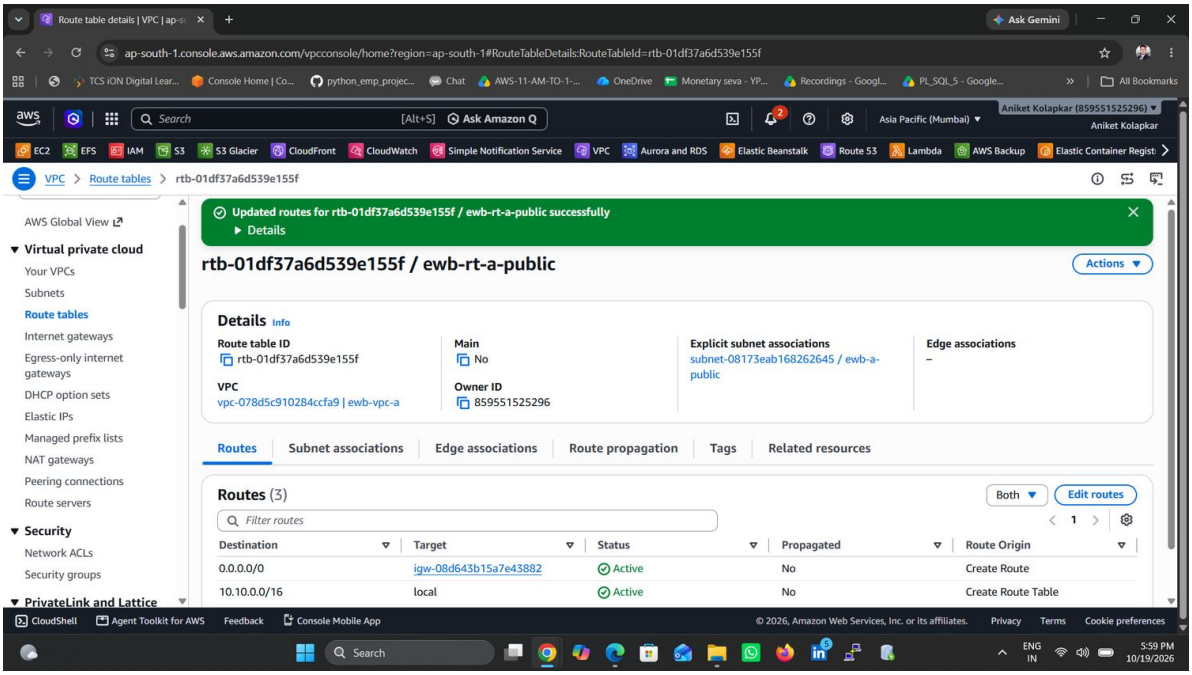
Evidence file: 48.png

Screenshot 49 — Configure routes and review EC2 instance settings



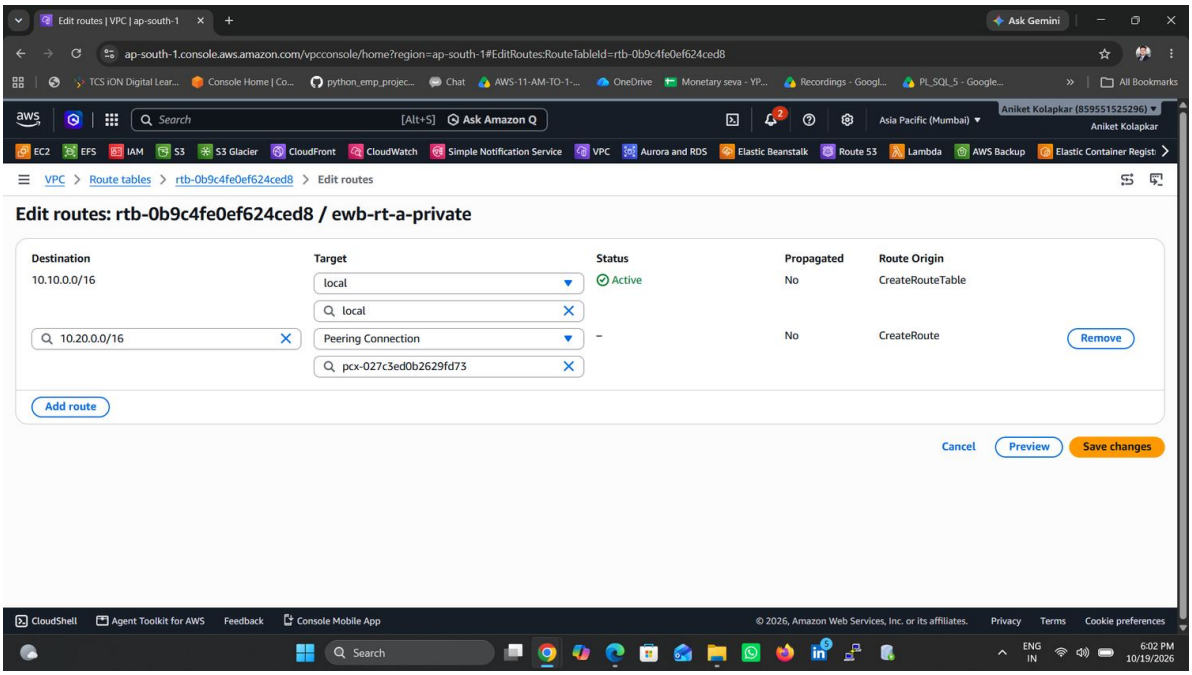
Evidence file: 49.png

Screenshot 50 — Configure routes and review EC2 instance settings



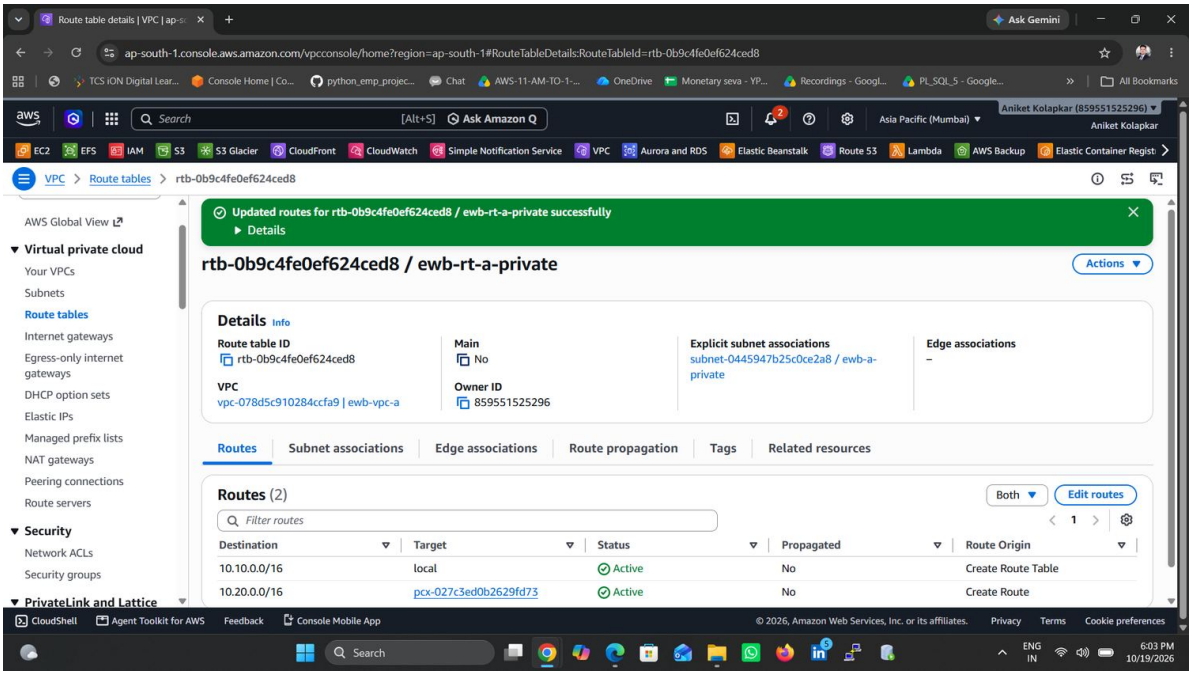
Evidence file: 50.png

Screenshot 51 — Configure routes and review EC2 instance settings



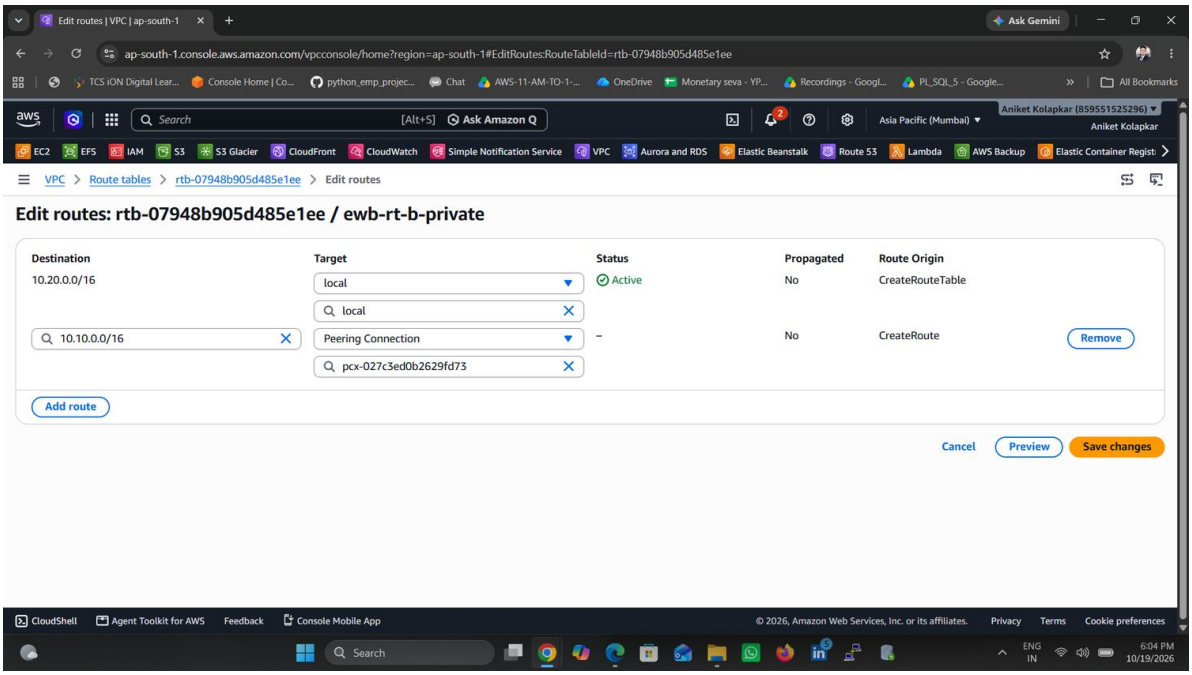
Evidence file: 51.png

Screenshot 52 — Configure routes and review EC2 instance settings



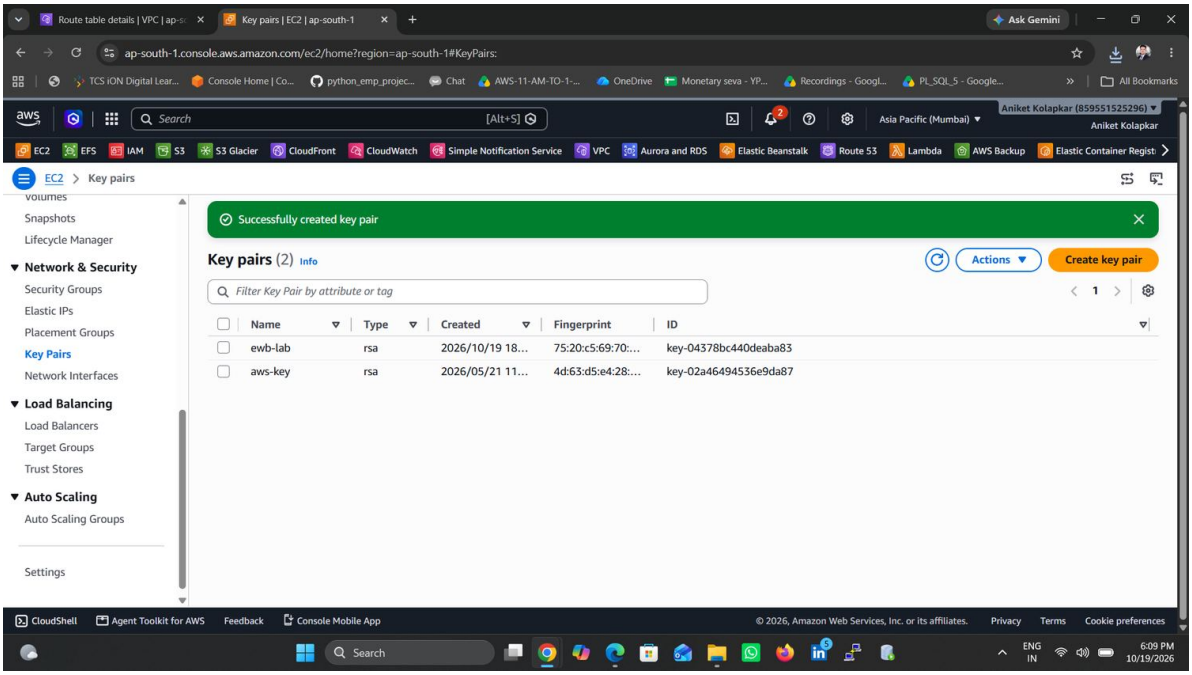
Evidence file: 52.png

Screenshot 53 — Configure routes and review EC2 instance settings



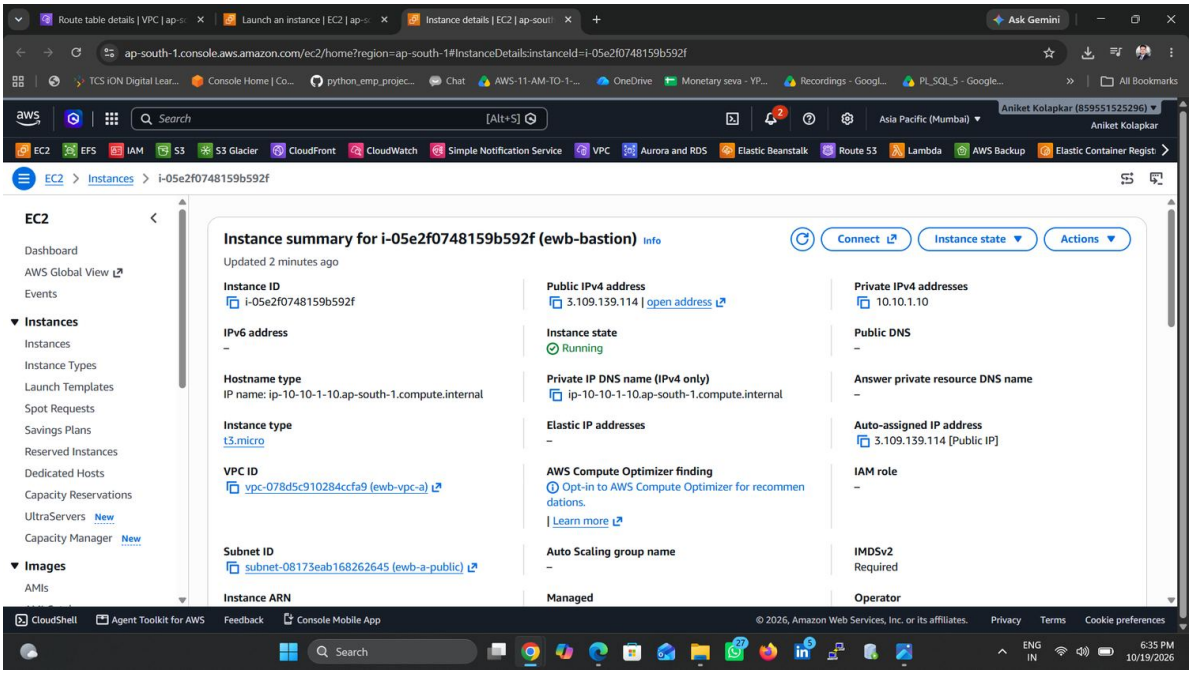
Evidence file: 53.png

Screenshot 54 — Configure routes and review EC2 instance settings



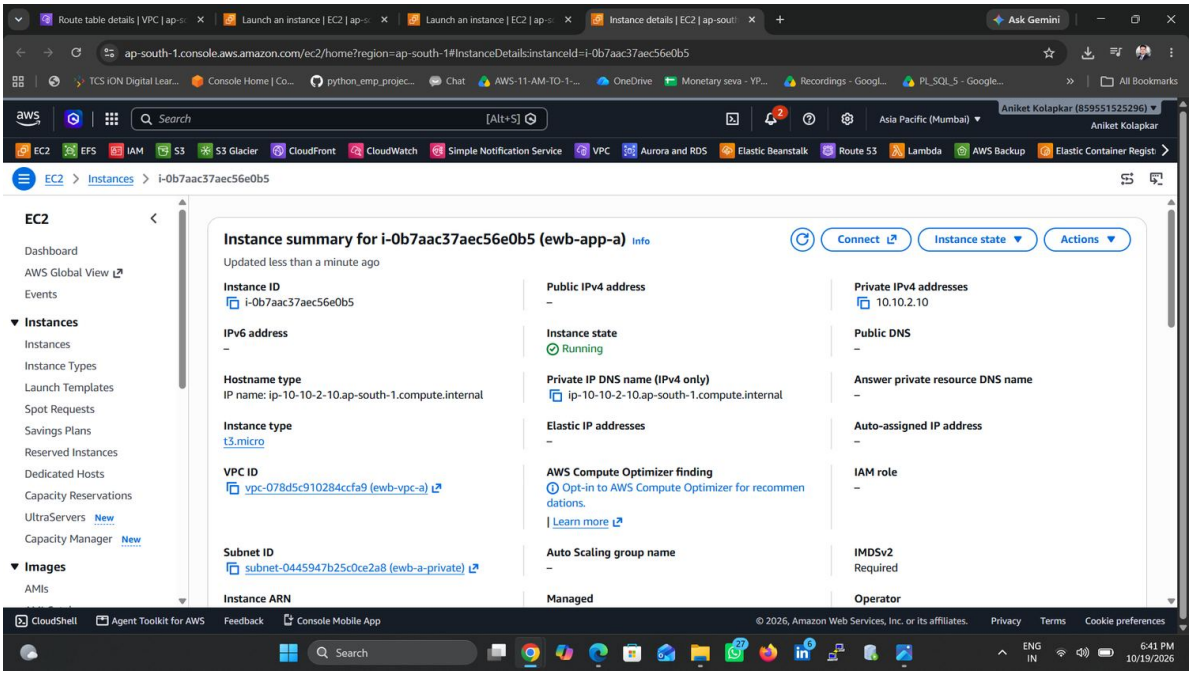
Evidence file: 54.png

Screenshot 55 — Connect to private EC2 hosts and run connectivity commands



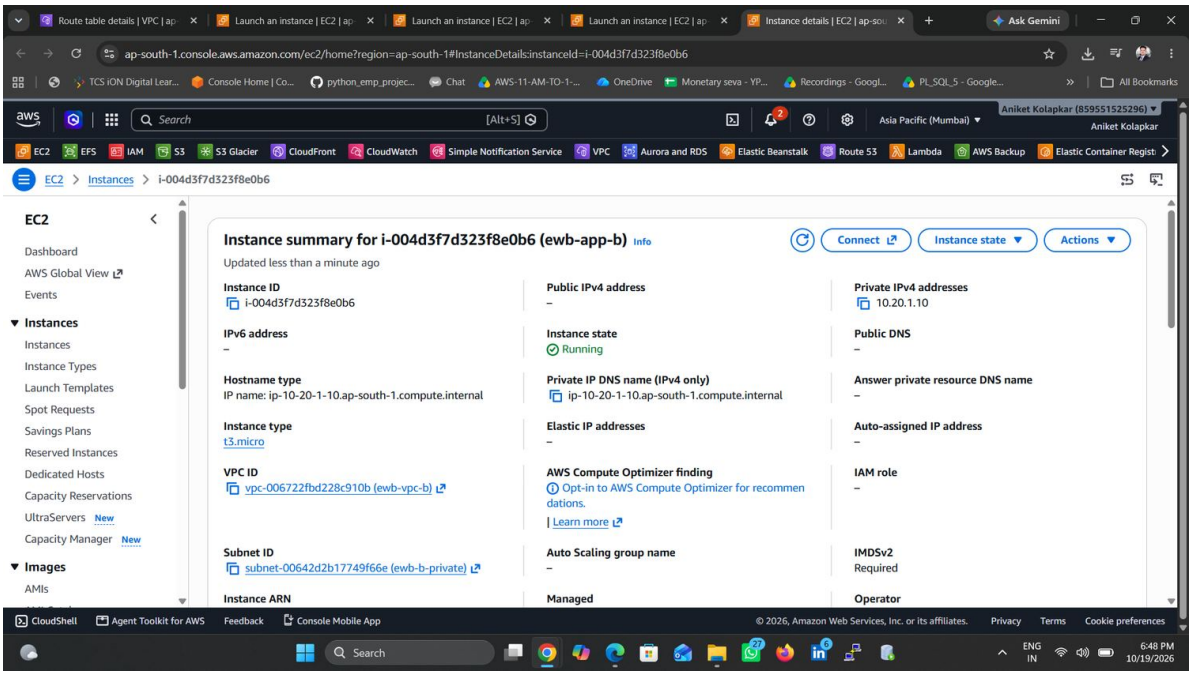
Evidence file: 55.png

Screenshot 56 — Connect to private EC2 hosts and run connectivity commands



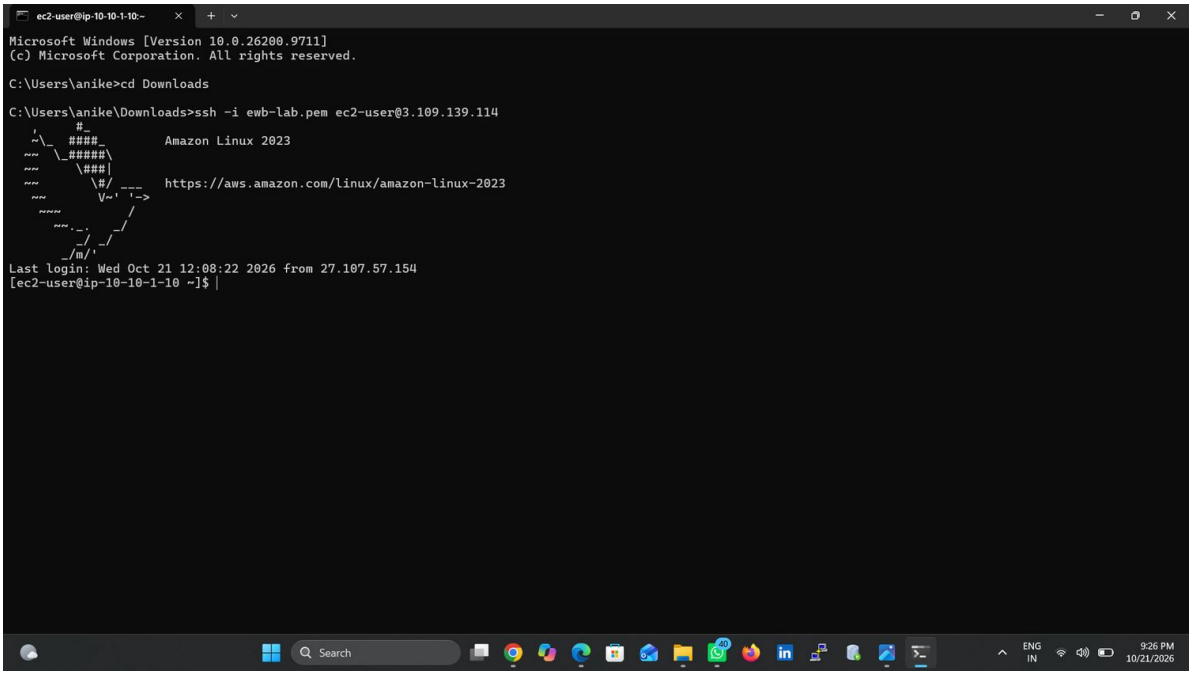
Evidence file: 56.png

Screenshot 57 — Connect to private EC2 hosts and run connectivity commands



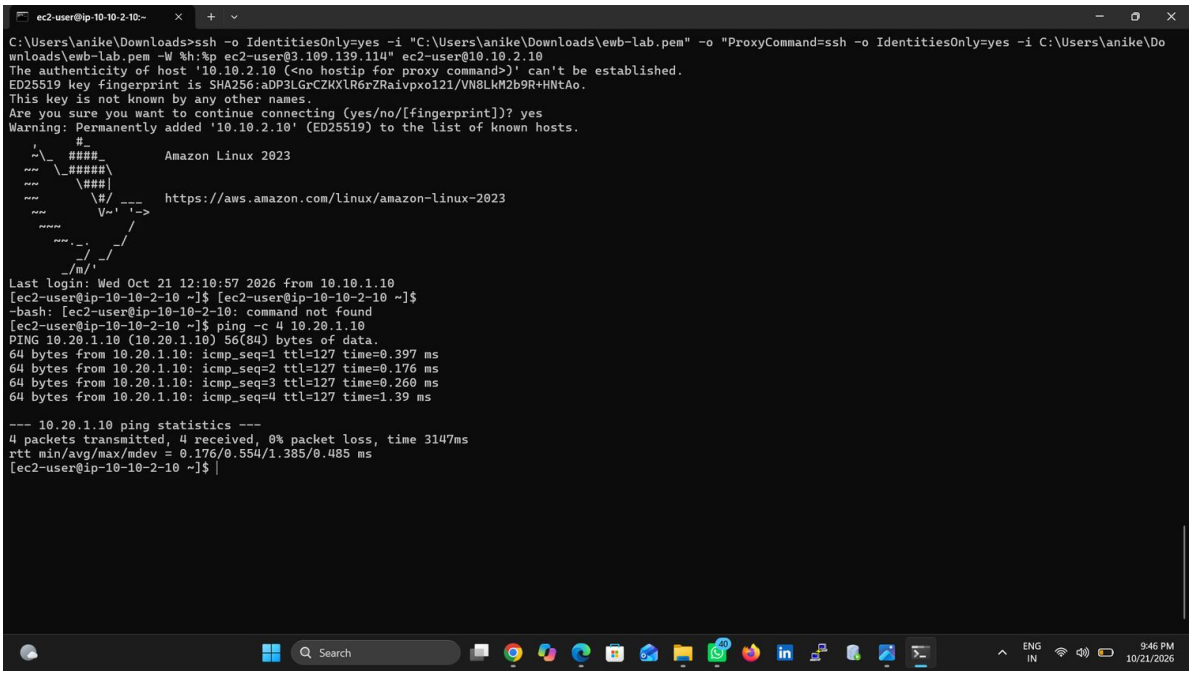
Evidence file: 57.png

Screenshot 58 — Connect to private EC2 hosts and run connectivity commands



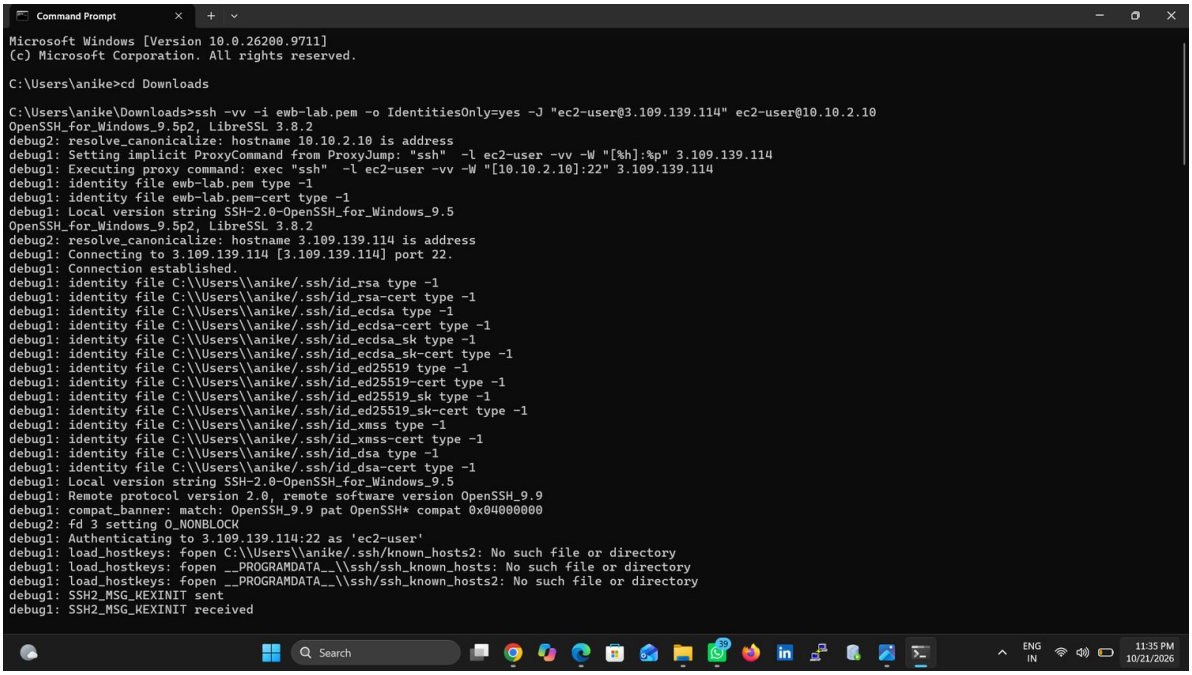
Evidence file: 58.png

Screenshot 59 — Connect to private EC2 hosts and run connectivity commands



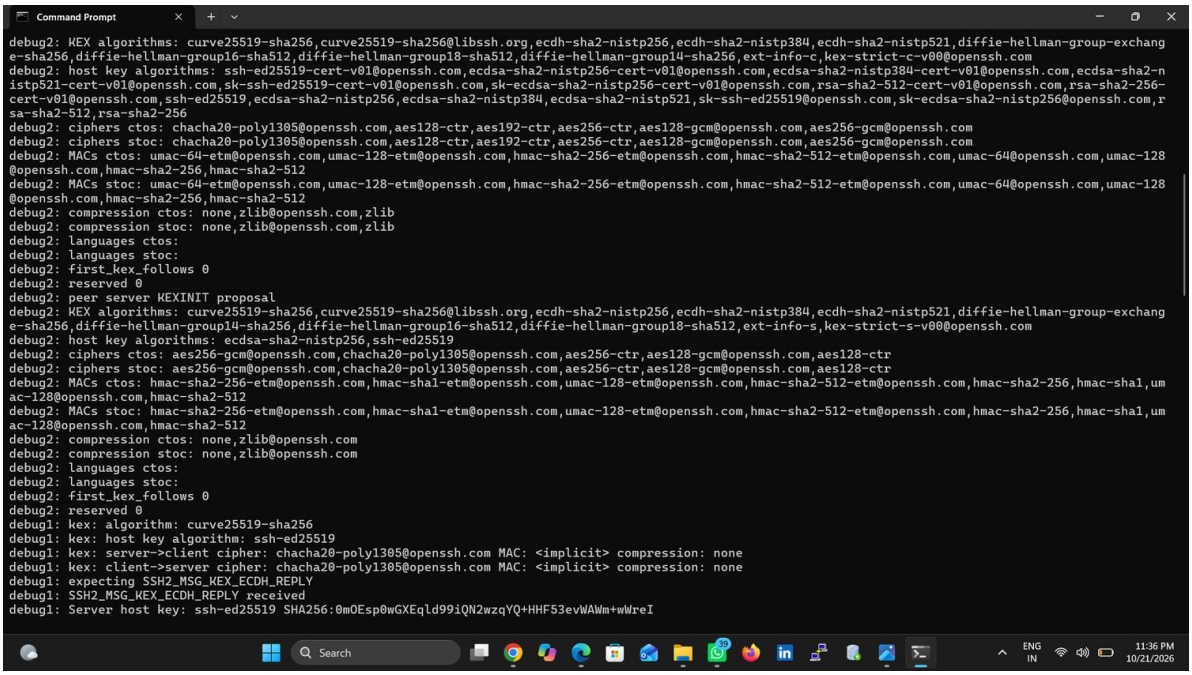
Evidence file: 59.png

Screenshot 60 — Connect to private EC2 hosts and run connectivity commands



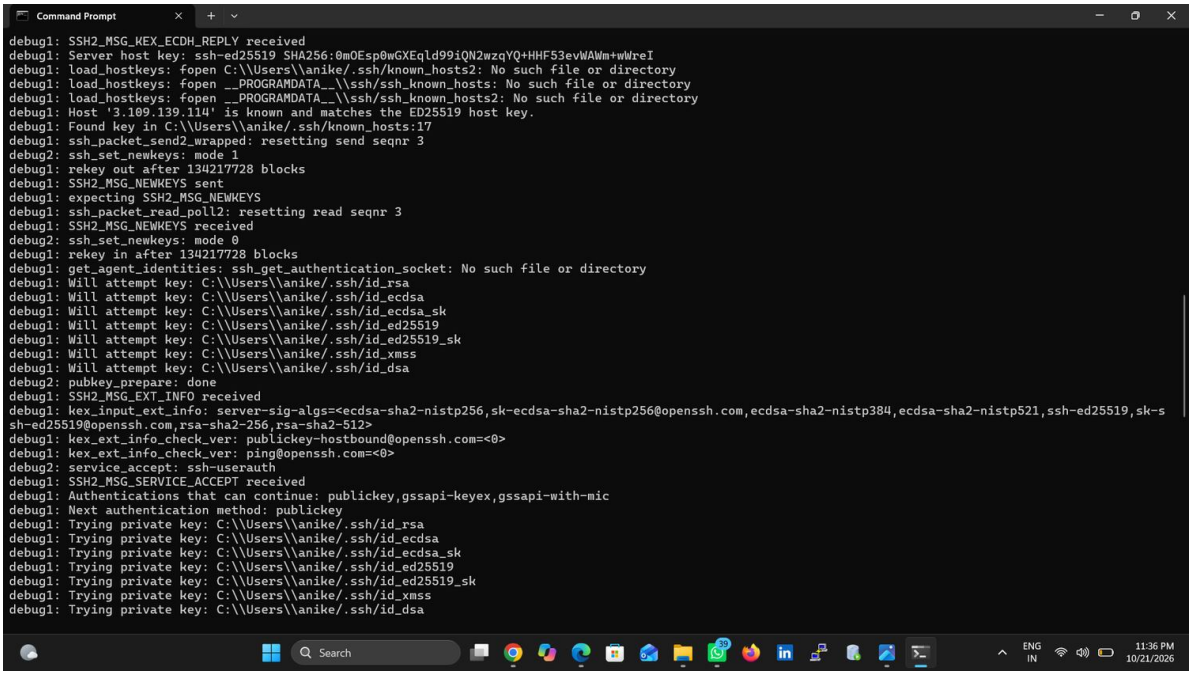
Evidence file: 60.png

Screenshot 61 — Connect to private EC2 hosts and run connectivity commands



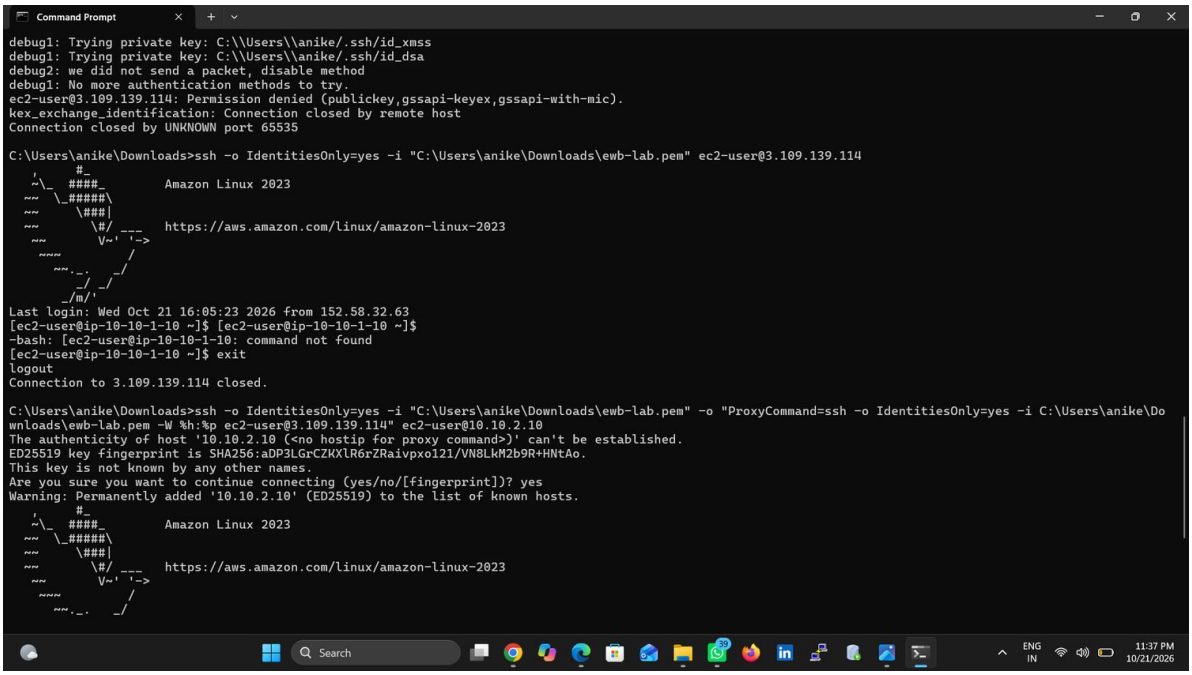
Evidence file: 61.png

Screenshot 62 — Connect to private EC2 hosts and run connectivity commands



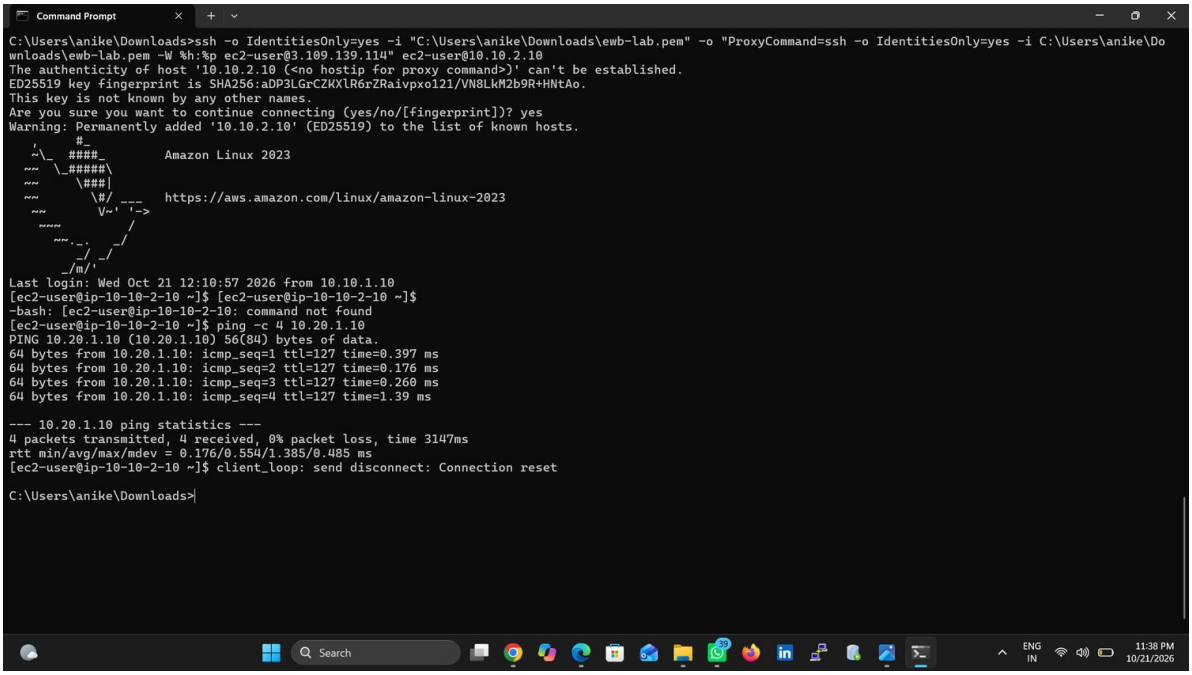
Evidence file: 62.png

Screenshot 63 — Connect to private EC2 hosts and run connectivity commands



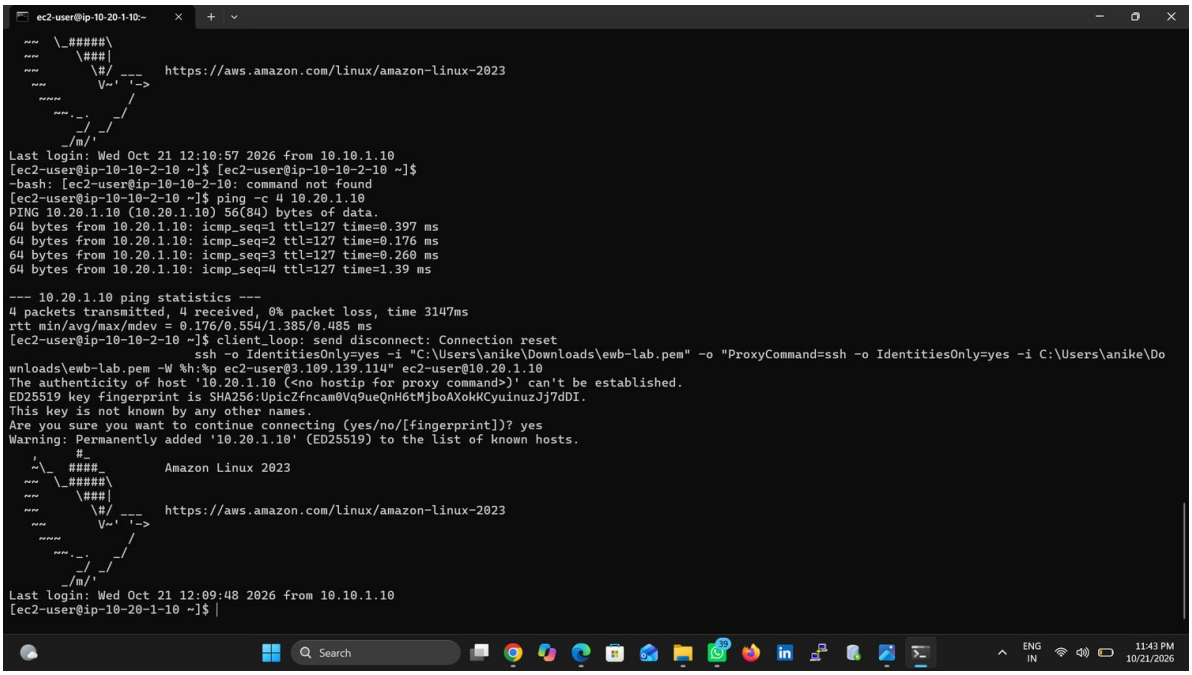
Evidence file: 63.png

Screenshot 64 — Connect to private EC2 hosts and run connectivity commands



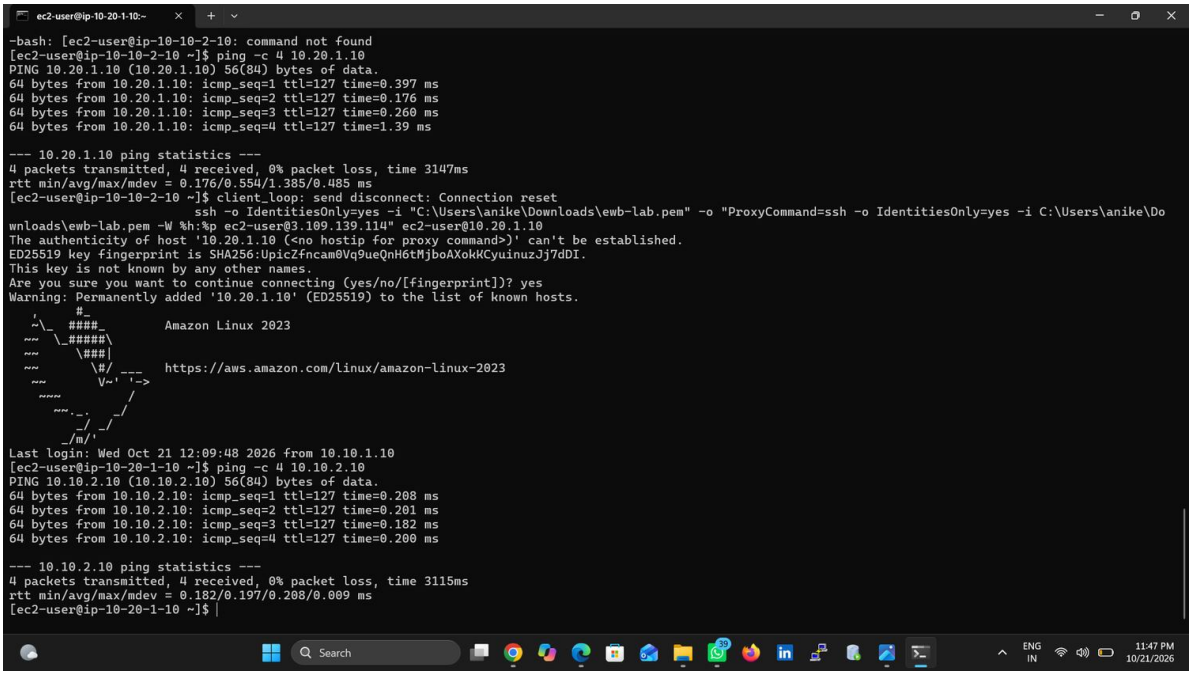
Evidence file: 64.png

Screenshot 65 — Connect to private EC2 hosts and run connectivity commands



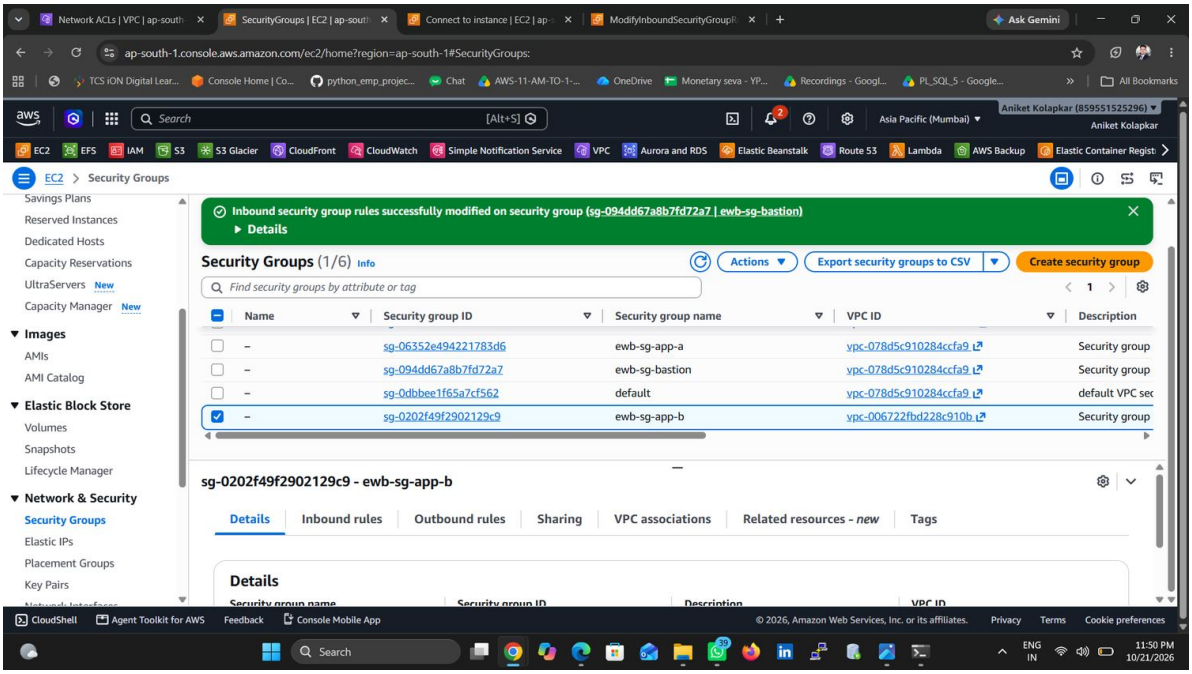
Evidence file: 65.png

Screenshot 66 — Connect to private EC2 hosts and run connectivity commands



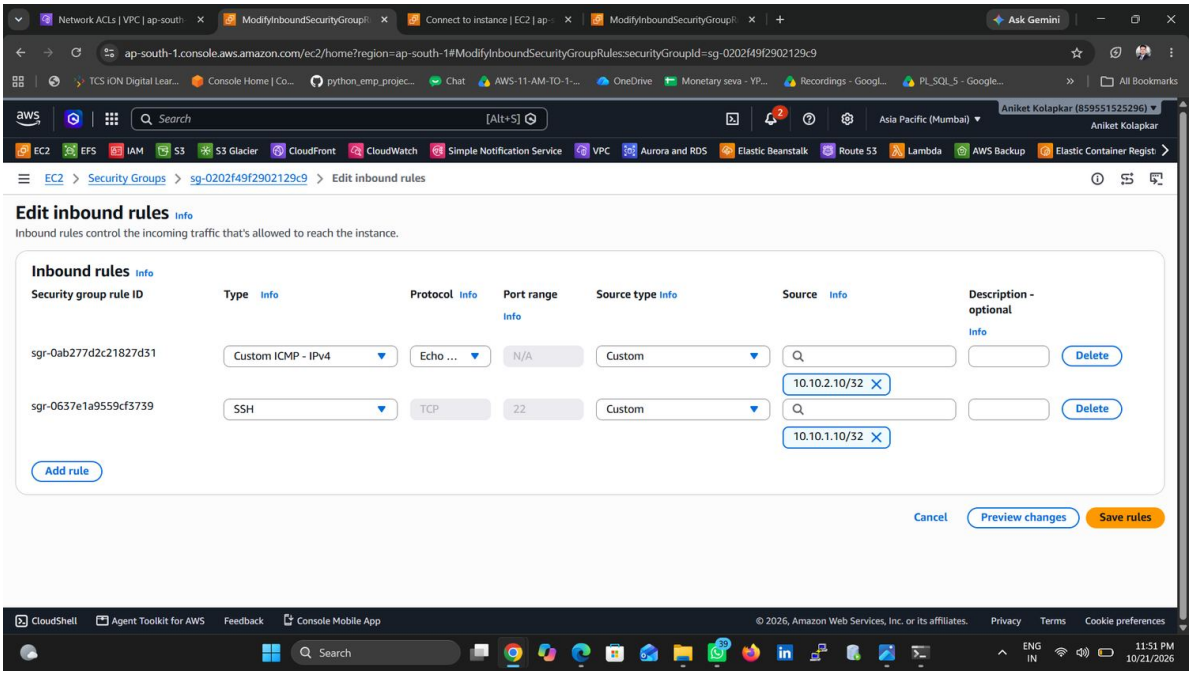
Evidence file: 66.png

Screenshot 67 — Review security-group rules and test traffic behavior



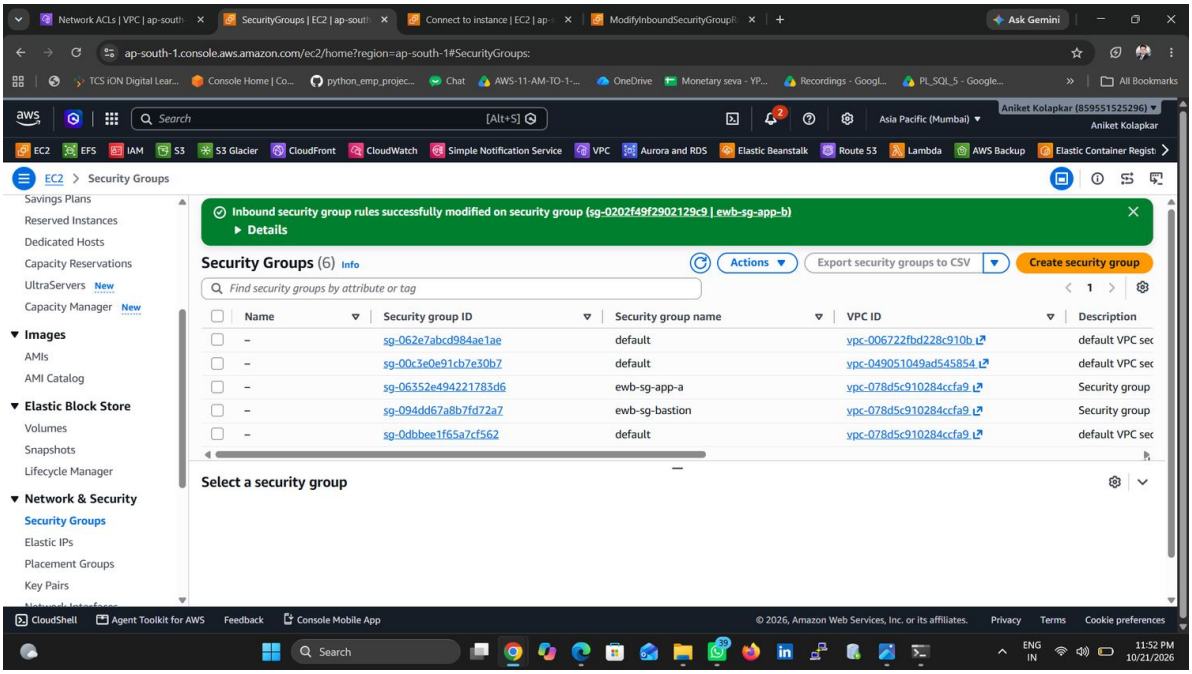
Evidence file: 67.png

Screenshot 68 — Review security-group rules and test traffic behavior



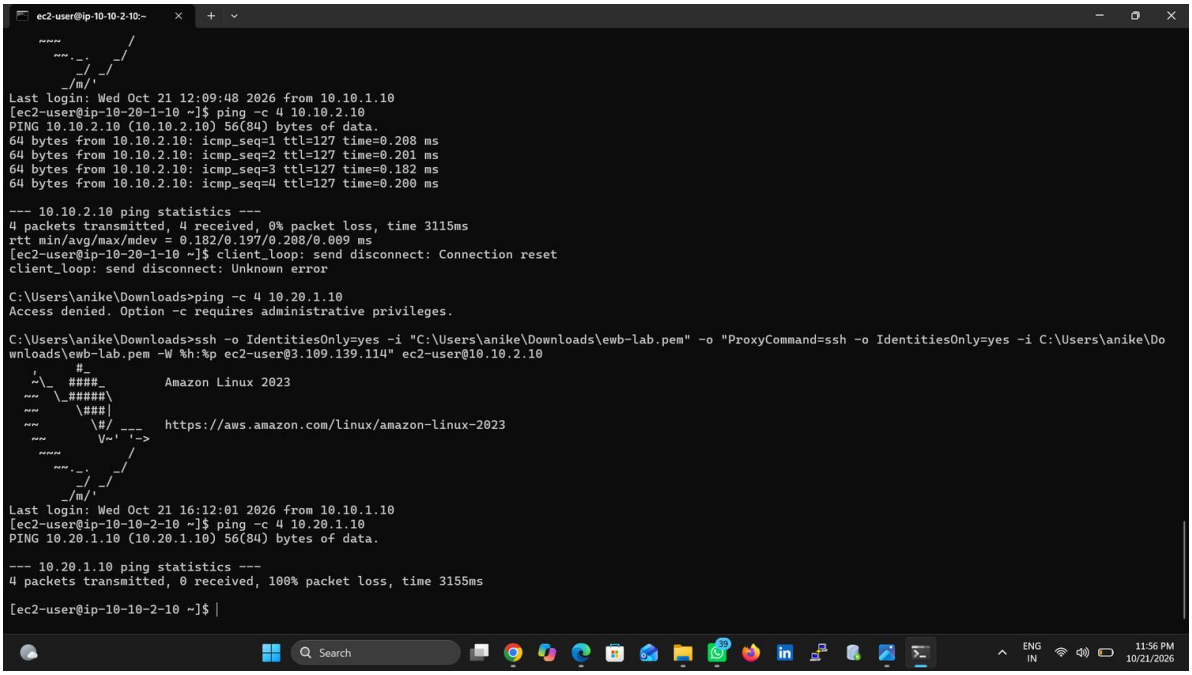
Evidence file: 68.png

Screenshot 69 — Review security-group rules and test traffic behavior



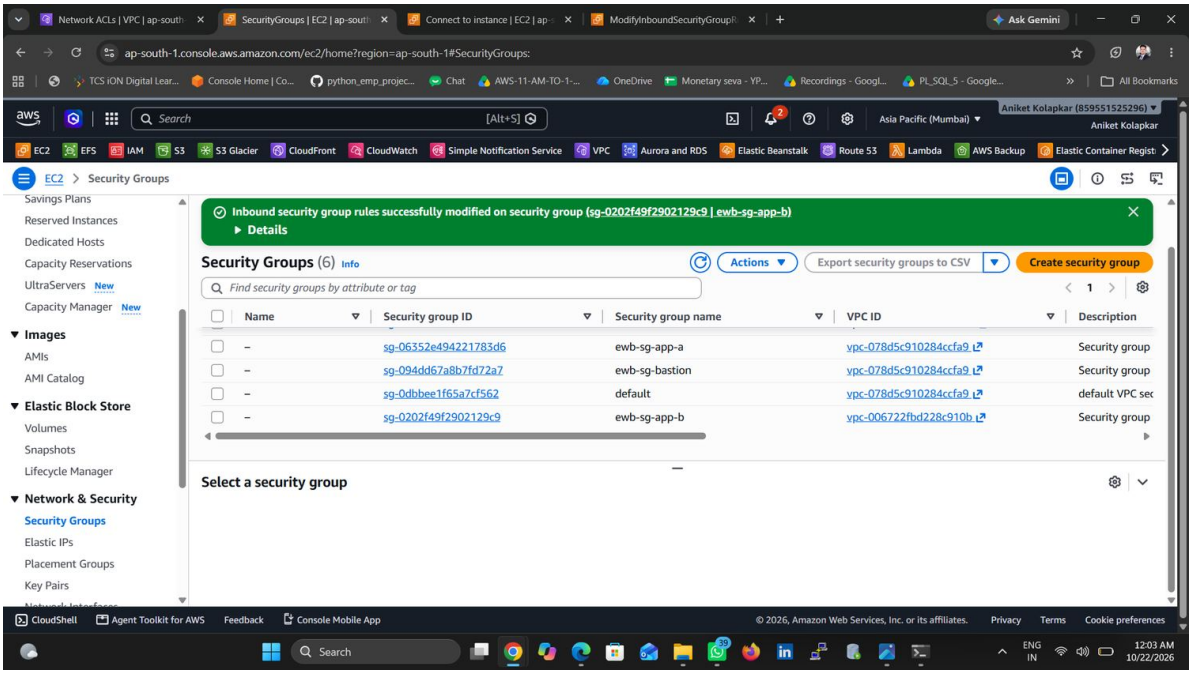
Evidence file: 69.png

Screenshot 70 — Review security-group rules and test traffic behavior



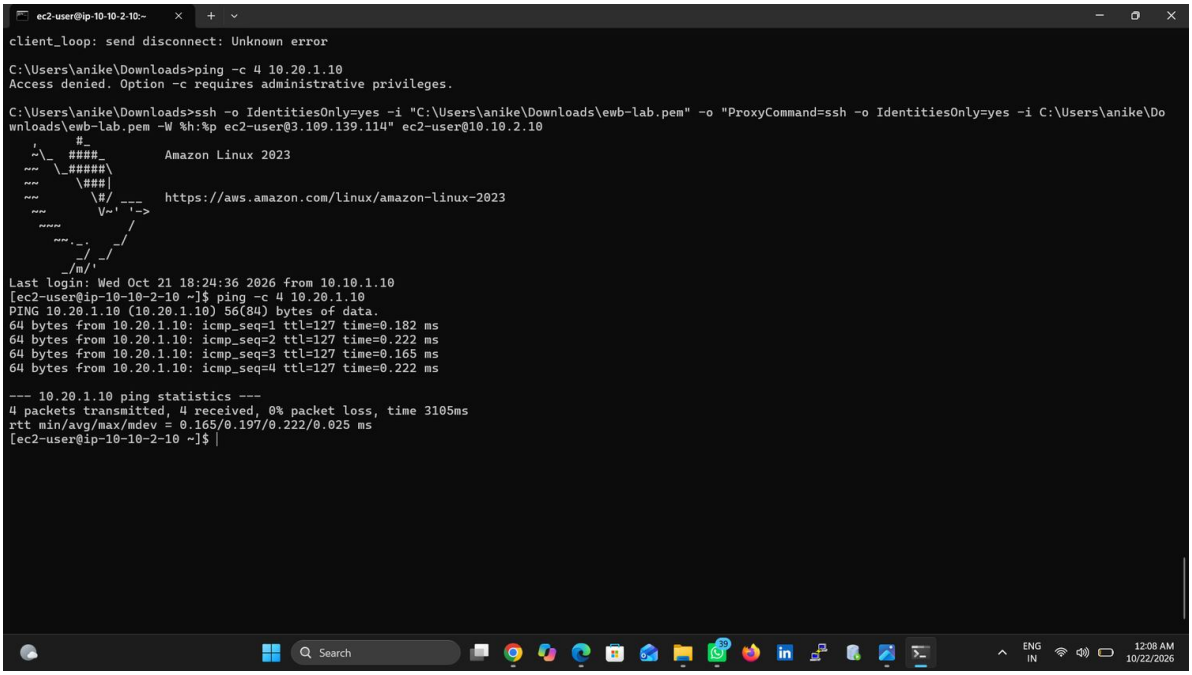
Evidence file: 70.png

Screenshot 71 — Review security-group rules and test traffic behavior



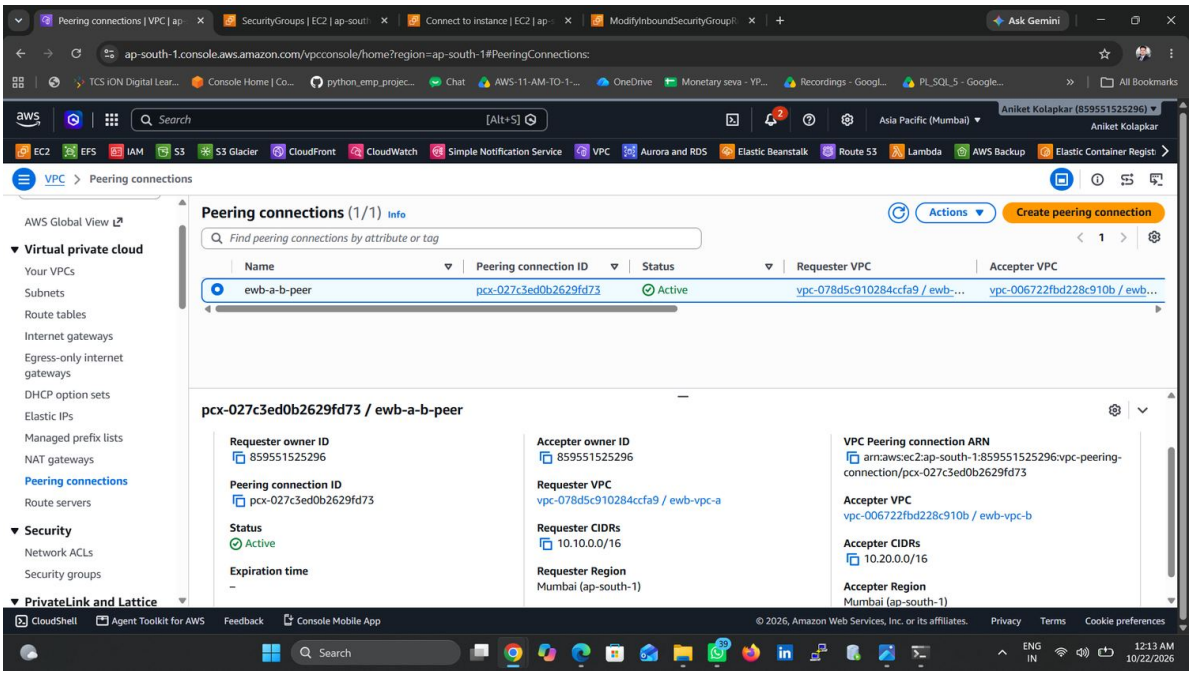
Evidence file: 71.png

Screenshot 72 — Review security-group rules and test traffic behavior



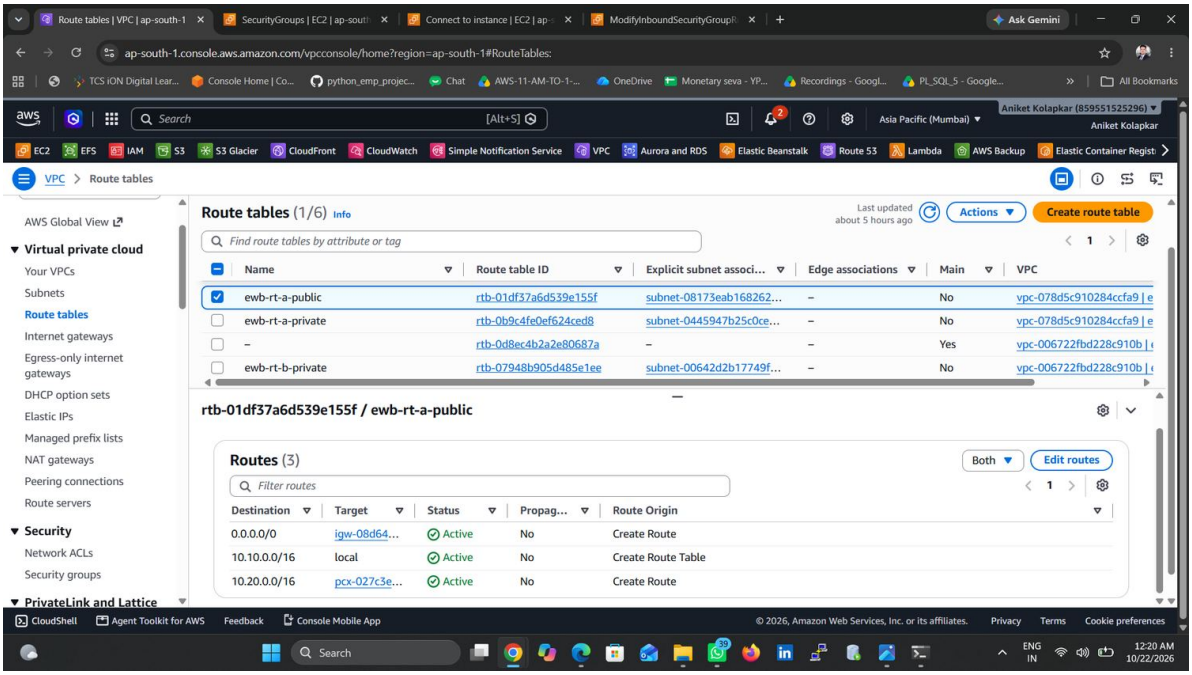
Evidence file: 72.png

Screenshot 73 — Inspect route tables, subnet associations, and connectivity configuration



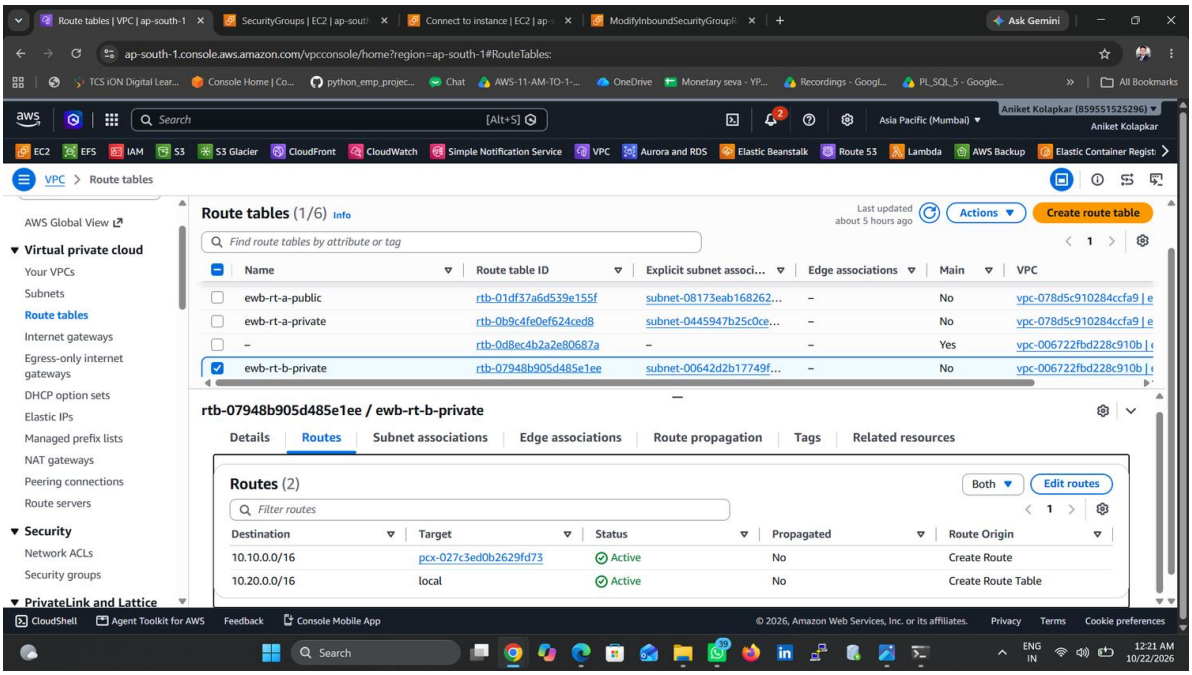
Evidence file: 73.png

Screenshot 74 — Inspect route tables, subnet associations, and connectivity configuration



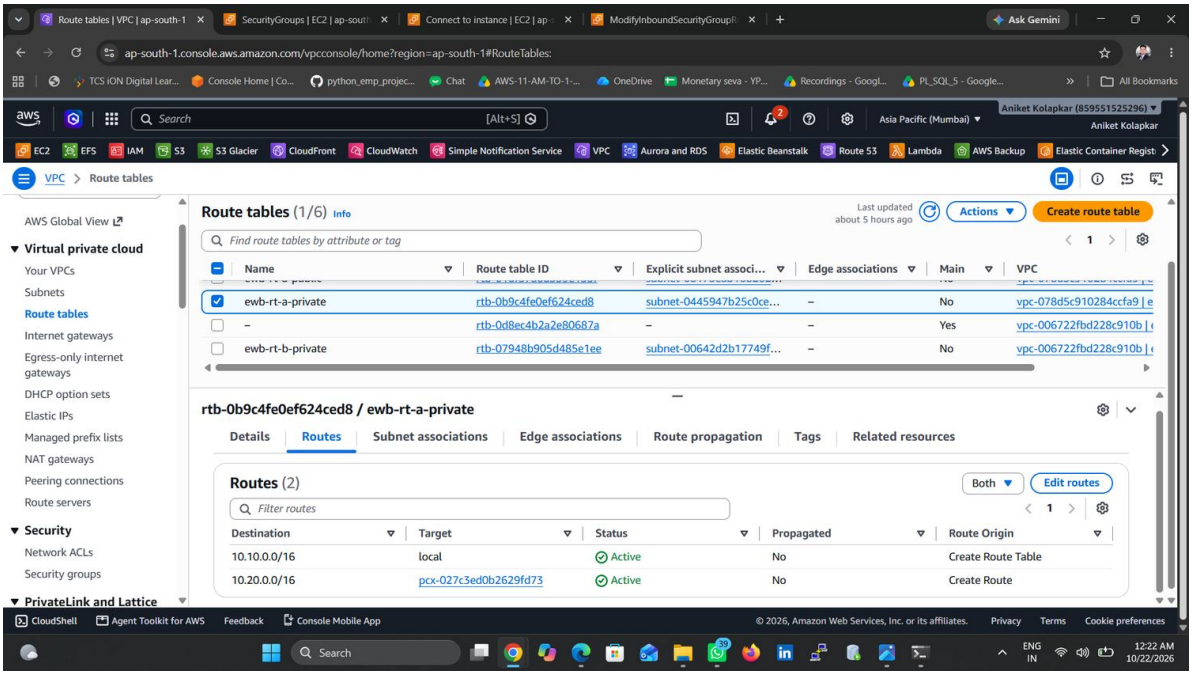
Evidence file: 74.png

Screenshot 75 — Inspect route tables, subnet associations, and connectivity configuration



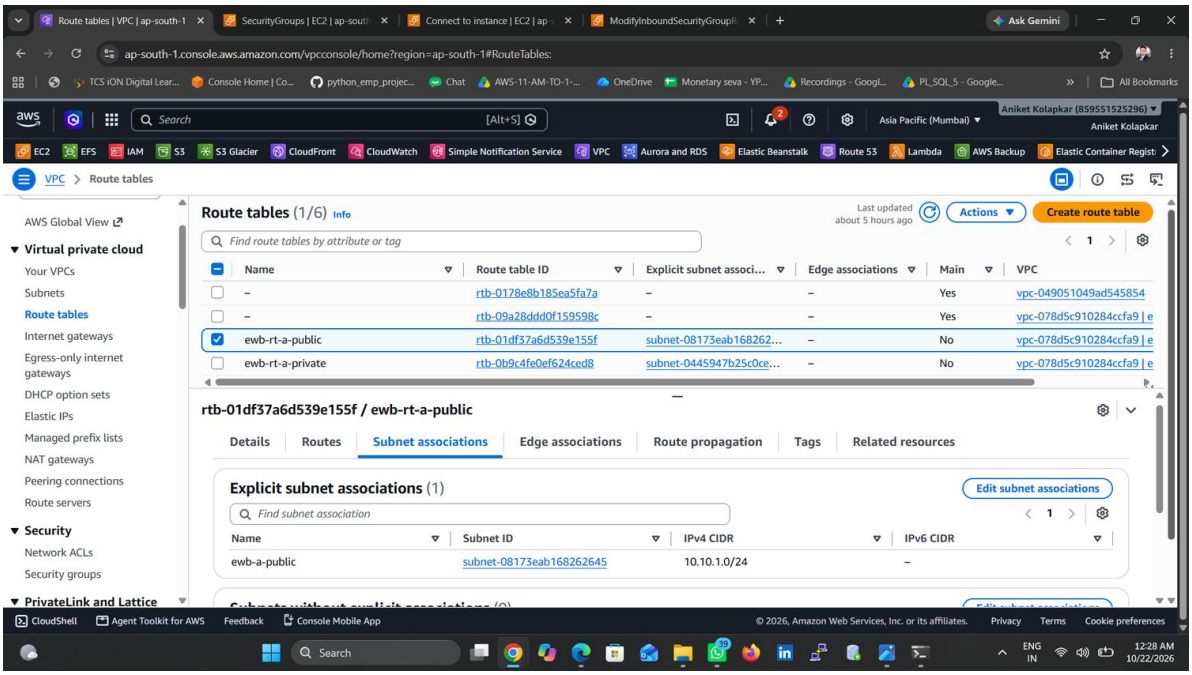
Evidence file: 75.png

Screenshot 76 — Inspect route tables, subnet associations, and connectivity configuration



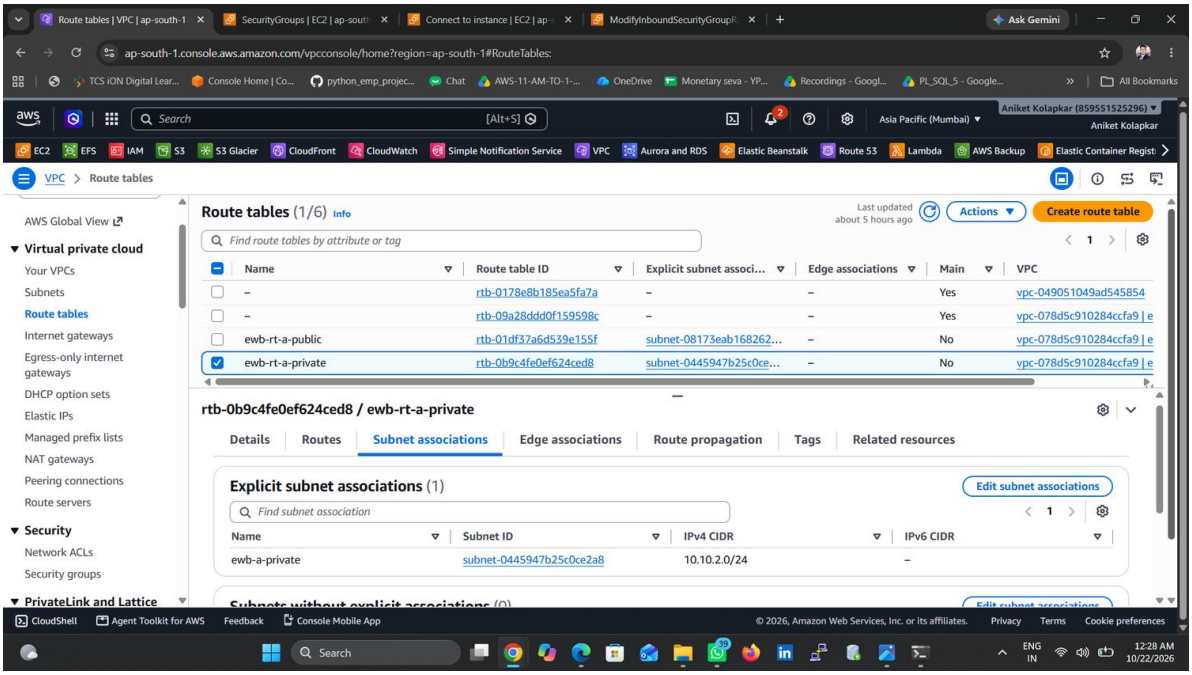
Evidence file: 76.png

Screenshot 77 — Inspect route tables, subnet associations, and connectivity configuration



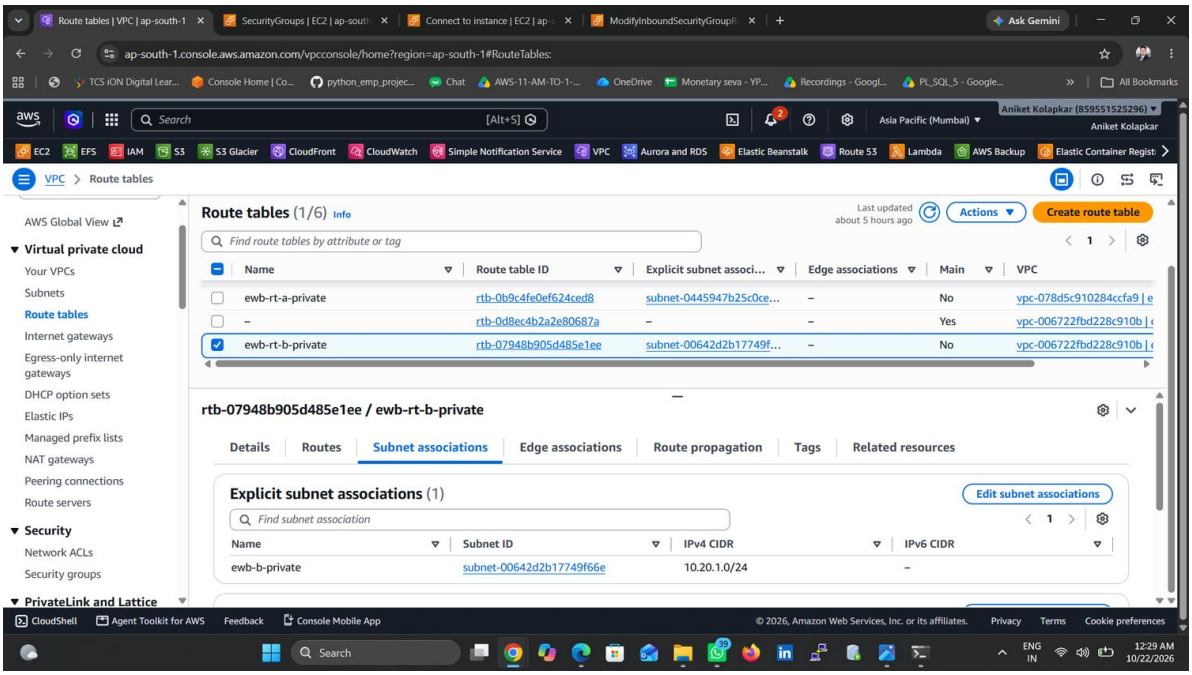
Evidence file: 77.png

Screenshot 78 — Inspect route tables, subnet associations, and connectivity configuration



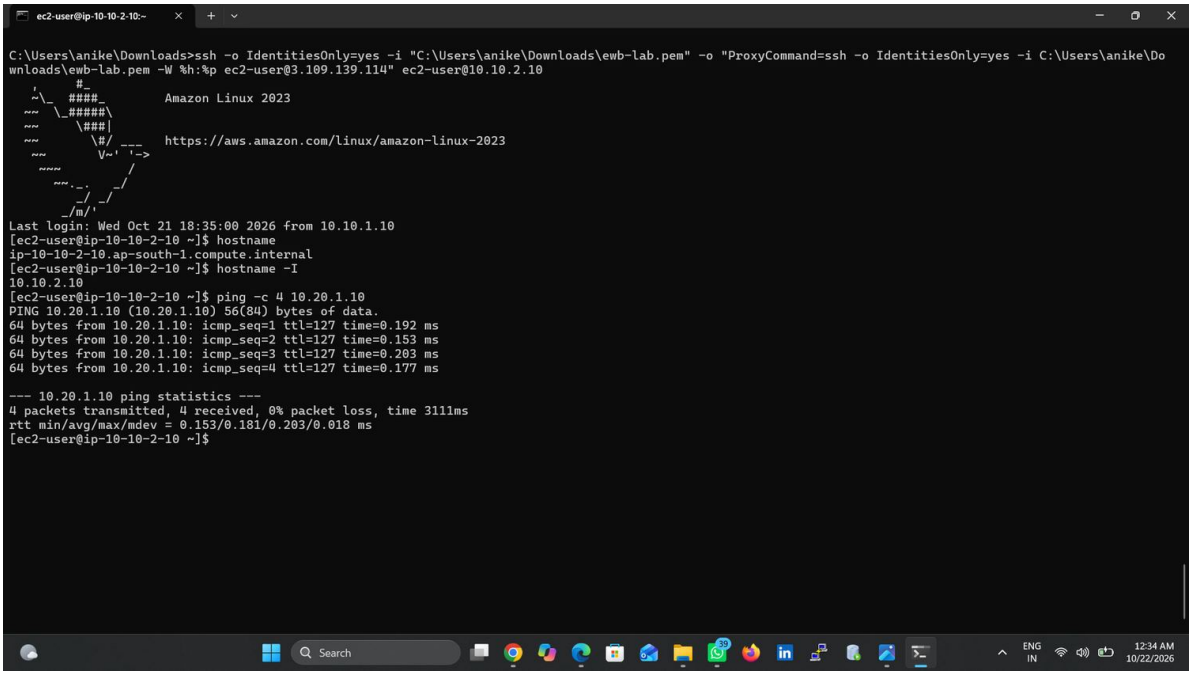
Evidence file: 78.png

Screenshot 79 — Inspect route tables, subnet associations, and connectivity configuration



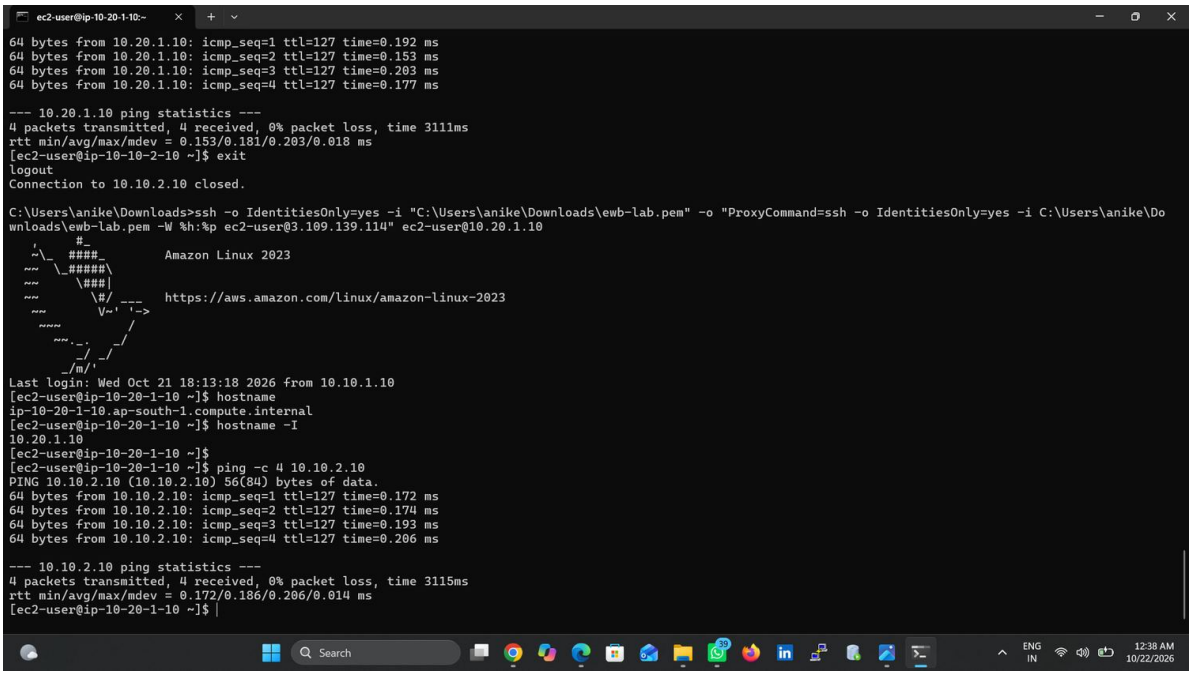
Evidence file: 79.png

Screenshot 80 — Inspect route tables, subnet associations, and connectivity configuration



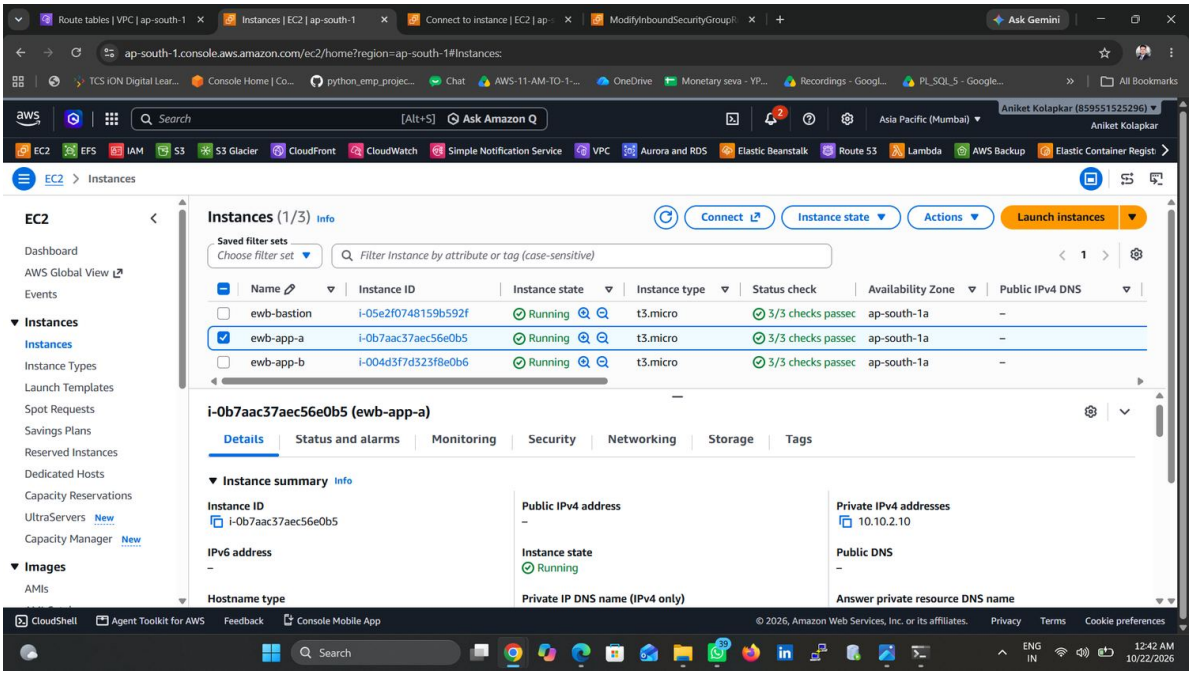
Evidence file: 80.png

Screenshot 81 — Inspect route tables, subnet associations, and connectivity configuration



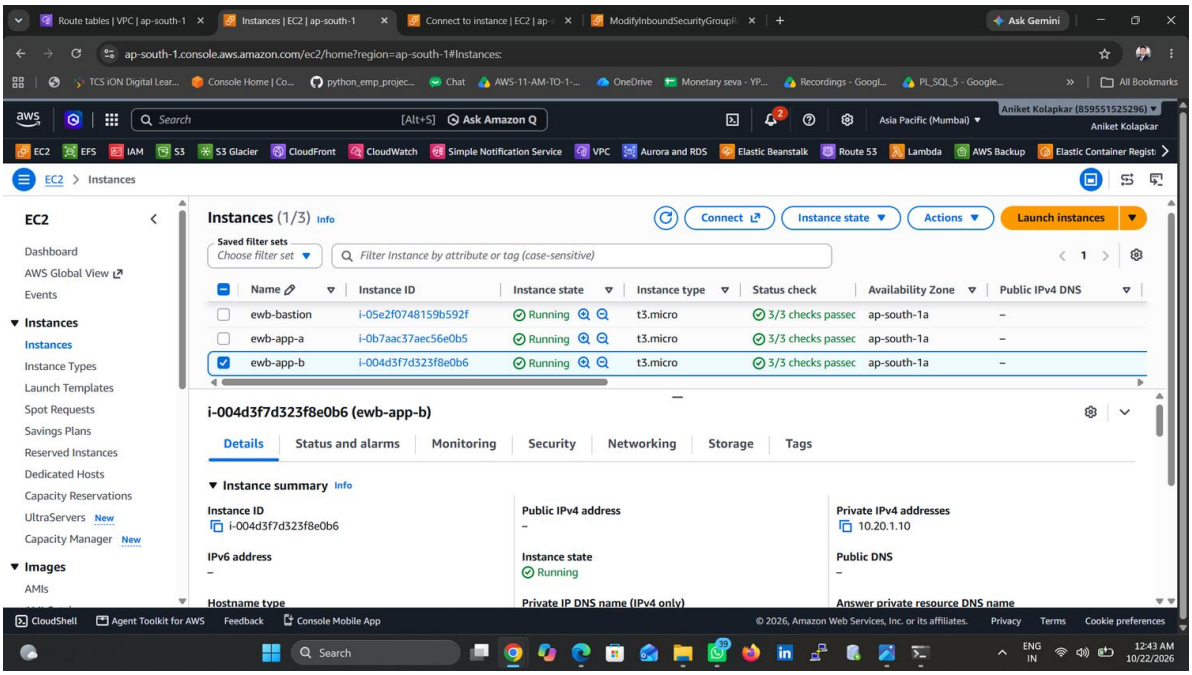
Evidence file: 81.png

Screenshot 82 — Inspect route tables, subnet associations, and connectivity configuration



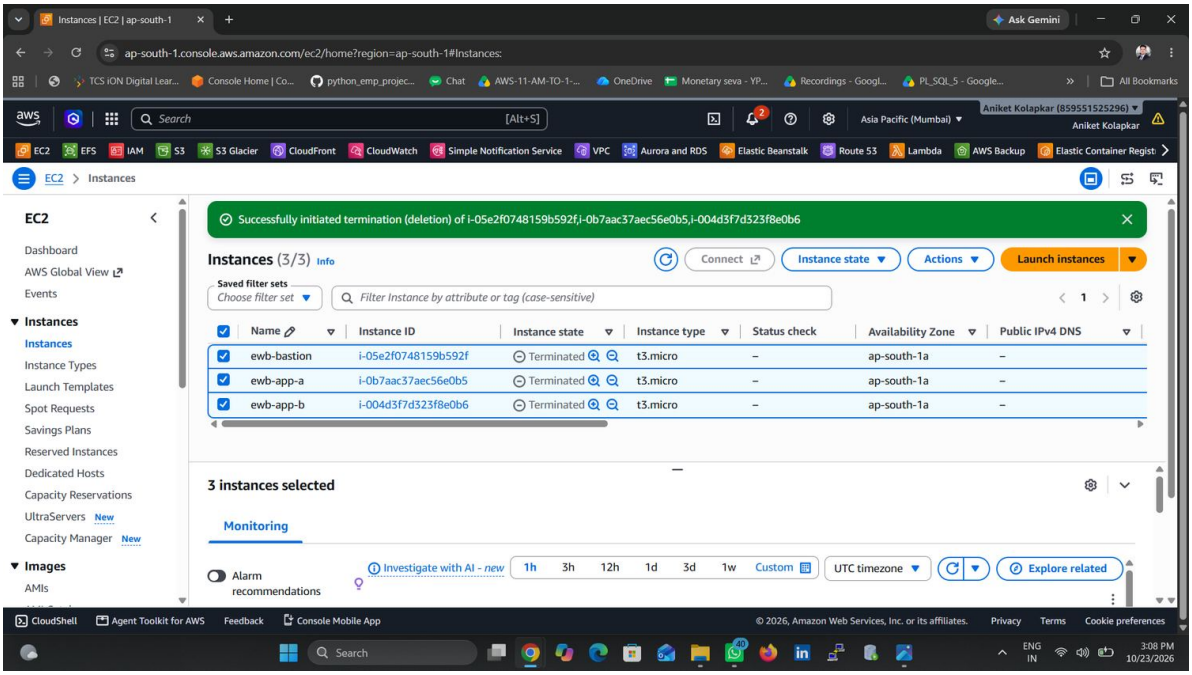
Evidence file: 82.png

Screenshot 83 — Inspect route tables, subnet associations, and connectivity configuration



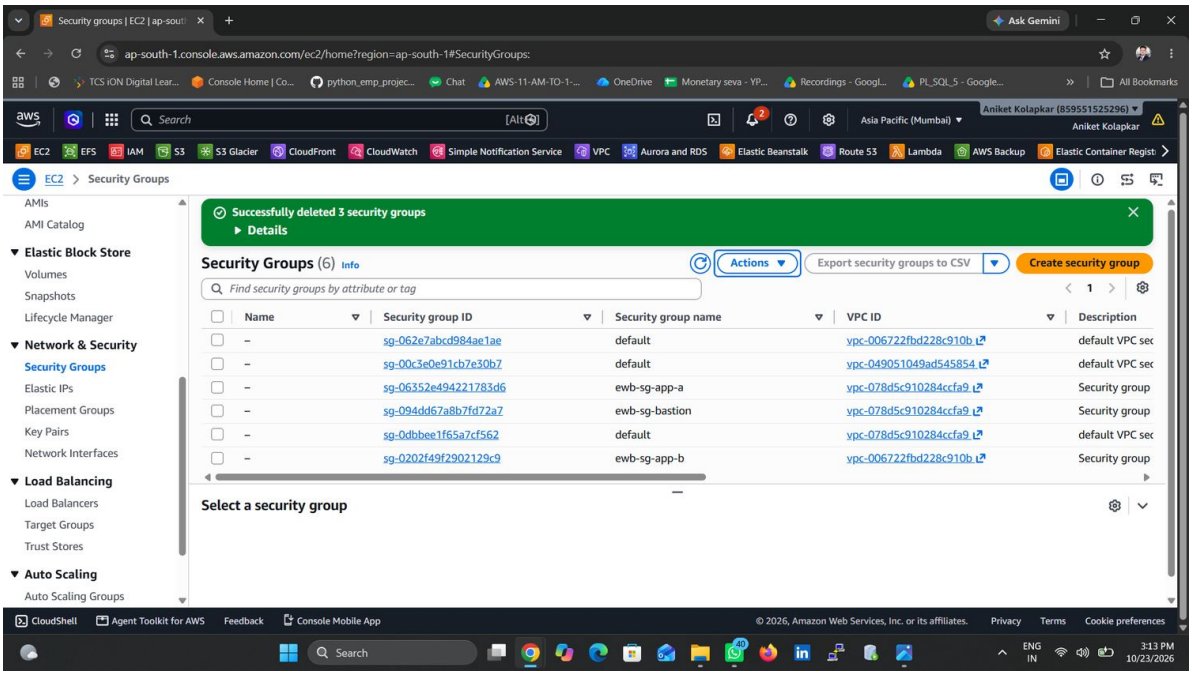
Evidence file: 83.png

Screenshot 84 — Cleanup: remove peering, security groups, and Internet Gateway



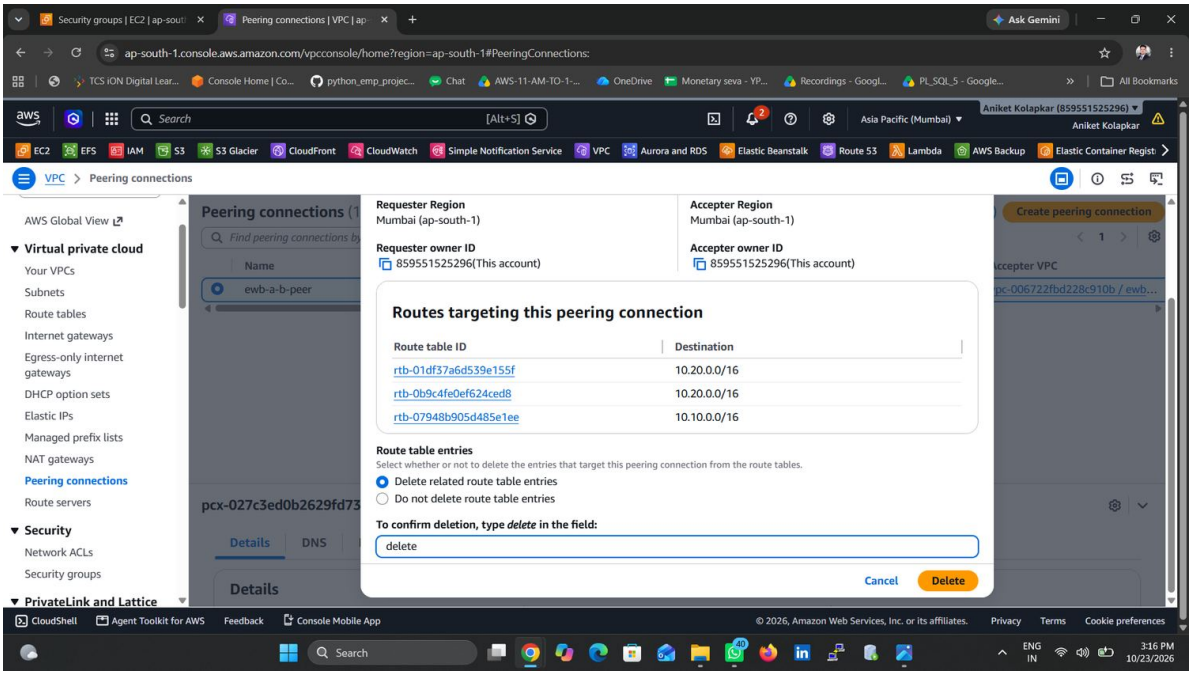
Evidence file: 84.png

Screenshot 85 — Cleanup: remove peering, security groups, and Internet Gateway



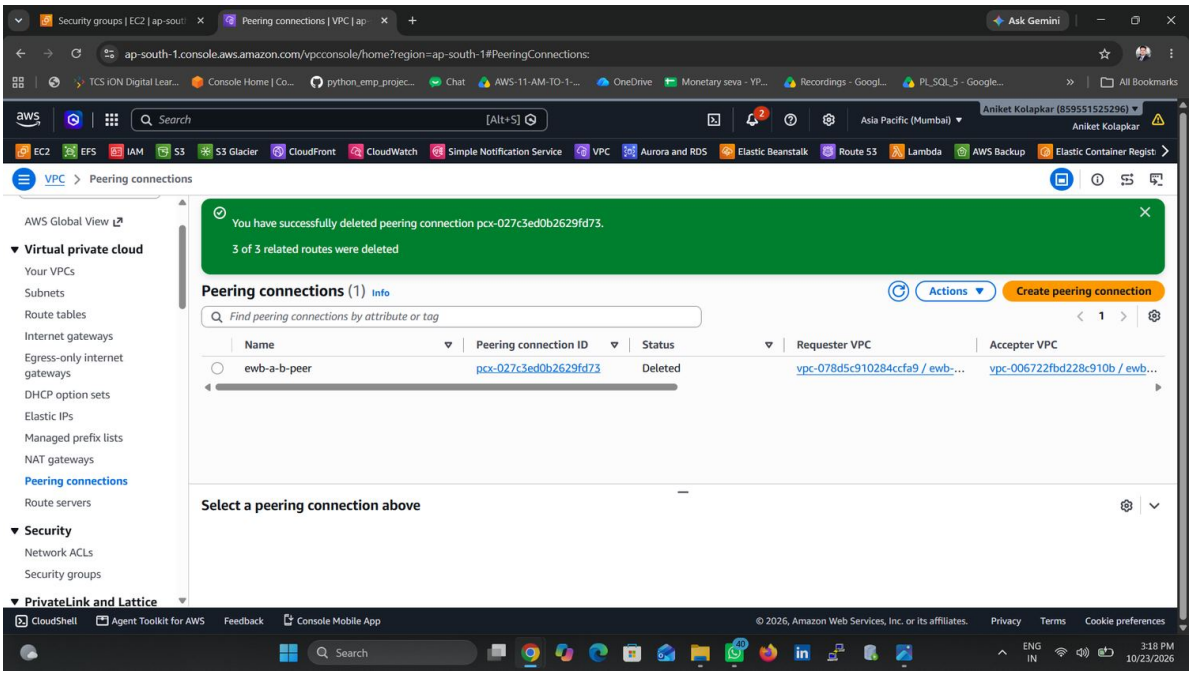
Evidence file: 85.png

Screenshot 86 — Cleanup: remove peering, security groups, and Internet Gateway



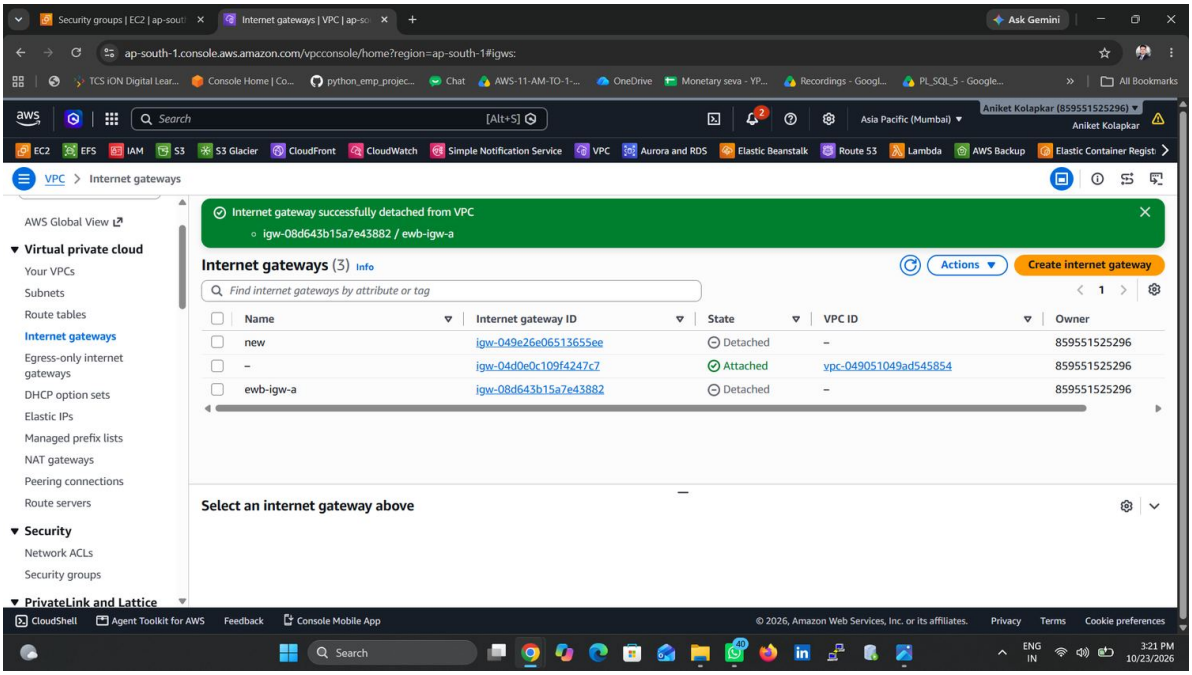
Evidence file: 86.png

Screenshot 87 — Cleanup: remove peering, security groups, and Internet Gateway



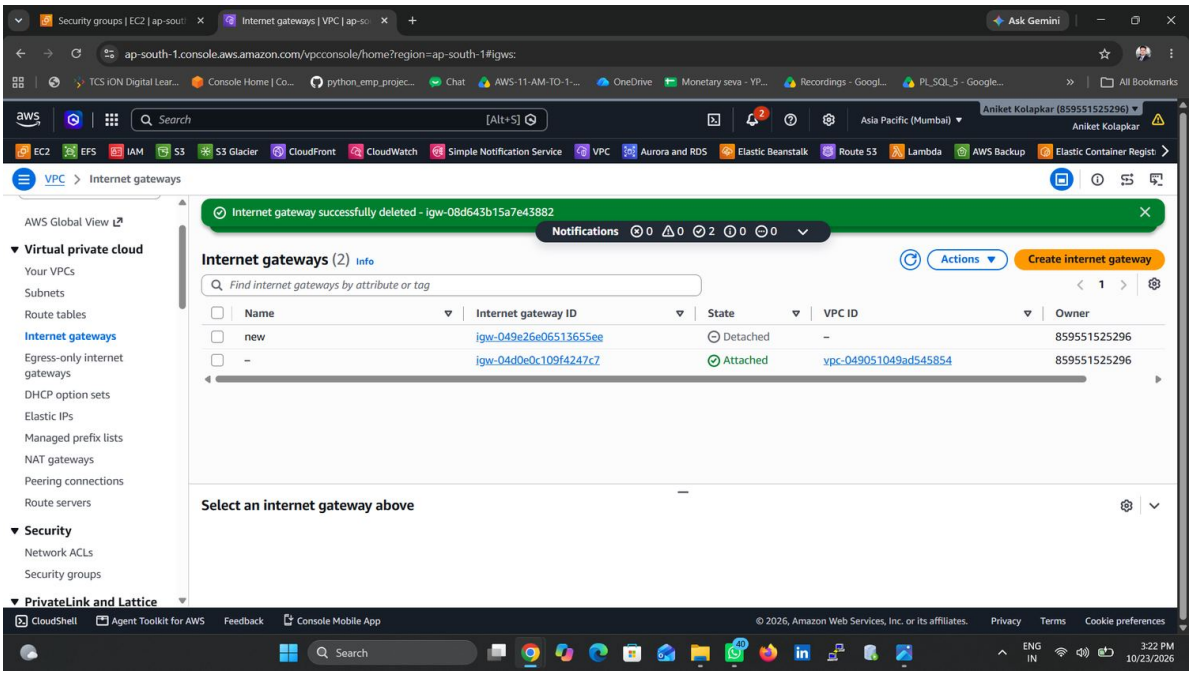
Evidence file: 87.png

Screenshot 88 — Cleanup: remove peering, security groups, and Internet Gateway



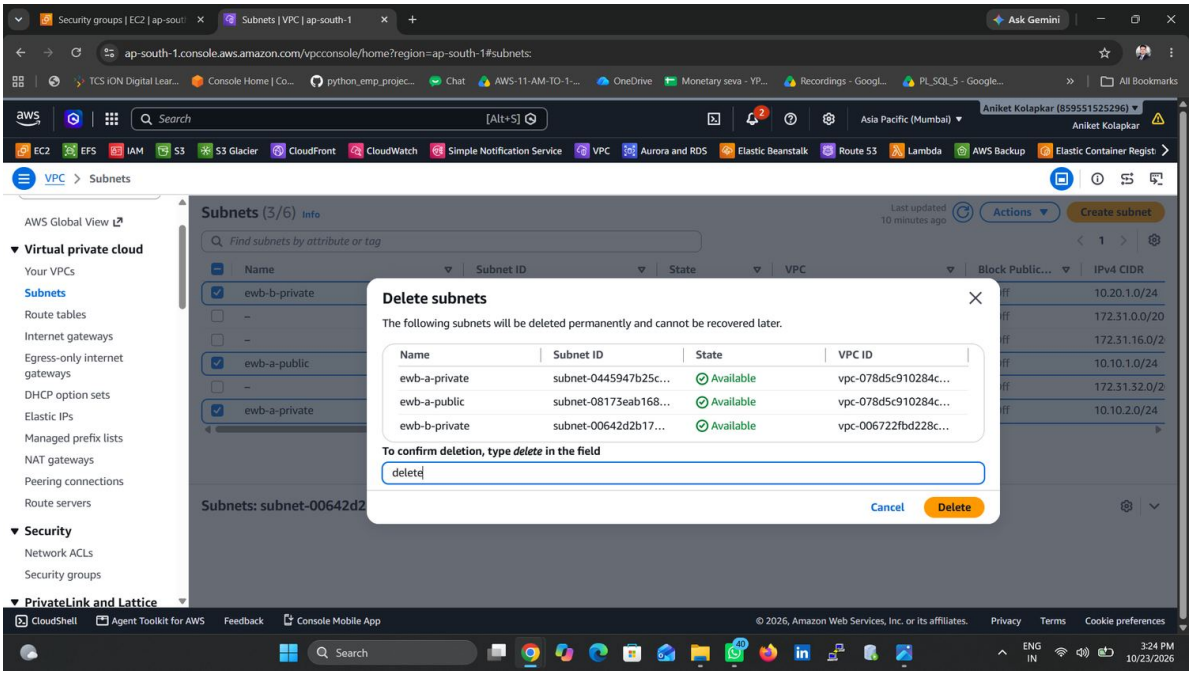
Evidence file: 88.png

Screenshot 89 — Cleanup: remove peering, security groups, and Internet Gateway



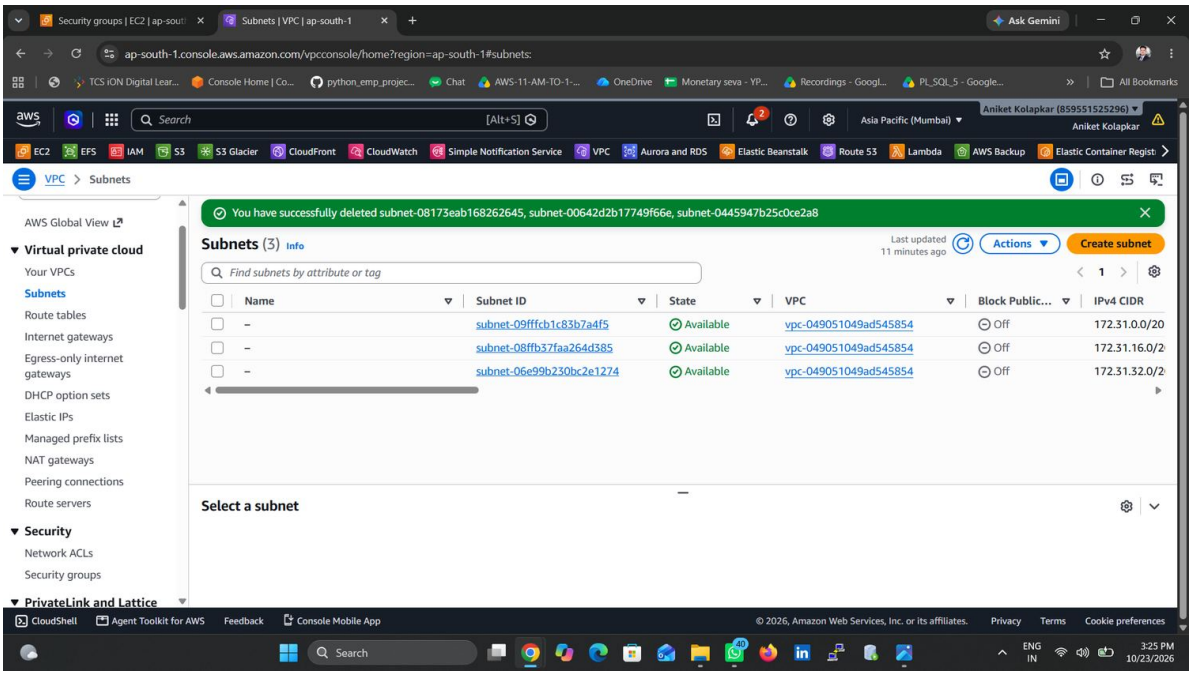
Evidence file: 89.png

Screenshot 90 — Cleanup: remove peering, security groups, and Internet Gateway



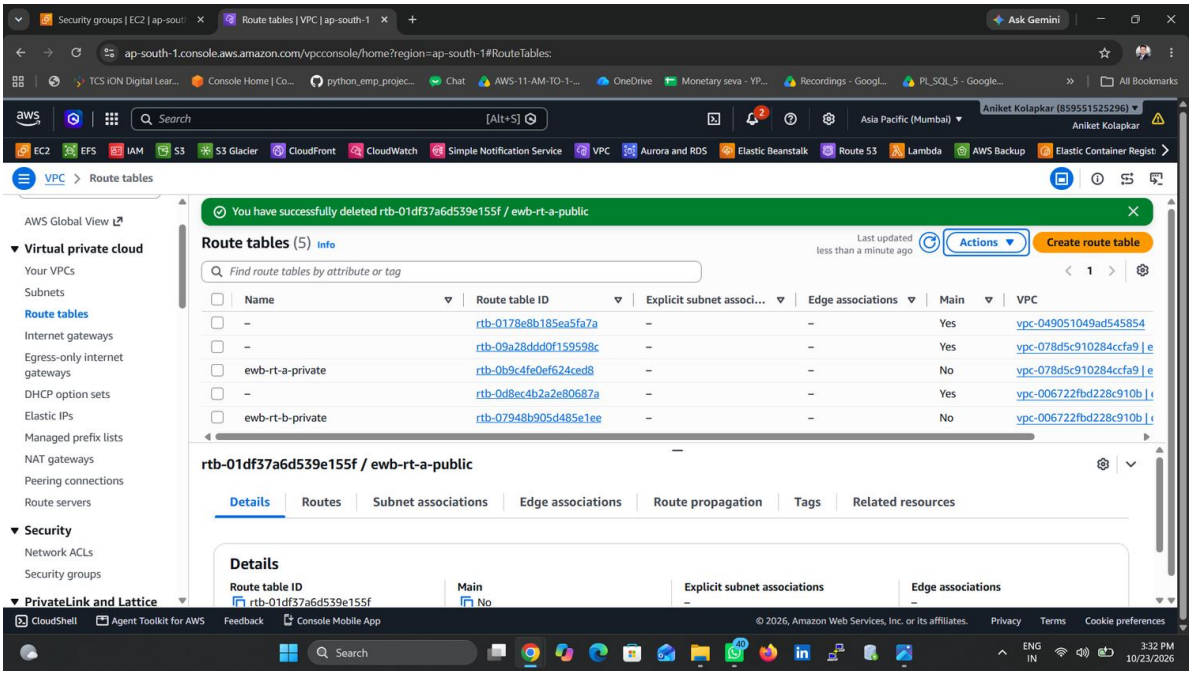
Evidence file: 90.png

Screenshot 91 — Cleanup: remove peering, security groups, and Internet Gateway



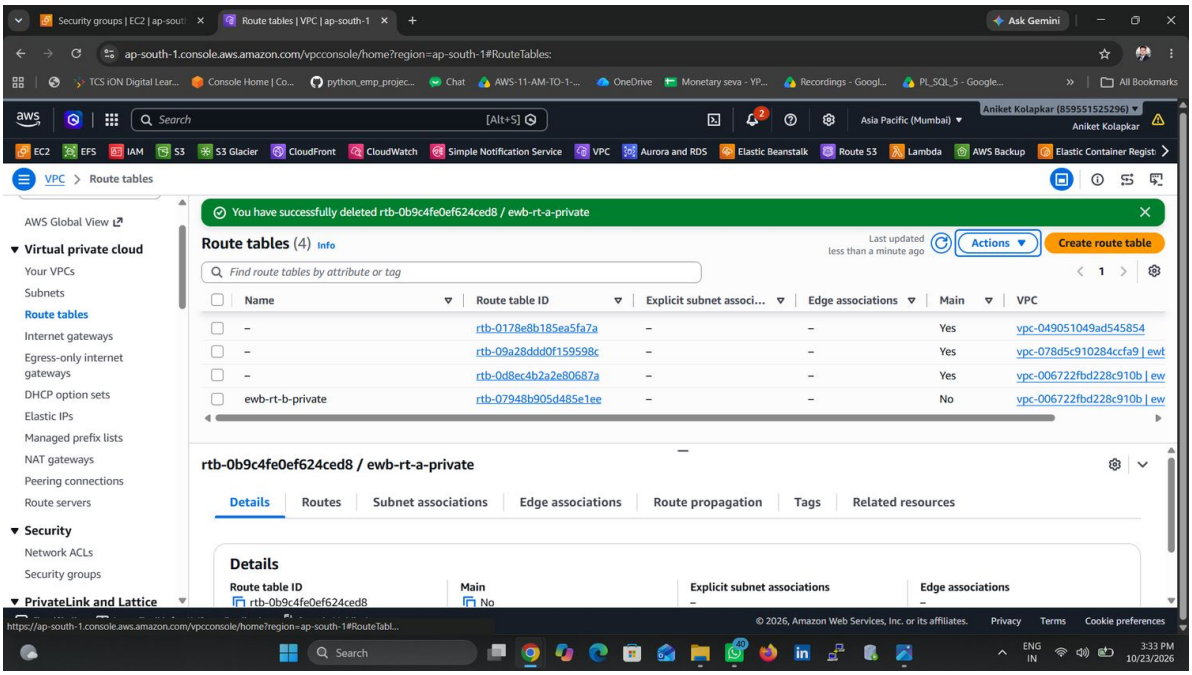
Evidence file: 91.png

Screenshot 92 — Cleanup: verify and remove route table entries



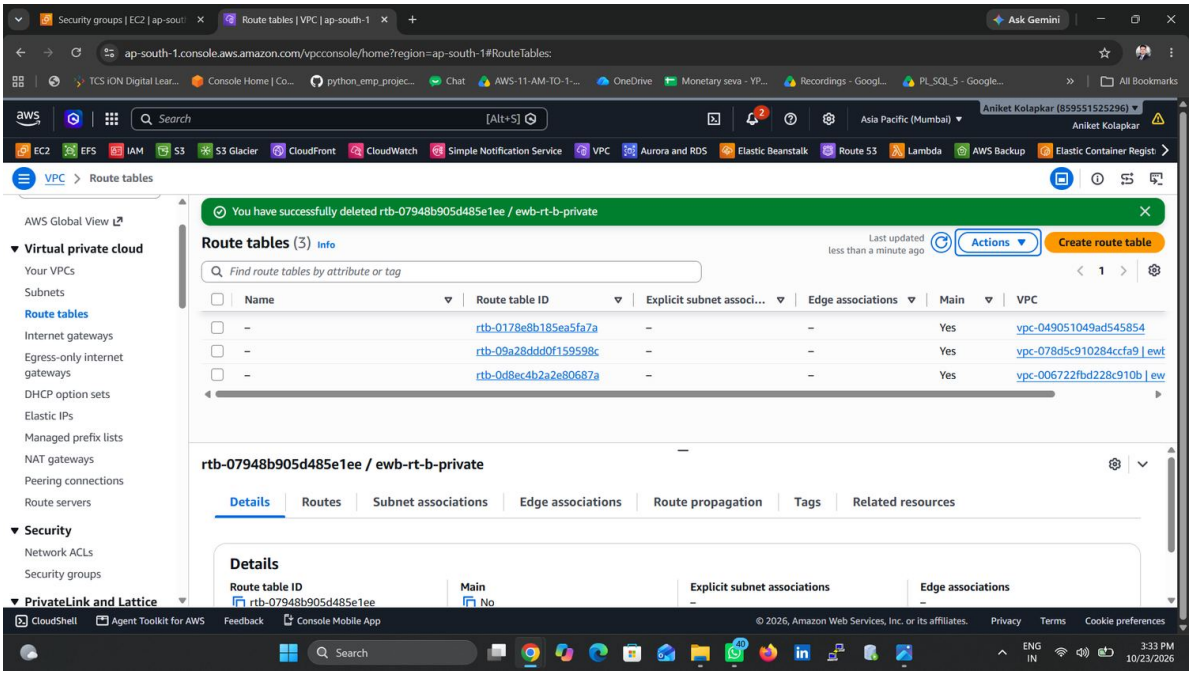
Evidence file: 92.png

Screenshot 93 — Cleanup: verify and remove route table entries



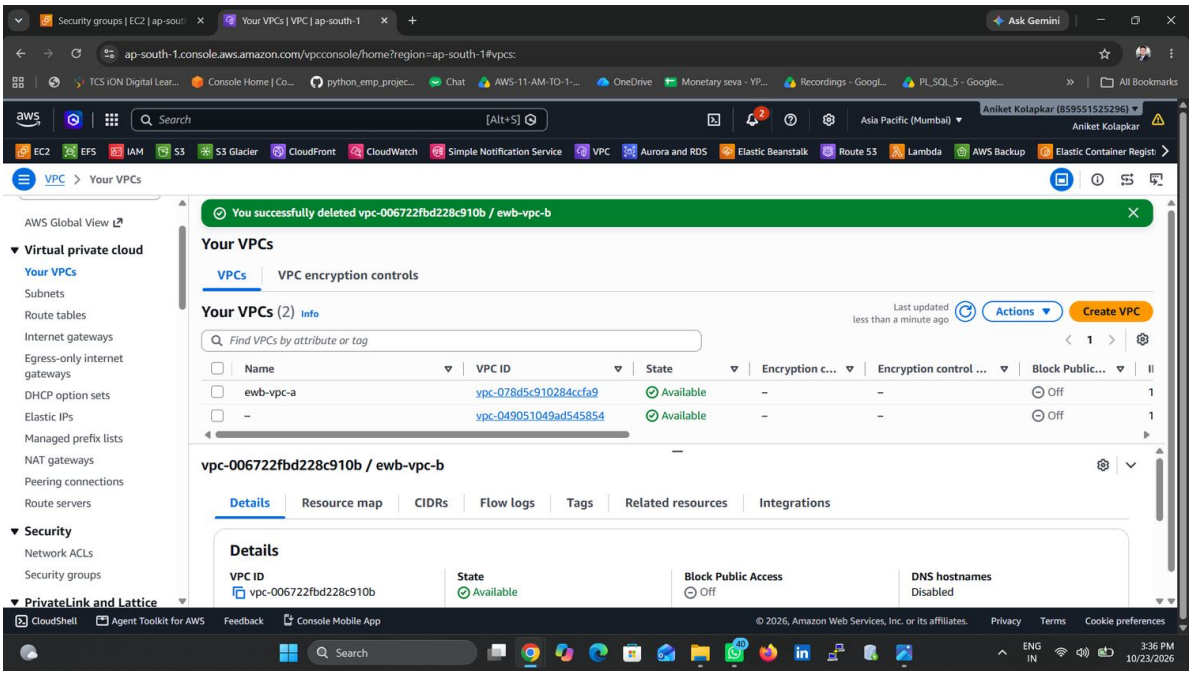
Evidence file: 93.png

Screenshot 94 — Cleanup: verify and remove route table entries



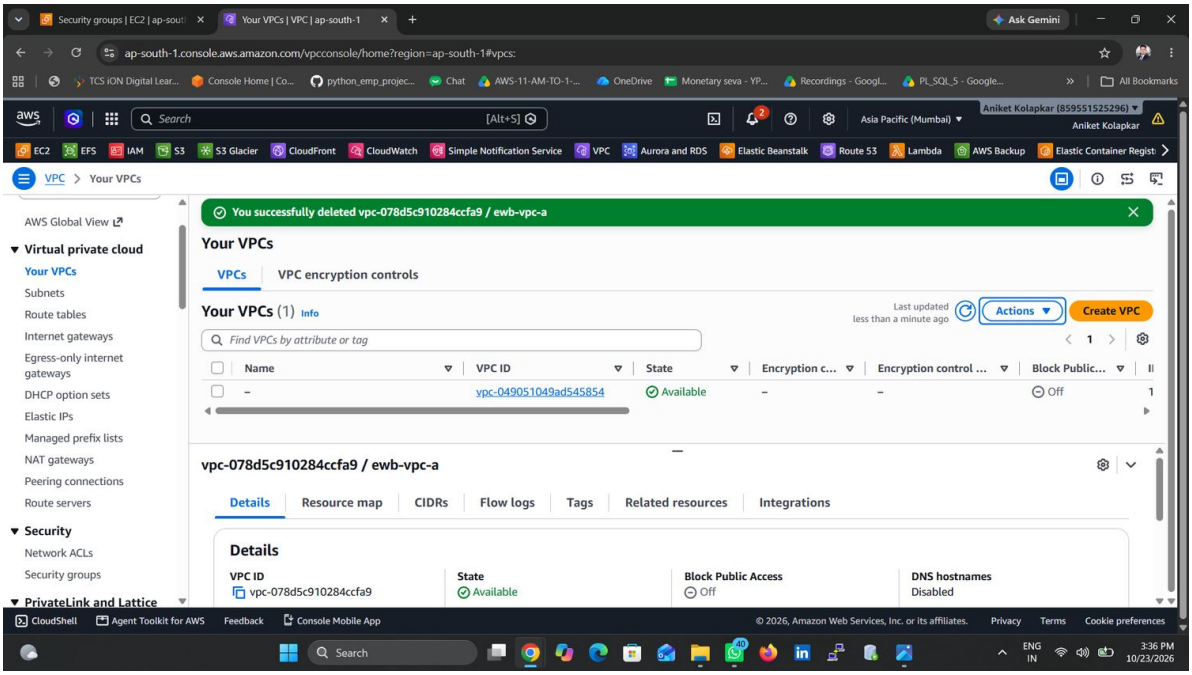
Evidence file: 94.png

Screenshot 95 — Cleanup: verify deletion of project VPC resources



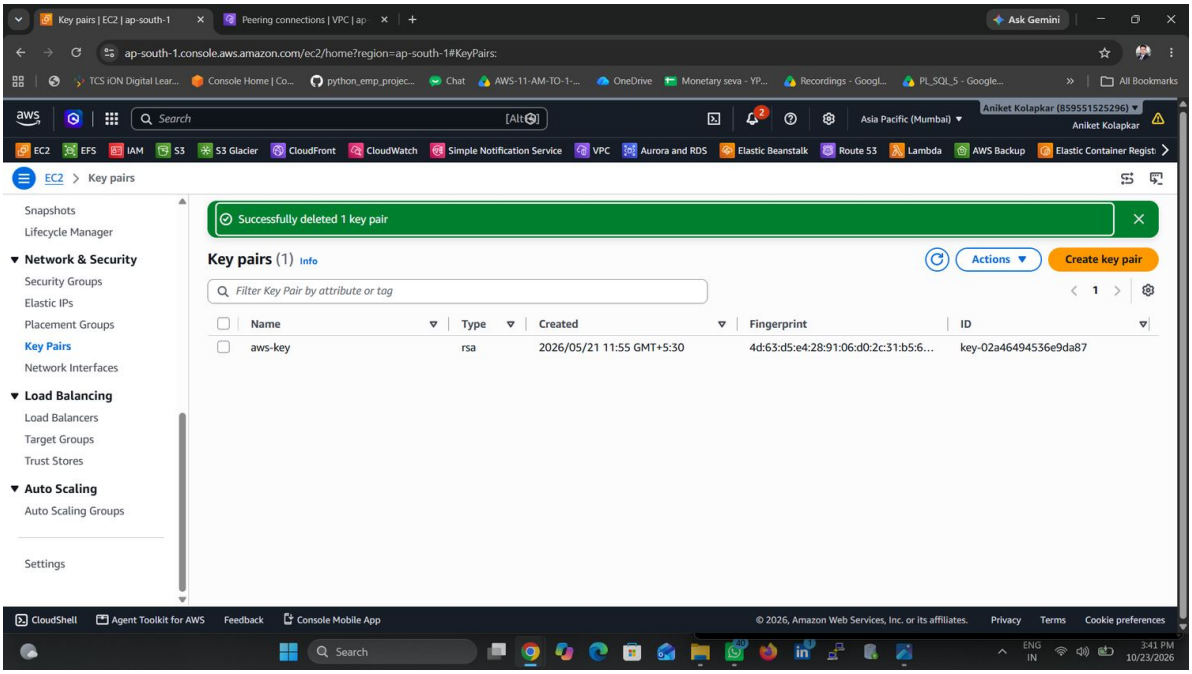
Evidence file: 95.png

Screenshot 96 — Cleanup: verify deletion of project VPC resources



Evidence file: 96.png

Screenshot 97 — Cleanup: verify deletion of project VPC resources



Evidence file: 97.png